Power Systems

Previous Year Questions • 1991–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Power Systems (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

Important Links & Resources

Visit our Website

prepfusion.in

Solutions, practice & report

pyq.prepfusion.in

Subscribe: English Channel

youtube.com/@PrepFusion_GATE

Subscribe: Hindi Channel

youtube.com/@prepfusion-hindi

Join our Telegram group

t.me/All_About_Learning

Follow us on LinkedIn

linkedin.com/company/prepfusionee

Subscribe: Himanshu Agarwal

youtube.com/@HimanshuAgarwal_

Subscribe: Anish Saha

youtube.com/@AnishSaha_

Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

This book exists because of the people below, who built, reviewed and typeset its content.

Design & Layout Architect — Nawras Ahamed.

Technical Reviewers — Ayush Prakash Singh, Abhishek Bhattacharjee, Shujaut Yaseen, Yuva Kishore Gottapu, Manepalli Pavan Vignesh Kumar, Palak Agarwal, Devmalya Ghatak, Kasi Viswanath A, Suresh Ravanam, Aabhas Anuraag Suman and Piyush Raj.

Content Developers — A S Srimithra, Chinthada Lakshmi Prasad, Gantasala Ravi Teja, Guduru Vinayak Varma, Harshaditya Punera, Katikala Sarath, Pranitha Musunuru, Safdar Nazim, Sai Mahalakshmi Janjanam, Shaila Daniels, Sharvari Patwardhan, Sumedha Ghosh, Vinay K Gowda, Aayushi Ghosh, Chevuri Siva Naga Sai Abhi Chandar, Govind Sharma, Kasina Sai Surya Vamsi, Machanpally Karthik Kumar, Seeku Vamsi Krishna and Suprio Dutta.

Contents

Sl.	Topic	Pages
1	Basics of Power System	1
2	Power System Faults	7
3	Power System Stability	18
4	Power System Matrices and Load Flow	25
5	Performance and Models of Transmission Lines and Cables	39
6	Parameters of Transmission Lines	43
7	Power Factor Improvement and Voltage Control Techniques	48
8	Economic Load Dispatch	53
9	Overhead Transmission Lines (O.H.T.L.) and Underground Cables (UG Cable)	59
10	Distribution Systems, HVDC, Sag and Corona	62
11	Relays Circuit Breakers	67

PrepFusion

UNIT

I

Basics of Power System

Topics

1. 3 phase systems
2. Single Line Diagrams and p.u. analysis
3. AC and DC Generation Concepts

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 28	**MCQ:** 22	**MSQ:** 1	**NAT:** 5
1M: 16	**2M:** 11	**Descriptive:** 1	**Avg Weightage** ~2M	**Range:** 1M(2022)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q1.24.1 MSQ 2024 2M

Which of the following options is/are correct for the Automatic Generation Control (AGC) and Automatic Voltage Regulator (AVR) installed with synchronous generators?

- **A.** AGC response has a local effect on frequency while AVR response has a global effect on voltage.
- **B.** AGC response has a global effect on frequency while AVR response has a local effect on voltage.
- **C.** AGC regulates the field current of the synchronous generator while AVR regulates the generator's mechanical power input.
- **D.** AGC regulates the generator's mechanical power input while AVR regulates the field current of the synchronous generator.

Q1.23.1 NAT 2023 2M

A balanced delta connected load consisting of the series connection of one resistor ($R = 15\ \Omega$) and a capacitor ($C = 212.21\ \mu$F) in each phase is connected to three-phase, 50 Hz, 415 V supply terminals through a line having an inductance of $L = 31.83$ mH per phase, as shown in figure below.

Considering the change in the supply terminal voltage with loading to be negligible, the magnitude of the voltage across the terminals V_{AB} in volts is

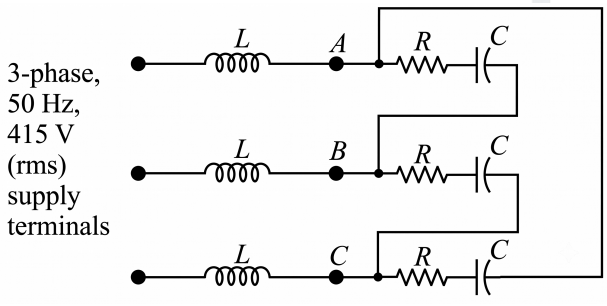


Fig. Q1.23.1

Q1.22.1 NAT 2022 1M

Two balanced three-phase loads, as shown in the figure, are connected to a $100\sqrt{3}$ V, three-phase, 50 Hz main supply. Given $Z_1 = (18 + j24)\ \Omega$ and $Z_2 = (6 + j8)\ \Omega$. The ammeter reading, in amperes, is _____. (round off to nearest integer)

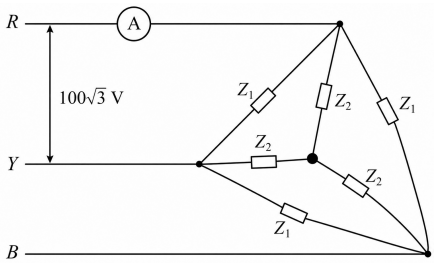


Fig. Q1.22.1

Q1.16.1 MCQ 2016 2M

In the given network $V_1 = 100\angle 0°$ V, $V_2 = 100\angle -120°$ V, $V_3 = 100\angle +120°$ V. The phasor current i (in Ampere) is

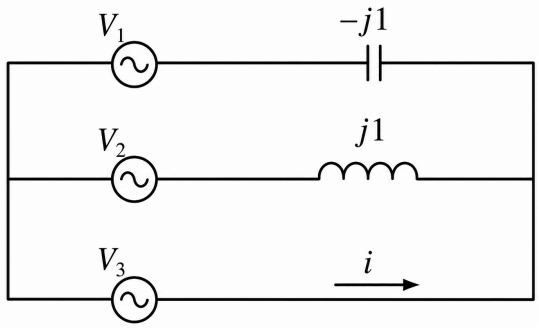


Fig. Q1.16.1

A. $173.2\angle -60°$ **B.** $173.2\angle 120°$
C. $100.0\angle -60°$ **D.** $100.0\angle 120°$

Q1.15.1 MCQ 2015 1M

Base load power plants are:

P: wind farms.
Q: run-of-river plants.
R: nuclear power plants.
S: diesel power plants.

A. P, Q and S only **B.** P, R and S only
C. P, Q and R only **D.** Q and R only

Q1.14.1 NAT 2014 1M

A three phase star-connected load is drawing power at a voltage of 0.9 pu and 0.8 power factor lagging. The three phase base power and base current are 100 MVA and 437.38 A respectively. The line-to-line load voltage in kV is _____.

Q1.14.2 MCQ 2014 1M

The line A to neutral voltage is $10\angle 15°$ V for a balanced three phase star-connected load with phase sequence ABC. The voltage of line B with respect to line C is given by

A. $10\sqrt{3}\angle 105°$ V
B. $10\angle 105°$ V
C. $10\sqrt{3}\angle -75°$ V
D. $-10\sqrt{3}\angle 90°$ V

Q1.10.1 MCQ 2010 2M

For the power system shown in the figure below, the specifications of the components are:

G_1 : 25 kV, 100 MVA, $X = 9\%$
G_2 : 25 kV, 100 MVA, $X = 9\%$
T_1 : 25 kV/220 kV, 90 MVA, $X = 12\%$
T_2 : 220 kV/25 kV, 90 MVA, $X = 12\%$
Line 1 : 220 kV, $X = 150\ \Omega$

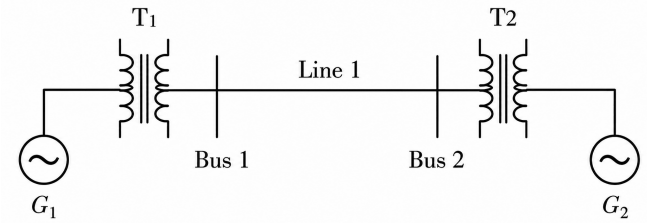
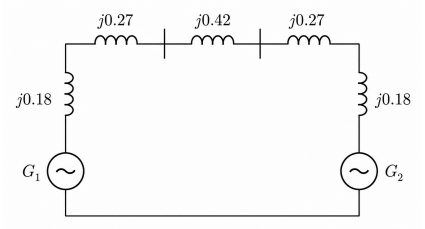
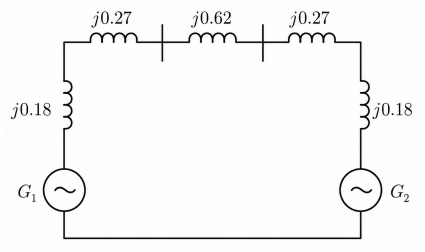


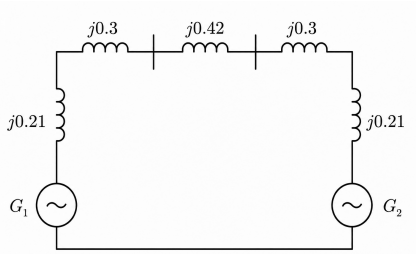
Fig. Q1.10.1

Choose 25 kV as the base voltage at generator G_1, and 200 MVA as the MVA base. The impedance diagram is

A.


B.


C.

D.


Q1.09.1 MCQ 2009 1M

Out of the following plant categories

(i) Nuclear

(ii) Run-of-river

(iii) Pump Storage

(iv) Diesel

The base-load power plants are

A. (i) and (ii)
B. (ii) and (iii)
C. (i), (ii) and (iv)
D. (i), (iii) and (iv)

Q1.07.1 MCQ 2007 1M

Consider the transformer connections in a part of a power system shown in the figure. The nature of transformer connections and phase shifts are indicated for all but one transformer.

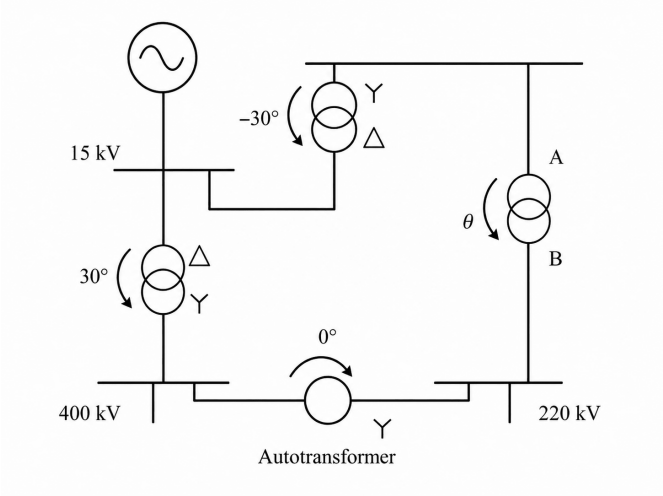


Fig. Q1.07.1

Which of the following connections, and the corresponding phase shift θ, should be used for the transformer between A and B?

A. Star–Star ($\theta = 0°$) **B.** Star–Delta ($\theta = -30°$)
C. Delta–Star ($\theta = 30°$) **D.** Star–Zigzag ($\theta = 30°$)

Q1.07.2 MCQ 2007 2M

A three-phase balanced star-connected voltage source with frequency ω rad/s is connected to a star-connected balanced load which is purely inductive. The instantaneous line currents and phase-to-neutral voltages are denoted by (i_a, i_b, i_c) and (v_{an}, v_{bn}, v_{cn}) respectively, and their rms values are denoted by V and I. If

$$R = \begin{bmatrix} v_{an} & v_{bn} & v_{cn} \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & 0 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

then the magnitude of R is

A. $3VI$ **B.** VI **C.** $0.7VI$ **D.** 0

Q1.06.1 MCQ 2006 2M

A generator is connected through a 20 MVA, 13.8/138 kV step down transformer, to a transmission line. At the receiving end of the line a load is supplied through a step down transformer of 10 MVA, 138/69 kV rating. A 0.72 pu load, evaluated on load side transformer ratings as base values, is supplied from the above system. If system base values of 10 MVA and 69 kV in load circuit are chosen, the value of the load (in per unit) in generator circuit will be

A. 36 **B.** 1.44 **C.** 0.72 **D.** 0.18

Q1.05.1 MCQ 2005 1M

The p.u. parameters for a 500 MVA machine on its own base are: Inertia $M = 20$ p.u., Reactance $X = 2$ p.u.. The p.u. values of inertia and reactance on 100 MVA common base, respectively, are

A. 4, 0.4 **B.** 100, 10 **C.** 4, 10 **D.** 100, 0.4

Q1.04.1 MCQ 2004 1M

The rated voltage of a 3-phase power system is given as

A. peak line to line voltage **B.** rms phase voltage
C. rms line to line voltage **D.** peak phase voltage

Q1.04.2 MCQ 2004 1M

The phase sequence of the 3-phase system shown in the figure is

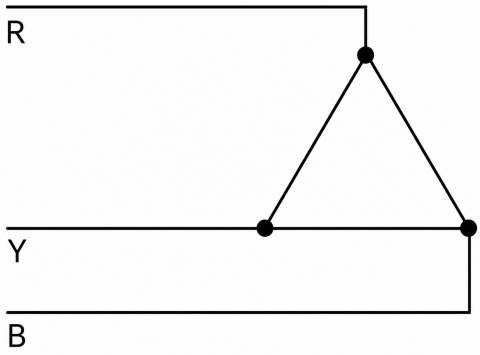


Fig. Q1.04.2

A. RYB **B.** RBY **C.** BRY **D.** YBR

Q1.04.3 MCQ 2004 2M

A new generator with $E_g = 1.4\angle 30°$ pu and $X_s = 1.0$ pu is connected to a bus of an existing power system represented by the Thevenin equivalent $E_{th} = 0.9\angle 0°$ pu in series with $Z_{th} = 0.25\angle 90°$ pu. The magnitude of the bus voltage V_t, in pu, is:

A. 0.990 **B.** 0.973 **C.** 0.963 **D.** 0.900

Q1.04.4 MCQ 2004 1M

In thermal power plants, the pressure in the working fluid cycle is developed by

A. condenser **B.** super heater
C. feed water pump **D.** turbine

Q1.04.5 MCQ 2004 1M

For harnessing low variable water heads, the suitable hydraulic turbine with high percentage of reaction and runner adjustable vanes is

A. Kaplan **B.** Francis **C.** Pelton **D.** Impeller

Q1.01.1 MCQ 2001 1M

A 50 Hz balanced three-phase, Y-connected supply is connected to a balanced three-phase Y-connected load. If the instantaneous phase-a supply voltage is $v_a = V\cos(\omega t)$ and the phase-a load current is $i_a = I\cos(\omega t - \phi)$ the instantaneous three-phase power is

A. a constant with a magnitude of $VI\cos\phi$

B. a constant with a magnitude of $\frac{3}{2}VI\cos\phi$

C. time-varying with an average value of $\frac{3}{2}VI\cos\phi$ and a frequency of 100 Hz

D. time-varying with an average value of $VI\cos\phi$ and a frequency of 50 Hz

Q1.01.2 MCQ 2001 2M

A 75 MVA, 10 kV synchronous generator has $X_d = 0.4$ p.u.. The value of X_d (in p.u.) on a base of 100 MVA, 11 kV is

A. 0.578 **B.** 0.279 **C.** 0.412 **D.** 0.44

Q1.00.1 MCQ 2000 1M

For given base voltage and base volt-amperes, the per unit impedance value of an element is x. What will be the per unit impedance value of this element when the voltage and volt-ampere bases are both doubled?

A. 0.5x **B.** 2x **C.** 4x **D.** x

Q1.00.2 NAT 2000 5M

For perfectly balanced operation a certain three-phase AC power electronic circuit generates odd harmonic currents of order five and seven in the three phases of the AC mains. Identify which of these harmonics form a positive-sequence system and which form a negative-sequence system.

Q1.00.3 MCQ 2000 1M

In a thermal power plant, the feed water coming to the economizer is heated using

A. direct heat in the furnace **B.** H.P. steam
C. L.P. steam **D.** flue gases

Q1.99.1 MCQ 1999 1M

The load carrying capability of a long AC transmission line is:

A. always limited by the conductor size

B. limited by stability considerations

C. reduced at a low ambient temperatures

D. decreased by the use of bundled conductors of single conductors

Q1.99.2 MCQ 1999 2M

An industrial consumer has a daily load pattern of 2000 kW, 0.8 lag for 12 hours, and 1000 kW UPF for 12 hours. The load factor is:

A. 0.5 **B.** 0.75 **C.** 0.6 **D.** 2.0

Q1.97.1 NAT 1997 2M

The synchronous reactance of a 200 MVA, 10 kV, 3-phase, 50 Hz generator is 1.0 p.u. at its own base. Its p.u. reactance at 100 MVA, 20 kV base will be _____

Q1.96.1 MCQ 1996 2M

Which material is used in controlling chain reaction in a nuclear reactor?

A. Thorium **B.** Heavy water **C.** Boron **D.** Beryllium

Q1.95.1 MCQ 1995 1M

In order to have a lower cost of electrical energy generation,

A. The load factor and diversity factor should be low

B. The load factor should be low but diversity factor should be high

C. The load factor should be high but diversity factor should be low

D. The load factor and diversity factor should be high

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q1.24.1 B,D	Q1.23.1 415	Q1.22.1 20
Q1.16.1 A	Q1.15.1 C	Q1.14.1 117 to 120
Q1.14.2 C	Q1.10.1 B	Q1.09.1 A
Q1.07.1 A	Q1.07.2 D	Q1.06.1 C
Q1.05.1 D	Q1.04.1 C	Q1.04.2 B
Q1.04.3 B	Q1.04.4 C	Q1.04.5 A
Q1.01.1 B	Q1.01.2 D	Q1.00.1 A
Q1.00.2 *	Q1.00.3 D	Q1.99.1 B
Q1.99.2 B	Q1.97.1 0.125	Q1.96.1 C
Q1.95.1 D		

PrepFusion

UNIT

II

Power System Faults

Topics

1. Types of Symmetrical Faults
2. Reactors in Power System
3. Transient Behaviour of Alternator during Fault
4. Balanced vs Unbalanced Systems
5. Symmetrical Components
6. Sequence Networks
7. Types of Unsymmetrical Fault

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 53	**MCQ:** 33	**MSQ:** 1	**NAT:** 19
1M: 15	**2M:** 34	**Descriptive:** 4	**Avg Weightage** ~3M	**Range:** 1M(2026)–4M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

The initial three-phase voltage phasors $(\bar{V}_A, \bar{V}_B, \bar{V}_C)$ at a bus of a power network are as shown in **Case-1**. Due to a disturbance, the bus voltage phasors changed in phase by a small angle $\Delta\theta$, and the magnitudes remained the same as depicted in **Case-2**.

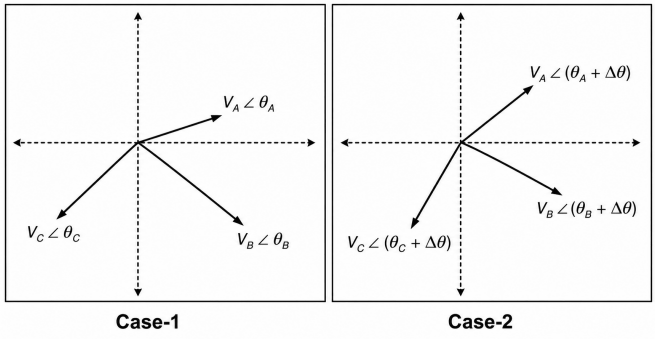


Fig. Q2.26.1

Which one of the following statements is correct about the zero sequence components?

A. The zero sequence components in Case-1 and Case-2 have the same phase angle and magnitude

B. The magnitude of the zero sequence component in Case-1 is greater than that in Case-2

C. The magnitude of the zero sequence component in Case-2 is greater than that in Case-1

D. The zero sequence components in Case-1 and Case-2 have the same magnitude but different phase angles

Q2.24.1 MSQ 2024 2M

For a two-phase network, the phase voltages V_p and V_q are related to the sequence voltages V_α and V_β by $\begin{bmatrix} V_p \\ V_q \end{bmatrix} = \mathbf{S} \begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix}$. The possible option(s) for the matrix **S** is/are:

☐ **A.** $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

☐ **B.** $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

☐ **C.** $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

☐ **D.** $\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$

Q2.23.1 NAT 2023 2M

The two-bus power system shown in figure (i) has one alternator supplying a synchronous motor load through a Y-Δ transformer. The positive, negative and zero-sequence diagrams of the system are shown in figures (ii), (iii) and (iv), respectively. All reactances in the sequence diagrams are in p.u. If a bolted line-to-line fault (fault impedance = zero) between phases '*b*' and '*c*' at bus 1, neglecting the pre-fault currents, the magnitude of the fault current (from phase '*b*' to '*c*') in p.u. is _____ (Round off to 2 decimal places).

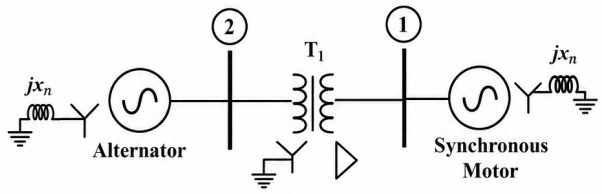


figure (i): Single-line diagram of the power system

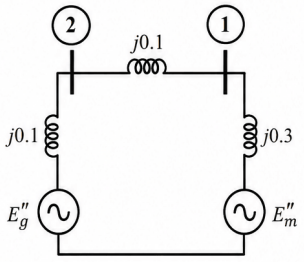


figure (ii): Positive-sequence network

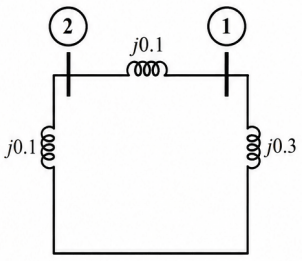


figure (iii): Negative-sequence network

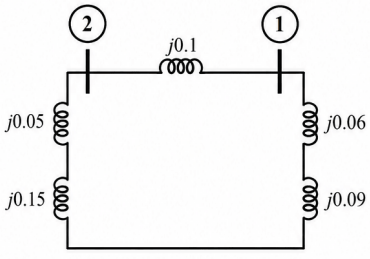


figure (iv): Zero-sequence network

Fig. Q2.23.1

Q2.22.1 MCQ 2022 1M

The valid positive, negative and zero sequence impedances (in p.u.), respectively, for a 220 kV, fully transposed three-phase transmission line, from the given choices are

A. 1.1, 0.15 and 0.08 **B.** 0.15, 0.15 and 0.35
C. 0.2, 0.2 and 0.2 **D.** 0.1, 0.3 and 0.1

Q2.21.1 MCQ 2021 2M

Suppose I_A, I_B and I_C are a set of unbalanced current phasors in a three-phase system. The phase-B zero-sequence current $I_{B0} = 0.1\angle 0°$ p.u. If phase-A current $I_A = 1.1\angle 0°$ p.u and phase-C current $I_C = (1\angle 120° + 0.1)$ p.u, then I_B in p.u is

A. $1\angle 240° - 0.1\angle 0°$ **B.** $1.1\angle 240° - 0.1\angle 0°$
C. $1.1\angle -120° + 0.1\angle 0°$ **D.** $1\angle -120° + 0.1\angle 0°$

Q2.20.1 NAT 2020 2M

A cylindrical rotor synchronous generator has steady-state synchronous reactance of 0.7 p.u. and subtransient reactance of 0.2 p.u. It is operating at $(1 + j0)$ p.u. terminal voltage with an internal emf of $(1 + j0.7)$ p.u. Following a three-phase solid short circuit fault at the terminal of the generator, the magnitude of the subtransient internal emf (rounded off to 2 decimal places) is _____ p.u.

Q2.19.1 NAT 2019 2M

A 30 kV, 50 Hz, 50 MVA generator has the positive, negative, and zero sequence reactances of 0.25 pu, 0.15 pu, and 0.05 pu, respectively. The neutral of the generator is grounded with a reactance such that the fault current for a bolted LG fault and that of a bolted three-phase fault at the generator terminal are equal. The value of grounding reactance in ohms (round off to one decimal place) is _____.

Q2.18.1 NAT 2018 1M

The series impedance matrix of a short three-phase transmission line in phase coordinates is

$$\begin{bmatrix} Z_s & Z_m & Z_m \\ Z_m & Z_s & Z_m \\ Z_m & Z_m & Z_s \end{bmatrix}$$

If the positive sequence impedance is $(1 + j10)\ \Omega$, and the zero sequence impedance is $(4 + j31)\ \Omega$, then the imaginary part of Z_m (in Ω) is _____ (up to 2 decimal places).

Q2.18.2 NAT 2018 1M

The positive, negative and zero sequence impedances of a 125 MVA, three-phase, 15.5 kV, star-grounded, 50 Hz generator are $j0.1$ pu, $j0.05$ pu and $j0.01$ pu respectively on the machine rating base. The machine is unloaded and working at the rated terminal voltage. If the grounding impedance of the generator is $j0.01$ pu, then the magnitude of fault current for a b-phase to ground fault (in kA) is _____ (up to 2 decimal places).

Q2.18.3 MCQ 2018 2M

The positive, negative and zero sequence impedances of a three phase generator are Z_1, Z_2 and Z_0 respectively. For a line-to-line fault with fault impedance Z_f, the fault current is kI_f, where I_f is the fault current with zero fault impedance. The relation between Z_f and k is

A. $Z_f = \dfrac{(Z_1 + Z_2)(1 - k)}{k}$ **B.** $Z_f = \dfrac{(Z_1 + Z_2)(1 + k)}{k}$
C. $Z_f = \dfrac{(Z_1 + Z_2)k}{1 - k}$ **D.** $Z_f = \dfrac{(Z_1 + Z_2)k}{1 + k}$

Q2.17.1 NAT 2017 2M

The positive, negative, and zero sequence reactances of a wye-connected synchronous generator are 0.2 pu, 0.2 pu, and 0.1 pu, respectively. The generator is on open circuit with a terminal voltage of 1 pu. The minimum value of the inductive reactance, in pu, required to be connected between neutral and ground so that the fault current does not exceed 3.75 pu if a single line to ground fault occurs at the terminals is _____ (assume fault impedance to be zero). (Give the answer up to one decimal place.)

Q2.16.1 NAT 2016 1M

The magnitude of three-phase fault currents at buses A and B of a power system are 10 pu and 8 pu, respectively. Neglect all resistances in the system and consider the pre-fault system to be unloaded. The pre-fault voltage at all buses in the system is 1.0 pu. The voltage magnitude at bus B during a three-phase fault at bus A is 0.8 pu. The voltage magnitude at bus A during a three-phase fault at bus B, in pu, is _____.

Q2.16.2 NAT 2016 2M

The single line diagram of a balanced power system is shown in the figure. The voltage magnitude at the generator internal bus is constant and 1.0 p.u. The p.u. reactances of different components in the system are also shown in the figure. The infinite bus voltage magnitude is 1.0 p.u. A three phase fault occurs at the middle of line 2.

The ratio of the maximum real power that can be transferred during the pre-fault condition to the maximum real power that can be transferred under the faulted condition is _____.

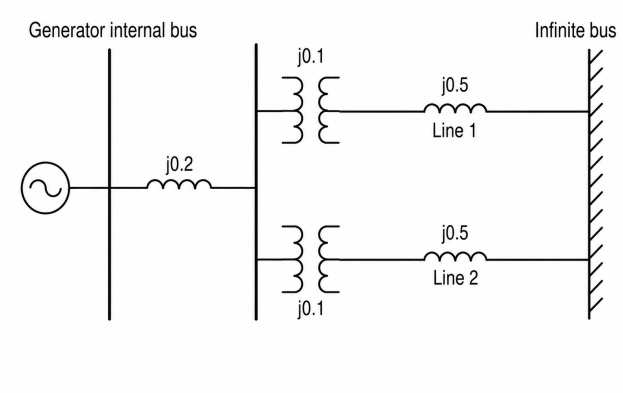


Fig. Q2.16.2

Q2.16.3 NAT 2016 2M

A 30 MVA, 3-phase, 50 Hz, 13.8 kV, star-connected synchronous generator has positive, negative and zero sequence reactances, 15%, 15% and 5% respectively. A reactance (X_n) is connected between the neutral of the generator and ground. A double line to ground fault takes place involving phases 'b' and 'c', with a fault impedance of $j0.1$ p.u. The value of X_n (in p.u.) that will limit the positive sequence generator current to 4270 A is _____.

Q2.16.4 NAT 2016 2M

Two identical unloaded generators are connected in parallel as shown in the figure. Both the generators are having positive, negative and zero sequence impedances of $j0.4$ p.u., $j0.3$ p.u. and $j0.15$ p.u., respectively. If the pre-fault voltage is 1 p.u., for a line-to-ground (L-G) fault at the terminals of the generators, the fault current, in p.u., is _____.

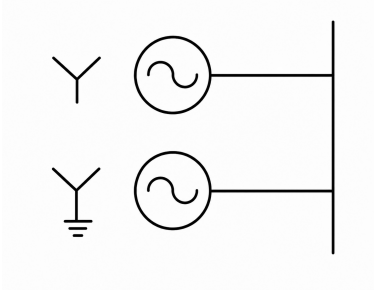


Fig. Q2.16.4

Q2.15.1 MCQ 2015 2M

A sustained three-phase fault occurs in the power system shown in the figure. The current and voltage phasors during the fault (on a common reference), after the natural transients have died down, are also shown. Where is the fault located?

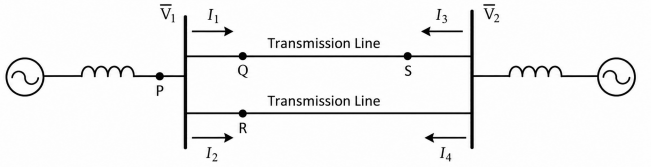


Fig. Q2.15.1

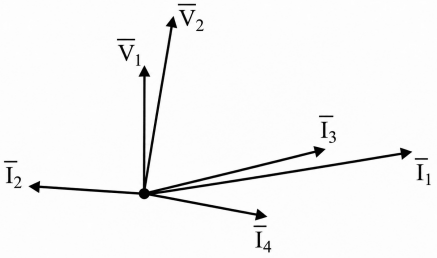


Fig. Q2.15.1

A. Location P **B.** Location Q **C.** Location R **D.** Location S

Q2.14.1 MCQ 2014 1M

Three-phase to ground fault takes place at locations F_1 and F_2 in the system shown in the figure.
If the fault takes place at location F_1, then the voltage and the current at bus A are V_{F1} and I_{F1} respectively. If the fault takes place at location F_2, then the voltage and the current at bus A are V_{F2} and I_{F2} respectively. The correct statement about voltages and currents during faults at F_1 and F_2 is

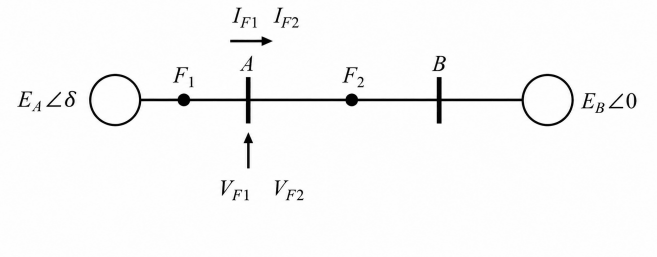


Fig. Q2.14.1

A. V_{F1} leads I_{F1} and V_{F2} leads I_{F2}

B. V_{F1} leads I_{F1} and V_{F2} lags I_{F2}

C. V_{F1} lags I_{F1} and V_{F2} leads I_{F2}

D. V_{F1} lags I_{F1} and V_{F2} lags I_{F2}

Q2.14.2 MCQ 2014 1M

A 2-bus system and corresponding zero sequence network are shown in the figure.

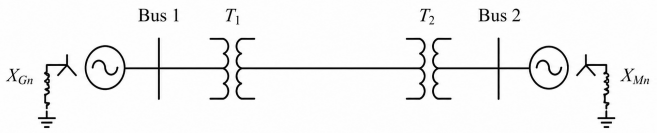


Fig. Q2.14.2

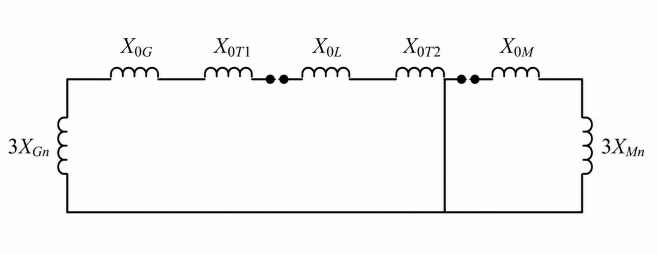


Fig. Q2.14.2

The transformers T_1 and T_2 are connected as

A. Y−Y and Y−Δ
B. Y_g−Y and Y_g−Δ
C. Y_g−Δ and Δ−Y_g
D. Δ−Y_g and Y−Y_g

Q2.14.3 MCQ 2014 2M

In an unbalanced three-phase system, $I_a = 1\angle(-90°)$ pu, $I_{b2} = 4\angle(-150°)$ pu, and $I_{c0} = 3\angle 90°$ pu. The magnitude of phase current I_b, in pu, is:

A. 1.00 **B.** 7.81 **C.** 11.53 **D.** 13.00

Q2.14.4 NAT 2014 2M

A three phase, 100 MVA, 25 kV generator has solidly grounded neutral. The positive, negative, and zero sequence reactances of the generator are 0.2 pu, 0.2 pu, and 0.05 pu, respectively, at the machine base quantities. If a bolted single phase to ground fault occurs at the terminal of the unloaded generator, the fault current in amperes immediately after the fault is _____.

Q2.14.5 MCQ 2014 1M

For a fully transposed transmission line

A. positive, negative and zero sequence impedances are equal

B. positive and negative sequence impedances are equal

C. zero and positive sequence impedances are equal

D. negative and zero sequence impedances are equal

Q2.12.1 MCQ 2012 1M

The sequence components of the fault current are as follows:

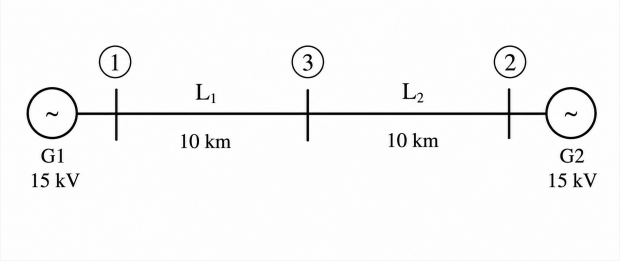
$I_{\text{positive}} = j1.5$ pu, $I_{\text{negative}} = -j0.5$ pu, $I_{\text{zero}} = -j1$ pu

The type of fault in the system is

A. LG **B.** LL **C.** LLG **D.** LLLG

Common Data Questions

Answer the questions Q2.11.1 and Q2.11.2 using the following data.
Two generator units G_1 and G_2 are connected by a 15 kV transmission line with a bus at the mid-point as shown below.



Generator data:

G_1 : 250 MVA, 15 kV, $X_1 = 25\%$

G_2 : 100 MVA, 15 kV, $X_2 = 10\%$

Transmission line data:

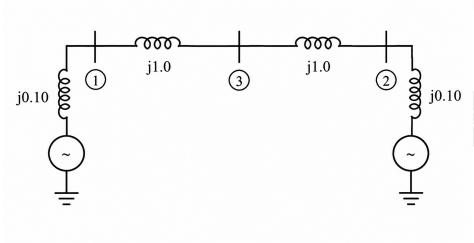
$L_1 = L_2 = 10$ km

$X_{line} = 0.225\ \Omega/km$

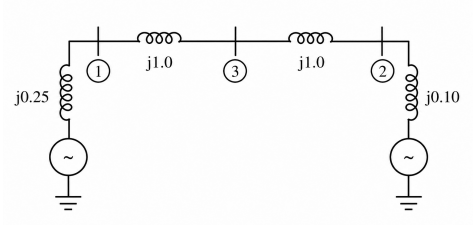
Q2.11.1 MCQ 2011 2M

For the above system shown in the figure, the positive sequence diagram with the p.u values on the 100 MVA common base is

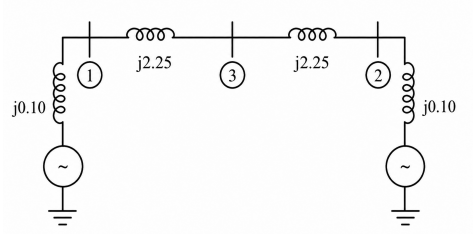
A.



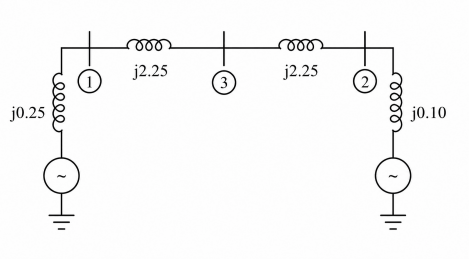
B.



C.



D.



Q2.11.2 MCQ 2011 2M

In the above system, the three-phase fault MVA at bus 3 is

A. 82.55 MVA **B.** 85.11 MVA
C. 170.91 MVA **D.** 181.82 MVA

Q2.10.1 MCQ 2010 1M

The zero-sequence circuit of the three-phase transformer shown in the figure is

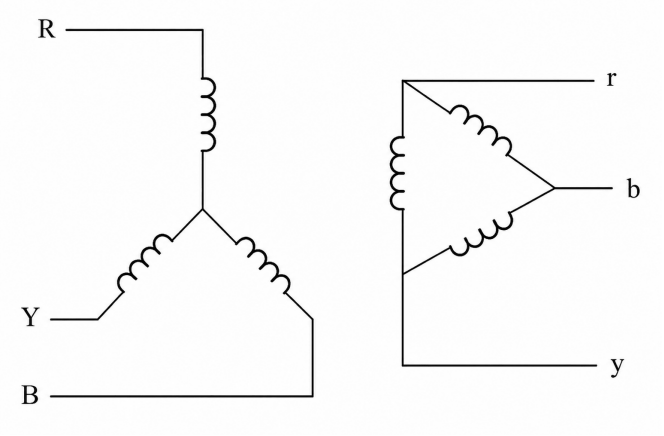
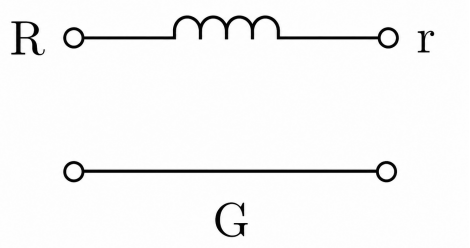
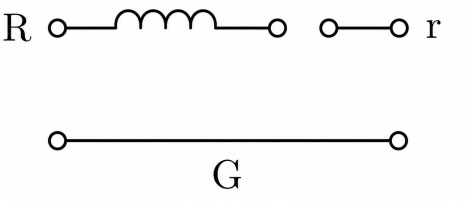


Fig. Q2.10.1

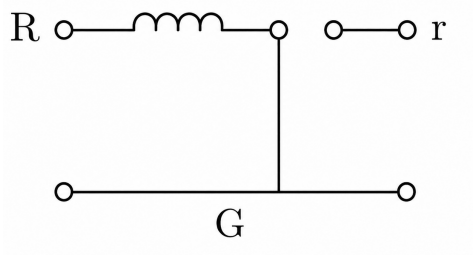
A.



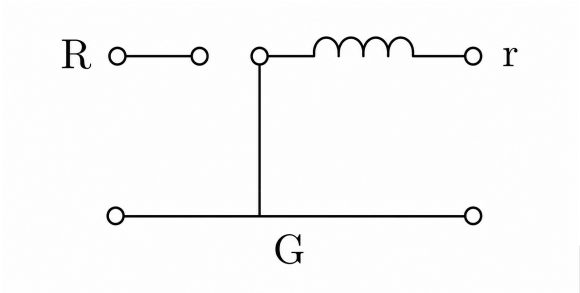
B.



C.



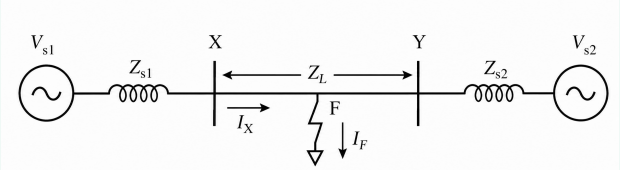
D.



Common Data Questions

Answer questions Q2.08.1, Q2.08.2 and Q2.08.3 based on the below data:

Consider the power system shown below.



Given, $V_{s1} = V_{s2} = 1.0 + j0.0$ pu, $Z_{s1} = Z_{s2} = 0.001 + j0.01$ pu, $Z_L = 0.006 + j0.06$ pu, base MVA = 100, voltage base = 400 kV (line-to-line), and nominal frequency = 50 Hz. A symmetrical three-phase fault occurs at the centre of the line (point F) at time t_0. The positive-sequence impedance from source S_1 to point F is $0.004 + j0.04$ pu. The reference voltage for phase a is defined as $v(t) = V_m \cos(\omega t)$. The phase-$a$ fault current from bus X has a negative decaying dc offset with maximum initial magnitude. Assume negative-sequence impedances are equal to positive-sequence impedances and zero-sequence impedances are three times the positive-sequence impedances.

Q2.08.1 MCQ 2008 2M

Using the common data given, the instant (t_0) of the fault will be

A. 4.682 ms **B.** 9.667 ms **C.** 14.667 ms **D.** 19.667 ms

Q2.08.2 MCQ 2008 2M

Using the common data given, the rms value of the ac component of fault current (I_X) will be

A. 3.59 kA **B.** 5.07 kA **C.** 7.18 kA **D.** 10.15 kA

Q2.08.3 MCQ 2008 2M

Using the common data given, instead of the three-phase fault, if a single line-to-ground fault occurs on phase a at point F with zero fault impedance, then the rms value of the ac component of fault current (I_X) for phase a will be

A. 4.97 pu **B.** 7.0 pu **C.** 14.93 pu **D.** 29.85 pu

Q2.07.1 MCQ 2007 2M

Suppose we define a sequence transformation between "a-b-c" and "p-n-o" variables as follows:

$$\begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix} = k \begin{bmatrix} 1 & 1 & 1 \\ \alpha^2 & \alpha & 1 \\ \alpha & \alpha^2 & 1 \end{bmatrix} \begin{bmatrix} f_p \\ f_n \\ f_o \end{bmatrix}$$

where $\alpha = e^{j\frac{2\pi}{3}}$ and k is a constant.
Now, if it is given that

$$\begin{bmatrix} V_p \\ V_n \\ V_o \end{bmatrix} = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 2.0 \end{bmatrix} \begin{bmatrix} i_p \\ i_n \\ i_o \end{bmatrix}$$

and

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = Z \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix},$$

then

A. $Z = \frac{k^2}{3} \begin{bmatrix} 1.0 & -0.5 & -0.5 \\ -0.5 & 1.0 & -0.5 \\ -0.5 & -0.5 & 1.0 \end{bmatrix}$

B. $Z = \begin{bmatrix} 1.0 & 0.5 & 0.5 \\ 0.5 & 1.0 & 0.5 \\ 0.5 & 0.5 & 1.0 \end{bmatrix}$

C. $Z = 3k^2 \begin{bmatrix} 1.0 & 0.75 & 0.5 \\ 0.5 & 1.0 & 0.75 \\ 0.75 & 0.5 & 1.0 \end{bmatrix}$

D. $Z = \begin{bmatrix} 1.0 & 0.5 & 0.75 \\ 0.75 & 1.0 & 0.5 \\ 0.5 & 0.75 & 1.0 \end{bmatrix}$

Q2.06.1 MCQ 2006 2M

Three identical star connected resistors of 1.0 pu are connected to an unbalanced 3 phase supply. The load neutral is isolated. The symmetrical components of the line voltages in pu are $V_{ab1} = X\angle\theta_1$ and $V_{ab2} = Y\angle\theta_2$ respectively. If all the pu calculations are with the respective base values, the phase to neutral sequence voltages are:

A. $V_{an1} = X\angle(\theta_1 + 30°), \quad V_{an2} = Y\angle(\theta_2 - 30°)$

B. $V_{an1} = X\angle(\theta_1 - 30°), \quad V_{an2} = Y\angle(\theta_2 + 30°)$

C. $V_{an1} = \frac{X}{\sqrt{3}}\angle(\theta_1 - 30°), \quad V_{an2} = \frac{Y}{\sqrt{3}}\angle(\theta_2 - 30°)$

D. $V_{an1} = \frac{X}{\sqrt{3}}\angle(\theta_1 - 60°), \quad V_{an2} = \frac{Y}{\sqrt{3}}\angle(\theta_2 - 60°)$

Common Data Questions

Answer the questions Q2.05.1 and Q2.05.2 based on the data below:
At a 220 kV substation of a power system, it is given that the three-phase fault level is 4000 MVA and the single-line-to-ground fault level is 5000 MVA. Neglecting the resistance and the shunt susceptances of the system.

Q2.05.1 MCQ 2005 2M

The positive sequence driving point reactance at the bus is

A. 2.5 Ω **B.** 4.033 Ω **C.** 5.5 Ω **D.** 12.1 Ω

Q2.05.2 MCQ 2005 2M

The zero-sequence driving point reactance at the bus is

A. 2.2 Ω **B.** 4.84 Ω **C.** 18.18 Ω **D.** 22.72 Ω

Q2.05.3 MCQ 2005 2M

For a transposed overhead transmission line with self reactance $x_s = 0.4\,\Omega$/km and mutual reactance $x_m = 0.1\,\Omega$/km, the positive-sequence reactance x_1 and zero-sequence reactance x_0, respectively, in Ω/km are:

A. 0.3, 0.2 **B.** 0.5, 0.2 **C.** 0.5, 0.6 **D.** 0.3, 0.6

Q2.04.1 MCQ 2004 2M

A 3-phase generator rated at 110 MVA, 11 kV is connected through circuit breakers to a transformer. The generator is having direct-axis sub-transient reactance $X''_d = 19\%$, transient reactance $X'_d = 26\%$ and synchronous reactance $X_d = 130\%$. The generator is operating at no load and rated voltage when a three-phase short-circuit fault occurs between the breakers and the transformer. The magnitude of initial symmetrical rms current in the breakers will be

A. 4.44 kA **B.** 22.20 kA **C.** 30.39 kA **D.** 38.45 kA

Q2.04.2 MCQ 2004 2M

A 3-phase transmission line supplies a Δ-connected load Z. The conductor 'c' of the line develops an open-circuit fault as shown in the figure. The currents in the lines are as shown on the diagram. The positive-sequence current component in line 'a' will be

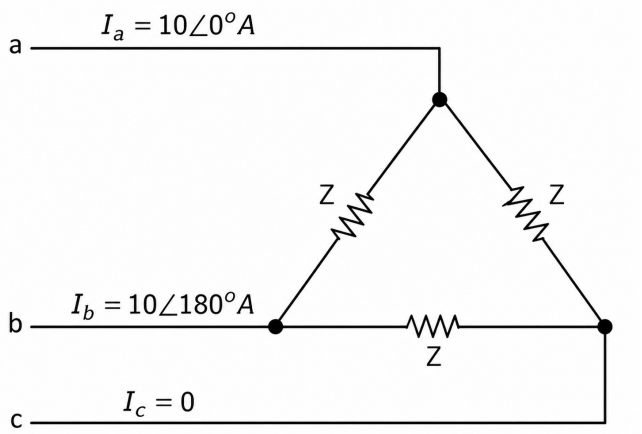


Fig. Q2.04.2

A. $5.78\angle(-30°)$ **B.** $5.78\angle 90°$
C. $6.33\angle 90°$ **D.** $10.00\angle(-30°)$

Q2.04.3 MCQ 2004 2M

A 500 MVA, 50 Hz, 3-phase turbo-generator produces power at 22 kV. The generator is Y-connected and its neutral is solidly grounded. The sequence reactances are $X_1 = X_2 = 0.15$ pu and $X_0 = 0.05$ pu.It is operating at rated voltage and disconnected from the rest of the system (no load). The magnitude of the sub-transient line current for a single line-to-ground fault at the generator terminal in pu will be

A. 2.851 **B.** 3.333 **C.** 6.667 **D.** 8.553

Q2.03.1 MCQ 2003 2M

A 20-MVA, 6.6-kV, 3-phase alternator is connected to a 3-phase transmission line. The per-unit positive-sequence, negative-sequence and zero-sequence impedances of the alternator are $j0.1$, $j0.1$, $j0.04$ respectively. The neutral

of the alternator is connected to ground through an inductive reactor of $j0.05$ pu. The per-unit positive-sequence, negative-sequence and zero-sequence impedances of the transmission line are $j0.1$, $j0.1$, $j0.3$ respectively. All per-unit values are based on the machine ratings. A solid ground fault occurs at one phase of the far end of the transmission line. The voltage of the alternator neutral with respect to ground during the fault is

A. 513.8 V **B.** 889.9 V **C.** 1112.0 V **D.** 642.2 V

Q2.03.2 MCQ 2003 2M

A three-phase alternator generating unbalanced voltages is connected to an unbalanced load through a 3-phase transmission line as shown in figure below. The neutral of the alternator and the star point of the load are solidly grounded. The phase voltages of the alternator are

$$E_a = 10\angle 0°\text{V}, \quad E_b = 10\angle -90°\text{V}, \quad E_c = 10\angle 120°\text{V}$$

The positive-sequence component of the load current is

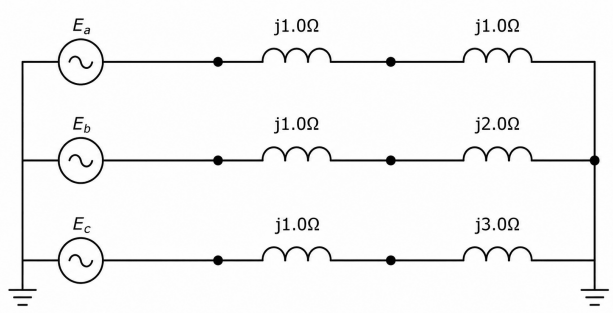


Fig. Q2.03.2

A. $1.310\angle -107°$A **B.** $0.332\angle -120°$A
C. $0.996\angle -120°$A **D.** $3.510\angle -81°$A

Q2.02.1 MCQ 2002 2M

A generator is connected to a transformer which feeds another transformer through a short feeder (see Fig.2.16). The zero sequence impedance values are expressed in pu on a common base and are indicated in Fig.2.16. The Thevenin equivalent zero sequence impedance at point B is

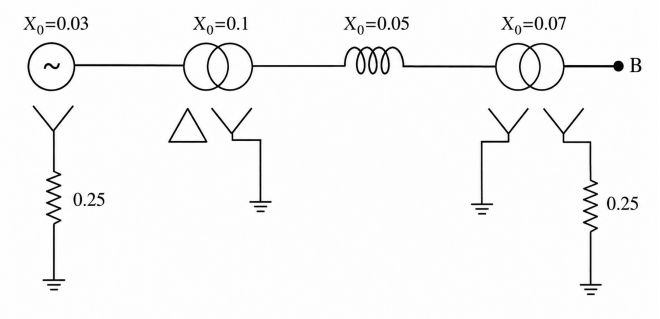


Fig. Q2.02.1

A. $0.8 + j0.6$ **B.** $0.75 + j0.22$
C. $0.75 + j0.25$ **D.** $1.5 + j0.25$

Q2.01.1 MCQ 2001 1M

X_d, X'_d and X''_d are steady-state d-axis synchronous reactance, transient d-axis reactance and sub-transient d-axis reactance of a synchronous machine respectively. Which of the following statements is true?

A. $X_d > X'_d > X''_d$ **B.** $X''_d > X'_d > X_d$
C. $X'_d > X''_d > X_d$ **D.** $X_d > X''_d > X'_d$

Q2.01.2 MCQ 2001 2M

A 50 Hz alternator is rated 500 MVA, 20 kV, with X_d = 1.0 pu and X''_d = 0.2 pu. It supplies a purely resistive load of 400 MW at 20 kV. The load is connected directly across the generator terminals when a symmetrical fault occurs at the load terminals. The initial rms current in the generator in per unit is

A. 7.22 **B.** 6.4 **C.** 3.22 **D.** 2.2

Q2.01.3 NAT 2001 5M

A single line-to-ground fault occurs on an unloaded generator in phase a. The positive, negative and zero sequence impedances of the generator are $j0.25$pu, $j0.25$pu, $j0.15$pu respectively. The generator neutral is grounded through a reactance of $j0.05$ pu. The prefault generator terminal voltage is 1.0 pu.

(a) Draw the positive, negative and zero sequence networks for the fault.
(b) Draw the interconnection of the sequence networks for the fault analysis.
(c) Determine the fault current.

Q2.00.1 NAT 2000 5M

For the configuration shown in Fig. 14, the breaker connecting a large system to bus 2 is initially open. The system 3-phase fault level at bus 3 under this condition is not known. The system 3-phase fault level after closing the system breaker, at the bus 1 was found to be 5.0 p.u. What will be the new 3-phase fault level at system bus 3 after the interconnection?

All per unit values are on common bases. Prefault bus voltages are assumed to be 1.0 p.u. at all buses.

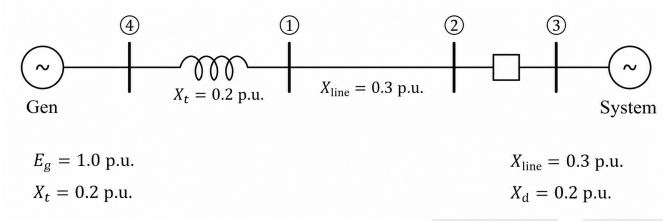


Fig. Q2.00.1

Q2.00.2 MCQ 2000 2M

The severity of line-to-ground and three-phase faults at the terminals of an unloaded synchronous generator is to be the same. If the terminal voltage is 1.0 p.u. and $X_1 = X_2 = j0.1$ p.u., $X_0 = j0.05$ p.u. for the alternator, then the required inductive reactance for neutral grounding is

A. 0.0166 p.u. **B.** 0.05 p.u. **C.** 0.1 p.u. **D.** 0.15 p.u.

Q2.99.1 NAT 1999 5M

Determine the magnitudes of the symmetrical components (I_{a0}, I_{a1}, I_{a2}) of the currents in a three wire (RYB) three phase system, when a short circuit occurs between R and Y phase wires, the fault current being 100 A.

Q2.98.1 MCQ 1998 2M

For the network shown in figure below, the zero sequence reactance's in p.u. are indicated. The zero sequence driving point reactance of node 3 is:

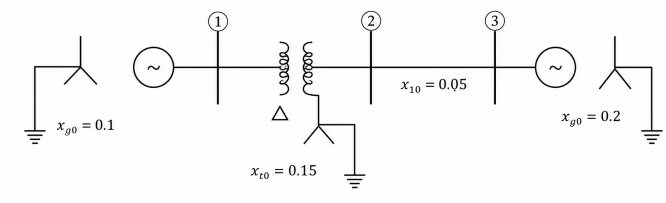


Fig. Q2.98.1

A. 0.12 **B.** 0.30 **C.** 0.10 **D.** 0.20

Q2.97.1 MCQ 1997 1M

For a fault at the terminals of a synchronous generator, the fault current is maximum for a

A. 3-phase fault **B.** 3-phase to ground fault
C. line-to-ground fault **D.** line-to-line fault

Q2.96.1 MCQ 1996 1M

For an unbalanced fault, with paths for zero-sequence currents, at the point of fault

A. the negative and zero sequence voltages are minimum

B. the negative and zero sequence voltages are maximum

C. the negative sequence voltage is minimum and zero sequence voltage is maximum

D. the negative sequence voltage is maximum and zero sequence voltage is minimum

Q2.95.1 NAT 1995 1M

The positive-sequence component of the voltage at the point of fault in a power system is zero for a _____ fault.

Q2.95.2 NAT 1995 5M

A three-phase star-connected alternator is rated 30 MVA, 13.8 kV and has $X_1 = 0.25$ p.u., $X_2 = 0.35$ p.u., $X_0 = 0.10$ p.u.. The neutral is solidly grounded. Determine the alternator line currents when a double-line-to-ground fault occurs at its terminals. Assume the alternator is unloaded and operating at rated voltage before the fault occurs.

Q2.94.1 NAT 1994 1M

In a power system, the three-phase fault MVA is always higher than the single-line-to-ground fault MVA at a bus. (True/False)

Q2.91.1 NAT 1991 2M

A fault occurring at the terminals of an unloaded synchronous generator operating at its rated voltage has resulted in the following values of current and voltage: $I_{a0} = j2.37$ p.u., $[I_{a1} = -j3.05$ p.u., $I_{a2} = j0.68$ p.u.$and V_{a0} = V_{a1} = V_{a2} = 0.237$ p.u.. The fault that has occurred is _____.

Q2.91.2 NAT 1991 2M

A 100 MVA generator with 10% reactance and a 200 MVA generator with 8% reactance (reactances on their own bases) are connected as shown in the figure. The fault level on Bus 1 is to be restricted to 1500 MVA. Calculate, on 100 MVA base,

(i) The reactance of bus-bar reactor X

(ii) Fault level of Bus 2

(iii) MVA rating of circuit breaker C

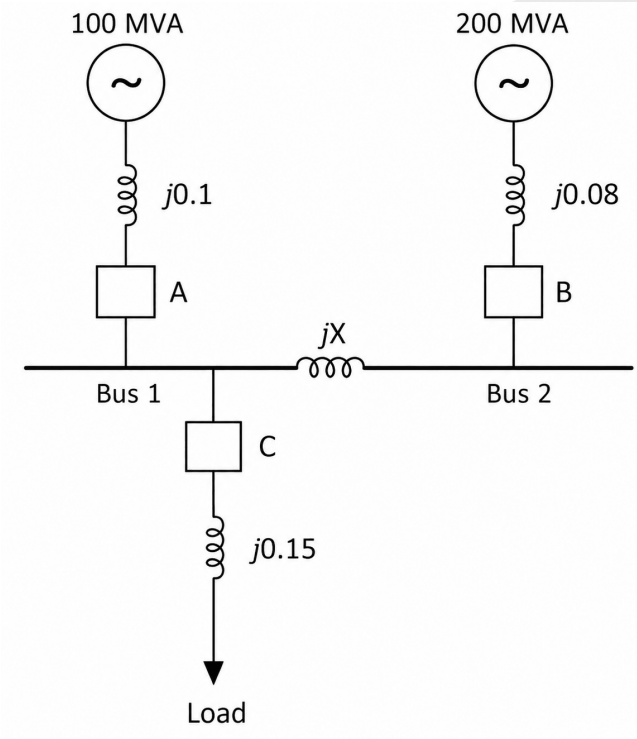


Fig. Q2.91.2

Q2.26.1 D	Q2.24.1 A,D	Q2.23.1 7.21
Q2.22.1 B	Q2.21.1 D	Q2.20.1 1.02
Q2.19.1 1.7 to 1.9	Q2.18.1 7	Q2.18.2 73 to 74
Q2.18.3 A	Q2.17.1 0.1	Q2.16.1 0.83 to 0.85
Q2.16.2 2.20 to 2.35	Q2.16.3 1.05 to 1.15	Q2.16.4 5.5 to 6.5
Q2.15.1 B	Q2.14.1 C	Q2.14.2 B
Q2.14.3 C	Q2.14.4 *	Q2.14.5 B
Q2.12.1 C	Q2.11.1 A	Q2.11.2 D
Q2.10.1 C	Q2.08.1 A	Q2.08.2 A
Q2.08.3 C	Q2.07.1 B	Q2.06.1 B
Q2.05.1 D	Q2.05.2 B	Q2.05.3 D
Q2.04.1 C	Q2.04.2 A	Q2.04.3 D
Q2.03.1 D	Q2.03.2 D	Q2.02.1 B
Q2.01.1 A	Q2.01.2 B	Q2.01.3 3.75
Q2.00.1 11.43	Q2.00.2 A	Q2.99.1 *
Q2.98.1 C	Q2.97.1 C	Q2.96.1 B
Q2.95.1 *	Q2.95.2 7.12 p.u.	Q2.94.1 FALSE
Q2.91.1 LLG	Q2.91.2 *	

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

UNIT

III

Power System Stability

Topics

1. Swing Equation - Motor and Generator
2. Inertia Constant
3. Transient Stability
4. Critical Clearing Time
5. Methods to Improve stability
6. System Stability Concepts
7. Equal Area Criterion

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 32	**MCQ:** 19	**MSQ:** 0	**NAT:** 13
1M: 8	**2M:** 17	**Descriptive:** 7	**Avg Weightage** ~2M	**Range:** 1M(2018)–2M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q3.25.1 MCQ 2025 2M

In the system shown below, the generator was initially supplying power to the grid. A temporary LLLG fault occurred at F very close to circuit breaker 1. The circuit breakers open to isolate the line. The fault self-clears. The circuit breakers reclose and restore the line. Which one of the following diagrams best indicates the rotor accelerating and decelerating areas?

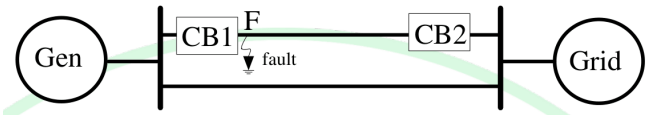
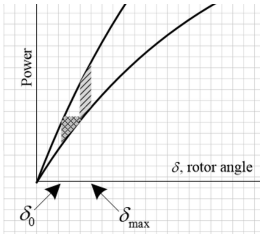
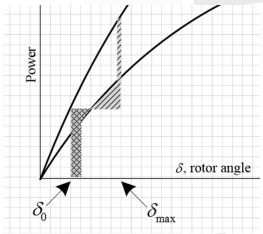


Fig. Q3.25.1

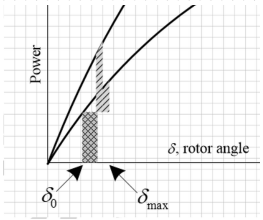
A.



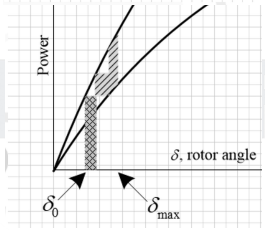
B.



C.



D.



Q3.24.1 NAT 2024 2M

The single line diagram of a lossless system is shown in the figure. The system is operating in steady-state at a stable equilibrium point with the power output of the generator being $P_{max} \sin \delta$, where δ is the load angle and the mechanical power input is $0.5P_{max}$. A fault occurs on line 2 such that the power output of the generator is less than $0.5P_{max}$ during the fault. After the fault is cleared by opening line 2, the power output of the generator is $\left\{P_{max}/\sqrt{2}\right\} \sin \delta$. If the critical fault clearing angle is $\pi/2$ radians, the accelerating area on the power angle curve is _____ times P_{max} (rounded off to 2 decimal places).

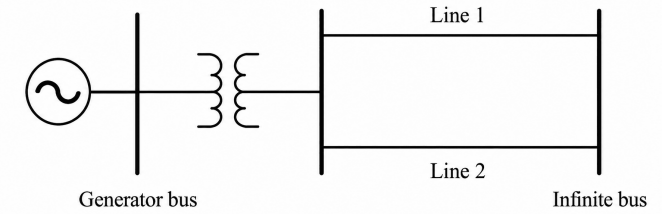


Fig. Q3.24.1

Q3.18.1 MCQ 2018 1M

Consider a lossy transmission line with V_1 and V_2 as the sending and receiving end voltages, respectively. Z and X are the series impedance and reactance of the line, respectively. The steady-state stability limit for the transmission line will be

A. greater than $\left|\frac{V_1V_2}{X}\right|$ **B.** less than $\left|\frac{V_1V_2}{X}\right|$

C. equal to $\left|\frac{V_1V_2}{X}\right|$ **D.** equal to $\left|\frac{V_1V_2}{Z}\right|$

Q3.17.1 NAT 2017 2M

The figure shows the single line diagram of a power system with a double circuit transmission line. The expression for electrical power is $1.5 \sin \delta$, where δ is the rotor angle. The system is operating at the stable equilibrium point with mechanical power equal to 1 pu. If one of the transmission line circuits is removed, the maximum value of δ, as the rotor swings, is 1.221 radian. If the expression for electrical power with one transmission line circuit removed is $P_{max} \sin \delta$, the value of P_{max}, in pu is _____. (Give the answer up to three decimal places.)

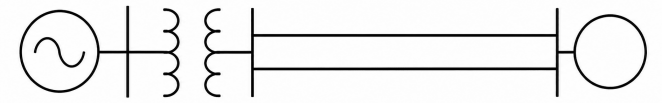


Fig. Q3.17.1

Q3.15.1 NAT 2015 2M

A 50 Hz generating unit has H-constant of 2 MJ/MVA. The machine is initially operating in steady state at synchronous speed, and producing 1 pu of real power. The initial value of the rotor angle δ is 5°, when a bolted three phase to ground short circuit fault occurs at the terminal of the generator. Assuming the input mechanical power to remain at 1 pu, the value of δ in degrees, 0.02 second after the fault is _____.

Q3.15.2 MCQ 2015 1M

The synchronous generator shown in the figure is supplying active power to an infinite bus via two short, lossless transmission lines, and is initially in steady state. The mechanical power input to the generator and the voltage magnitude E are constant. If one line is tripped at time t_1 by opening the circuit breakers at the two ends (although there is no fault), then it is seen that the generator undergoes a stable transient. Which one of the following waveforms of the rotor angle δ shows the transient correctly?

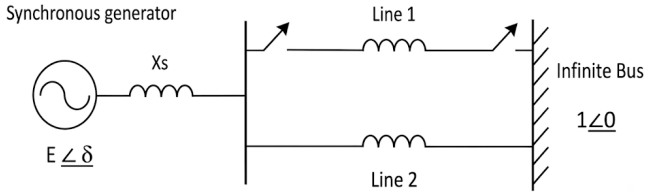
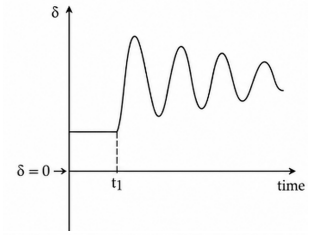
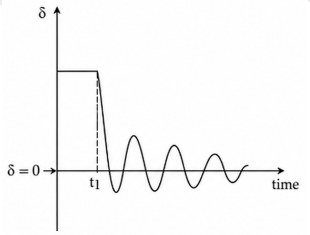


Fig. Q3.15.2

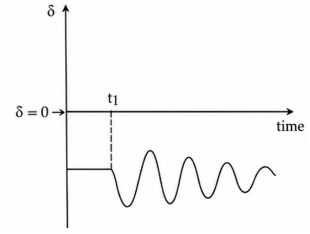
A.



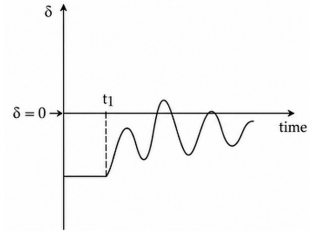
B.



C.



D.



Q3.14.1 NAT 2014 2M

The figure shows the single line diagram of a single machine infinite bus system.

The inertia constant of the synchronous generator is $H = 5$ MW-s/MVA. Frequency is 50 Hz. Mechanical power is 1 pu. The system is operating at the stable equilibrium point with rotor angle δ equal to 30°. A three phase short circuit fault occurs at a certain location on one of the circuits of the double circuit transmission line. During fault, electrical power in pu is $P_e = P_{\max} \sin\delta$. If the values of δ and $d\delta/dt$ at the instant of fault clearing are 45° and 3.762 radian/s respectively, then $P_{\max}$ (in pu) is _____.

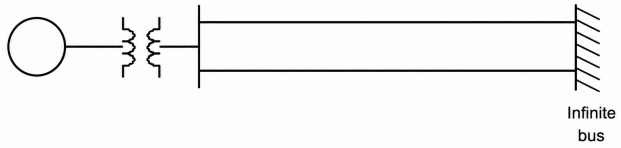


Fig. Q3.14.1

Q3.12.1 MCQ 2012 2M

A cylindrical rotor generator delivers 0.5 pu power in the steady-state to an infinite bus through a transmission line of reactance 0.5 pu. The generator no-load voltage is 1.5 pu and the infinite bus voltage is 1 pu. The inertia constant of the generator is 5 MW-s/MVA and the generator reactance is 1 pu. The critical clearing angle, in degrees, for a three-phase dead short circuit fault at the generator terminal is

A. 53.5 **B.** 60.2 **C.** 70.8 **D.** 79.6

Q3.09.1 MCQ 2009 2M

A 500 MW, 21 kV, 50 Hz, three-phase, 2-pole synchronous generator having a rated power factor of 0.9 has a moment of inertia of 27.5×10^3 kg-m^2. The inertia constant (H) will be

A. 2.44 s **B.** 2.71 s **C.** 4.88 s **D.** 5.42 s

Q3.08.1 MCQ 2008 2M

A lossless single-machine infinite-bus power system is shown below.

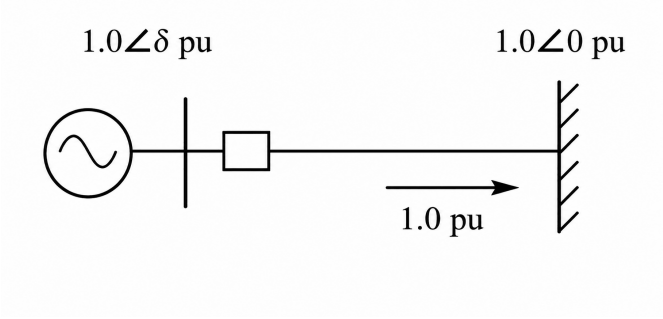


Fig. Q3.08.1

The synchronous generator transfers 1.0 per unit power to the infinite bus. The critical clearing time of the circuit breaker is 0.28 s. If another identical synchronous generator is connected in parallel to the existing generator and each generator is scheduled to supply 0.5 per unit power, then the critical clearing time of the circuit breaker will

A. Reduce but will be more than 0.14 s

B. Reduce to 0.14 s

C. Remain constant at 0.28 s

D. Increase beyond 0.28 s

Q3.07.1 MCQ 2007 2M

Consider a synchronous generator connected to an infinite bus by two identical parallel transmission lines. The transient reactance x' of the generator is 0.1 pu and the mechanical power input to it is constant at 1.0 pu. Due to some previous disturbance, the rotor angle (δ) is undergoing an undamped oscillation, with the maximum value of $\delta(t)$ equal to 130°. One of the parallel lines trips due to relay maloperation at an instant when $\delta(t) = 130°$ as shown in the figure. The maximum value of the per unit line reactance, x, such that the system does not lose synchronism subsequent to this tripping is

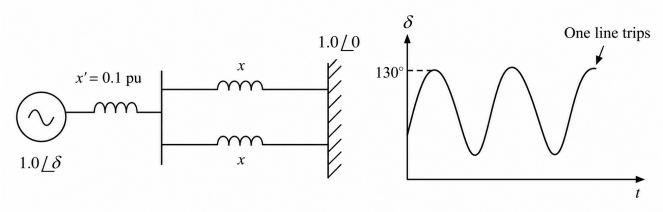


Fig. Q3.07.1

A. 0.87 **B.** 0.74 **C.** 0.67 **D.** 0.54

Common Data Questions

Answer the questions Q3.06.1 and Q3.06.2 based on the data below

A generator feeds power to an infinite bus through a double circuit transmission line. A Three-phase fault occurs at the middle point of one of the lines. The infinite bus voltage is 1 pu, the transient internal voltage of the generator is 1.1 pu and the equivalent transfer admittance during fault is 0.8 pu.

The 100 MVA generator has an inertia constant of 5 MJ/MVA and was delivering 1.0 pu power prior to the fault with rotor power angle of 30°. The system frequency is 50 Hz.

Q3.06.1 MCQ 2006 2M

The initial accelerating power (in pu) will be

A. 1.0 **B.** 0.6 **C.** 0.56 **D.** 0.4

Q3.06.2 MCQ 2006 2M

If the initial accelerating power is X pu, the initial acceleration in electrical deg/sec^2, and the inertia constant in MJ-sec/electrical deg respectively are

A. $31.4X$, 18 **B.** $1800X$, 0.056
C. $X/1800$, 0.056 **D.** $X/31.4$, 18

Q3.05.1 MCQ 2005 2M

A generator with constant 1.0 p.u. terminal voltage supplies power through a step-up transformer of 0.12 p.u. reactance and a double-circuit line to an infinite bus as shown below. The infinite bus voltage is maintained at 1.0 p.u. Neglecting the resistances and susceptances of the system, the steady state stability power limit of the system is 6.25 p.u. If one of the double-circuits is tripped, the resulting steady state stability power limit in p.u. will be

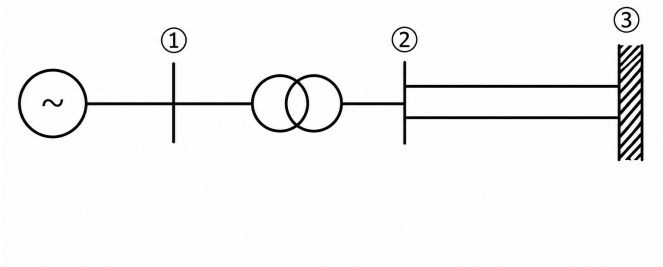


Fig. Q3.05.1

A. 12.5 p.u. **B.** 3.125 p.u. **C.** 10.0 p.u. **D.** 5.0 p.u.

Q3.04.1 MCQ 2004 2M

A 50 Hz, 4-pole, 500 MVA, 22 kV turbo-generator is delivering rated megavolt-amperes at 0.8 power factor. Suddenly a fault occurs reducing its electric power output by 40%. Neglect losses and assume constant power input to the shaft. The accelerating torque in the generator in MN-m at the time of the fault will be

A. 1.528 **B.** 1.018 **C.** 0.848 **D.** 0.509

Q3.03.1 MCQ 2003 1M

A round rotor generator having internal voltage $E_1 = 2.0$ pu and reactance $X_1 = 1.1$ pu is connected to a round rotor synchronous motor having internal voltage $E_2 = 1.3$ pu and reactance $X_2 = 1.2$ pu through a transmission line of reactance 0.5 pu. If the generator supplies 0.5 pu power, the rotor angle difference between the machines is:

A. 57.42° **B.** 1° **C.** 32.58° **D.** 122.58°

Q3.03.2 MCQ 2003 2M

A generator delivers power of 1.0 p.u. to an infinite bus through a purely reactive network. The maximum power that could be delivered by the generator is 2.0 p.u. A three-phase fault occurs at the terminals of the generator which reduces the generator output to zero. The fault is cleared after t_c second. The original network is then restored. The maximum swing of the rotor angle is found to be $\delta_{max} = 110°$. Then the rotor angle in electrical degrees at $t = t_c$ is

A. 55° **B.** 70° **C.** 69.14° **D.** 72.4°

Q3.02.1 MCQ 2002 1M

Consider a power system with three identical generators. The transmission losses are negligible. One generator (G1) has a speed governor which maintains its speed constant at the rated value, while the other generators (G2 and G3) have governors with a droop of 5%. If the load of the system is increased, then in steady state,

A. generation of G2 and G3 is increased equally while generation of G1 is unchanged.

B. generation of G1 alone is increased while generation of G2 and G3 is unchanged.

C. generation of G1, G2 and G3 is increased equally.

D. generation of G1, G2 and G3 is increased in the ratio 0.5 : 0.25 : 0.25.

Q3.02.2 NAT 2002 5M

A synchronous generator is to be connected to an infinite bus through a transmission line of reactance $X = 0.2$ pu, as shown in figure below. The generator data is as follows:

$$x' = 0.1 \text{ pu}, \quad E' = 1.0 \text{ pu}, \quad H = 5 \text{ MJ/MVA}$$

$$P_m = 0.0 \text{ pu}, \quad \omega_B = 2\pi \times 50 \text{ rad/s}$$

All quantities are expressed on a common base. The generator is initially running on open circuit with the frequency of the open-circuit voltage slightly higher than that of the infinite bus. If at the instant of switch closure

$$\delta = 0 \text{ and } \omega = \frac{d\delta}{dt} = \omega_{\text{init}}$$

compute the maximum value of ω_{init} so that the generator pulls into synchronism. Hint: Use the equation

$$\int \left(\frac{2H}{\omega_B}\right) \omega \, d\omega + P_e \, d\delta = 0$$

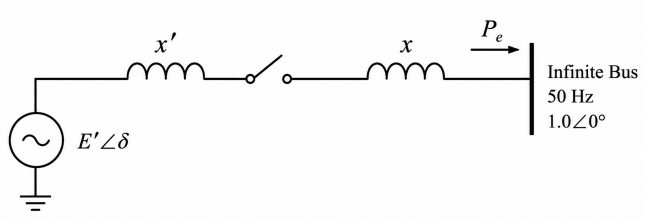


Fig. Q3.02.2

Q3.01.1 NAT 2001 5M

A synchronous generator is connected to an infinite bus through a lossless double circuit transmission line. The generator is delivering 1.0 per unit power at a load angle of 30° when a sudden fault reduces the peak power that can be transmitted to 0.5 per unit. After clearance of fault, the peak power that can be transmitted becomes 1.5 per unit. Find the critical clearing angle.

Q3.00.1 NAT 2000 5M

A synchronous generator, having a reactance of 0.15 p.u., is connected to an infinite bus through two identical parallel transmission lines having reactance of 0.3 p.u. each. In steady state, the generator is delivering 1 p.u. power to the infinite bus. For a three-phase fault at the receiving end of one line, calculate the accelerating power to be evaluated for the first step of the point-by-point method if the breaker clears the fault at the end of an interval. Also indicate how the accelerating power is evaluated if the breaker clears the fault at the middle of an interval.

(a) At the end of an interval

(b) At the middle of an interval

Q3.99.1 MCQ 1999 1M

Steady state stability of a power system is the ability of the power system to

A. maintain voltage at the rated voltage level

B. maintain frequency exactly at 50 Hz

C. maintain a spinning reserve margin at all times

D. maintain synchronism between machines and on external tie lines

Q3.98.1 MCQ 1998 2M

A power station consists of two synchronous generators A and B of ratings 250 MVA and 500 MVA with inertia 1.6 p.u. and 1 p.u., respectively on their own base MVA ratings. The equivalent p.u. inertia constant for the system on 100 MVA common base is:

A. 2.6 **B.** 0.615 **C.** 1.625 **D.** 9.0

Q3.98.2 NAT 1998 5M

An alternator is connected to an infinite bus as shown in figure below. It delivers 1.0 p.u. current at 0.8 p.f lagging at $V = 1.0$ p.u. The reactance X_d of the alternator is 1.2 p.u. Determine the active power output and the steady state power limit. Keeping the active power fixed, if the excitation is reduced, find the critical excitation corresponding to the stability limit.

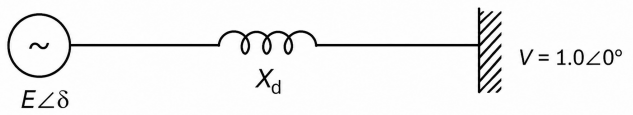


Fig. Q3.98.2

Q3.97.1 MCQ 1997 1M

A 100 MVA, 11 kV, three-phase, 50 Hz, 8-pole synchronous generator has an inertia constant H equal to 4 seconds. The stored energy in the rotor of the generator at synchronous speed will be $H = \frac{E}{G}$

A. 100 MJ **B.** 400 MJ **C.** 800 MJ **D.** 12.5 MJ

Q3.97.2 NAT 1997 5M

A synchronous motor is receiving 50% of the power. It is capable of receiving from an infinite bus. If the load on the motor is suddenly reduced to 80% of the previous value, determine the swing of the motor around its new equilibrium position.

Q3.97.3 NAT 1997 5M

Two power systems A and B each having a regulation (R) of 0.05 p.u. on their respective capacity bases and a stiffness coefficient of 0.75 p.u. are connected through a tie-line, initially carrying no power. The capacity of system A is 2000 MW and that of system B is 3000 MW. If there is an increase in load of 200 MW in system A, determine the steady-state change in power transfer.

Q3.96.1 MCQ 1996 1M

During a disturbance on a synchronous machine, the rotor swings from A to B, before finally settling down to a steady state at point C on the power-angle curve. The speed of the machine during oscillation is synchronous at point(s)

A. A and B **B.** A and C **C.** B and C **D.** Only at C

Q3.95.1 NAT 1995 5M

A lossless alternator supplies 50 MW to an infinite bus, the steady-state stability limit being 100 MW. Determine whether the alternator will remain stable if the input to the prime mover of the alternator is abruptly increased by 40 MW.

Q3.94.1 NAT 1994 2M

In a system, there are two generators operating in parallel. One generator, of rating 250 MVA, has an inertia constant of 6 MJ/MVA while the other generator of rating 150 MVA has an inertia constant of 4 MJ/MVA.The inertia constant for the combined system on 100 MVA common base is _____ MJ/MVA.

Q3.93.1 MCQ 1993 1M

The transient stability of the power system can be effectively improved by

A. Excitation control

B. Phase shifting transformer

C. Single pole switching circuit breakers

D. Increasing the turbine valve opening

Q3.91.1 NAT 1991 2M

A generator is supplying 1 per unit power to an infinite bus through the system shown in the figure. Following a fault at F, circuit breakers B_3 and B_4 open simultaneously. The $P - \delta$ relationships in three periods are given by: Pre-fault condition: $P = 2 \sin \delta$. During fault condition: $P = 0.2 \sin \delta$. After B_3, B_4 open: $P = 1.5 \sin \delta$. Calculate the critical angle δ before which breakers B_3 and B_4 must open so that synchronism is not lost. Also show it on a $P - \delta$ diagram.

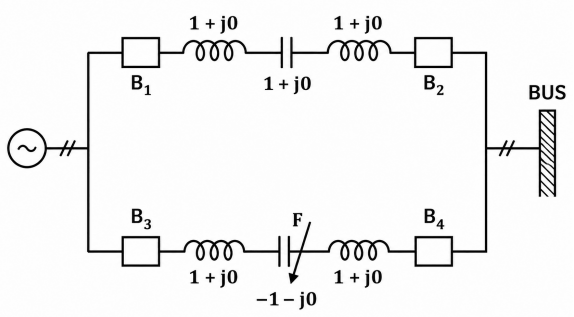


Fig. Q3.91.1

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q3.25.1 B	Q3.24.1 0.10 to 0.12	Q3.18.1 B
Q3.17.1 1.22	Q3.15.1 5.7 to 6.1	Q3.15.2 A
Q3.14.1 0.22 to 0.24	Q3.12.1 D	Q3.09.1 A
Q3.08.1 D	Q3.07.1 C	Q3.06.1 C
Q3.06.2 B	Q3.05.1 D	Q3.04.1 B
Q3.03.1 C	Q3.03.2 C	Q3.02.1 B
Q3.02.2 20.47 rad/sec	Q3.01.1 70.4°	Q3.00.1 *
Q3.99.1 D	Q3.98.1 D	Q3.98.2 *
Q3.97.1 B	Q3.97.2 17.32	Q3.97.3 *
Q3.96.1 A	Q3.95.1 UNSTABLE	Q3.94.1 21
Q3.93.1 C	Q3.91.1 62.71	

UNIT

IV

Power System Matrices and Load Flow

Topics

1. Bus Admittance Matrix
2. Bus Impedance Matrix
3. Fault Analysis using Z bus
4. Gauss-Seidel Load Flow Method
5. Newton-Raphson Load Flow Method
6. Complex Power concept - Effect of Bus parameters on Active and Reactive Power flow

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 59	**MCQ:** 44	**MSQ:** 2	**NAT:** 13
1M: 28	**2M:** 27	**Descriptive:** 4	**Avg Weightage** ~3M	**Range:** 1M(2019)–6M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MCQ 2026 1M

Consider a power system with N buses, of which P are generator buses and the remaining Q are load buses (where there is no generation).
Assume that there are no reactive power-limit violations at the generator buses.
What is the size of the Jacobian matrix in the Newton–Raphson load flow method?

A. $2N \times 2N$

B. $(2N - 1 - P) \times (2N - 1 - P)$

C. $(2N - Q) \times (2N - Q)$

D. $(P + Q) \times (P + Q)$

Q4.26.2 MSQ 2026 2M

For the balanced 3-phase transmission line shown, consider the following cases:
Case-1: $|V_1| = 1.1$ p.u., $|V_2| = 0.9$ p.u., $\bar{Z} = 0.75\angle 0°$ p.u. and $\theta_{12} = \theta_1 - \theta_2 = 0°$
Case-2: $|V_1| = 1.1$ p.u., $|V_2| = 0.9$ p.u., $\bar{Z} = 0.75\angle 90°$ p.u. and $\theta_{12} = \theta_1 - \theta_2 = 90°$

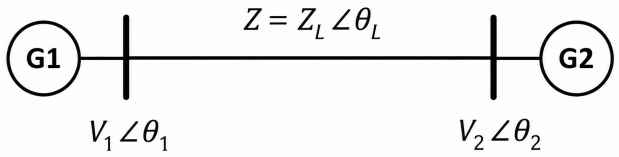


Fig. Q4.26.2

Which of the following statement(s) is/are correct about real power loss and reactive power loss in the line?

☐ **A.** Real power loss in Case-1 is more than that in Case-2

☐ **B.** Real power loss in Case-2 is more than that in Case-1

☐ **C.** Reactive power loss in Case-1 is more than that in Case-2

☐ **D.** Reactive power loss in Case-2 is more than that in Case-1

Q4.26.3 MCQ 2026 2M

A three-phase two-winding transformer has a voltage transformation ratio $\frac{V_P}{V_S} = 0.866 + j0.5$, where V_P is the primary side voltage in p.u., and V_S is the secondary side voltage in p.u. I_P and I_S represent the currents injected into the primary and secondary sides of the transformer, respectively. The admittance corresponding to the leakage impedance of the transformer referred to the secondary is y_t p.u. Neglect the magnetizing branch.

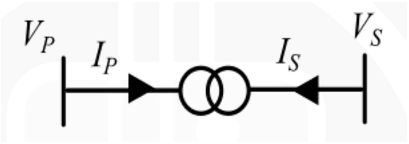


Fig. Q4.26.3

The Y_{bus} representation of this transformer is:

A. $\begin{bmatrix} I_P \\ I_S \end{bmatrix} = \begin{bmatrix} \frac{y_t}{0.866 + j0.5} & -\frac{y_t}{0.866 + j0.5} \\ -\frac{y_t}{0.866 + j0.5} & \frac{y_t}{0.866 + j0.5} \end{bmatrix} \begin{bmatrix} V_P \\ V_S \end{bmatrix}$

B. $\begin{bmatrix} I_P \\ I_S \end{bmatrix} = \begin{bmatrix} y_t & -y_t \\ -y_t & y_t \end{bmatrix} \begin{bmatrix} V_P \\ V_S \end{bmatrix}$

C. $\begin{bmatrix} I_P \\ I_S \end{bmatrix} = \begin{bmatrix} y_t & -\frac{y_t}{0.866 + j0.5} \\ -\frac{y_t}{0.866 + j0.5} & y_t \end{bmatrix} \begin{bmatrix} V_P \\ V_S \end{bmatrix}$

D. $\begin{bmatrix} I_P \\ I_S \end{bmatrix} = \begin{bmatrix} y_t & -\frac{y_t}{0.866 - j0.5} \\ -\frac{y_t}{0.866 + j0.5} & y_t \end{bmatrix} \begin{bmatrix} V_P \\ V_S \end{bmatrix}$

Q4.25.1 NAT 2025 1M

The bus impedance matrix of a 3-bus system (in pu) is

$$Z_{\text{bus}} = \begin{bmatrix} j0.059 & j0.061 & j0.038 \\ j0.061 & j0.093 & j0.066 \\ j0.038 & j0.066 & j0.110 \end{bmatrix}$$

A symmetrical fault (through a fault impedance of $j0.007$ pu) occurs at bus 2.
Neglecting pre-fault loading conditions, the voltage at bus 1, during the fault is _____ pu *(round off to three decimal places)*.

Q4.25.2 MCQ 2025 1M

During a power failure, a domestic household uninterrupted power supply (UPS) supplies AC power to a limited number of lights and fans in various rooms. As per a Newton–Raphson load-flow formulation, the UPS would be represented as a

A. Slack bus **B.** PQ bus **C.** PQV bus **D.** PV bus

Q4.24.1 MCQ 2024 1M

The figure shows the single line diagram of a 4-bus power network. Branches b_1, b_2, b_3, and b_4 have impedances $4z$, z, $2z$, and $4z$ per-unit (pu), respectively, where $z = r + jx$, with $r > 0$ and $x > 0$. The current drawn from each load bus (marked as arrows) is equal to I pu, where $I \neq 0$. If the network is to operate with minimum loss, the branch that should be opened is

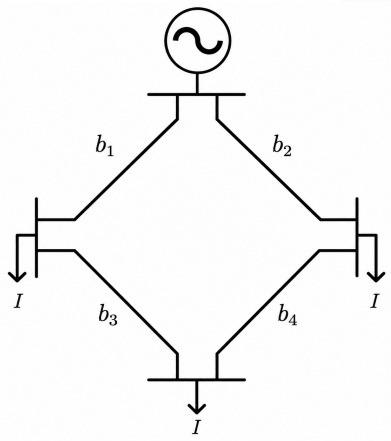


Fig. Q4.24.1

A. b_3 **B.** b_4 **C.** b_2 **D.** b_1

Q4.24.2 MCQ 2024 2M

For the three-bus lossless power network shown in the figure, the voltage magnitudes at all the buses are equal to 1 per unit (pu), and the differences of the voltage phase angles are very small. The line reactances are marked in the figure, where α, β, γ, and x are strictly positive. The bus injections P_1 and P_2 are in pu. If $P_1 = mP_2$, where $m > 0$, and the real power flow from bus 1 to bus 2 is 0 pu, then which one of the following option is correct?

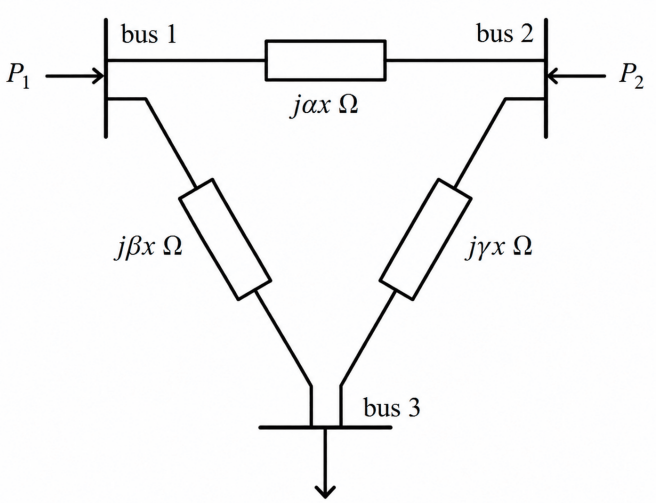


Fig. Q4.24.2

A. $\gamma = m\beta$ **B.** $\beta = m\gamma$ **C.** $\alpha = m\gamma$ **D.** $\alpha = m\beta$

Q4.23.1 MSQ 2023 1M

The bus admittance (Y_{bus}) matrix of a 3-bus power system is given below.

$$Y_{bus} = \begin{bmatrix} -j15 & j10 & j5 \\ j10 & -j13.5 & j4 \\ j5 & j4 & -j8 \end{bmatrix}$$

Considering that there is no shunt inductor connected to any of the buses, which of the following can NOT be true?

- ☐ **A.** Line charging capacitor of finite value is present in all three lines
- ☐ **B.** Line charging capacitor of finite value is present in line 2–3 only
- ☐ **C.** Line charging capacitor of finite value is present in line 2–3 only and shunt capacitor of finite value is present in bus 1 only
- ☐ **D.** Line charging capacitor of finite value is present in line 2–3 only and shunt capacitor of finite value is present in bus 3 only

Q4.23.2 MCQ 2023 2M

The three-bus power system shown in the figure has one alternator connected to bus 2 which supplies 200 MW and 40 MVAr power. Bus 3 is infinite bus having a voltage of magnitude $|V_3| = 1.0$ p.u. and angle of $-15°$. A variable

current source, $|I|\angle\phi$ is connected at bus 1 and controlled such that the magnitude of the bus 1 voltage is maintained at 1.05 p.u. and the phase angle of the source current, $\phi = \theta_1 \pm \frac{\pi}{2}$, where θ_1 is the phase angle of the bus 1 voltage. The three buses can be categorized for load flow analysis as

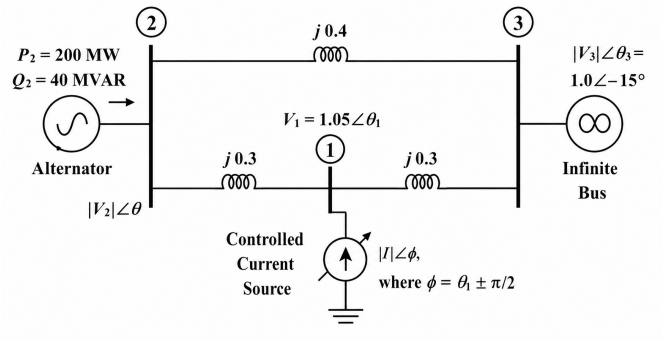


Fig. Q4.23.2

	Bus 1	Slack bus
A.	Bus 2	$P - \|V\|$ bus
	Bus 3	$P - Q$ bus
	Bus 1	$P - \|V\|$ bus
B.	Bus 2	$P - \|V\|$ bus
	Bus 3	Slack bus
	Bus 1	$P - Q$ bus
C.	Bus 2	$P - Q$ bus
	Bus 3	Slack bus
	Bus 1	$P - \|V\|$ bus
D.	Bus 2	$P - Q$ bus
	Bus 3	Slack bus

Q4.21.1 MCQ 2021 1M

Consider a power system consisting of N number of buses. Buses in this power system are categorized into slack bus, PV buses and PQ buses for load flow study. The number of PQ buses is N_L. The balanced Newton-Raphson method is used to carry out load flow study in polar form. H, S, M, and R are sub-matrices of the Jacobian matrix J as shown below:

$$\begin{bmatrix}\Delta P \\ \Delta Q\end{bmatrix} = J \begin{bmatrix}\Delta\delta \\ \Delta|V|\end{bmatrix}, \qquad \text{where} \qquad J = \begin{bmatrix}H & S \\ M & R\end{bmatrix}$$

The dimension of the sub-matrix M is

A. $N_L \times (N-1)$ **B.** $(N-1) \times (N-1-N_L)$

C. $N_L \times (N-1+N_L)$ **D.** $(N-1) \times (N-1+N_L)$

Q4.21.2 MCQ 2021 2M

A 3-Bus network is shown. Consider generators as ideal voltage sources. If rows 1, 2 and 3 of the Y_{Bus} matrix correspond to Bus 1, 2 and 3, respectively, then Y_{Bus} of the network is

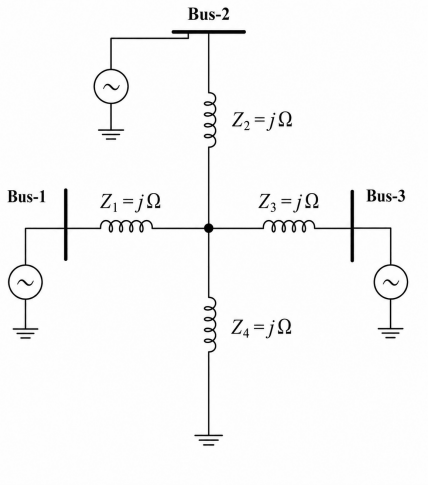


Fig. Q4.21.2

A. $\begin{bmatrix}-4j & j & j \\ j & -4j & j \\ j & j & -4j\end{bmatrix}$ **B.** $\begin{bmatrix}-4j & 2j & 2j \\ 2j & -4j & 2j \\ 2j & 2j & -4j\end{bmatrix}$

C. $\begin{bmatrix}-\frac{3}{4}j & \frac{1}{4}j & \frac{1}{4}j \\ \frac{1}{4}j & -\frac{3}{4}j & \frac{1}{4}j \\ \frac{1}{4}j & \frac{1}{4}j & -\frac{3}{4}j\end{bmatrix}$ **D.** $\begin{bmatrix}-\frac{1}{2}j & \frac{1}{4}j & \frac{1}{4}j \\ \frac{1}{4}j & -\frac{1}{2}j & \frac{1}{4}j \\ \frac{1}{4}j & \frac{1}{4}j & -\frac{1}{2}j\end{bmatrix}$

Q4.20.1 MCQ 2020 2M

Two buses, i and j, are connected with a transmission line of admittance Y, at the two ends of which there are ideal transformers with turns ratios as shown. Bus admittance matrix for the system is:

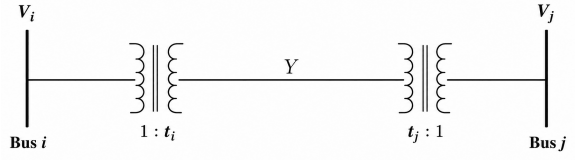


Fig. Q4.20.1

A. $\begin{bmatrix} -t_i t_j Y & t_j^2 Y \\ t_i^2 Y & -t_i t_j Y \end{bmatrix}$ **B.** $\begin{bmatrix} t_i t_j Y & -t_j^2 Y \\ -t_i^2 Y & t_i t_j Y \end{bmatrix}$

C. $\begin{bmatrix} t_i^2 Y & -t_i t_j Y \\ -t_i t_j Y & t_j^2 Y \end{bmatrix}$ **D.** $\begin{bmatrix} t_i t_j Y & -(t_i - t_j)^2 Y \\ -(t_i - t_j)^2 Y & t_i t_j Y \end{bmatrix}$

Q4.20.2 MCQ 2020 1M

Out of the following options, the most relevant information needed to specify the real power (P) at the PV buses in a load flow analysis is

A. solution of economic load dispatch

B. rated voltage of the generator

C. rated power output of the generator

D. base power of the generator

Q4.20.3 NAT 2020 2M

Bus 1 with voltage magnitude $V_1 = 1.1$ pu is sending reactive power Q_{12} towards bus 2 with voltage magnitude $V_2 = 1$ pu through a lossless transmission line of reactance X. Keeping the voltage at bus 2 fixed at 1 pu, magnitude of voltage at bus 1 is changed, so that the reactive power Q_{12} sent from bus 1 is increased by 20%. Real power flow through the line under both the conditions is zero. The new value of the voltage magnitude, V_1, in pu (rounded off to 2 decimal places), at bus 1 is _____.

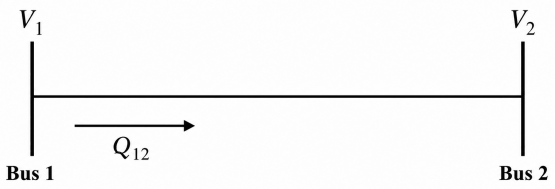


Fig. Q4.20.3

Q4.19.1 NAT 2019 1M

The Y_{bus} matrix of a two-bus power system having two identical parallel lines connected between them in pu is given as

$$Y_{bus} = \begin{bmatrix} -j8 & j20 \\ j20 & -j8 \end{bmatrix}$$

The magnitude of the series reactance of each line in pu (round off to one decimal place) is _____.

Q4.18.1 NAT 2018 1M

A 1000×1000 bus admittance matrix for an electric power system has 8000 non-zero elements. The minimum number of branches (transmission lines and transformers) in this system are _____ (up to 2 decimal places).

Q4.18.2 MCQ 2018 2M

Consider the two bus power system network with given loads as shown in the figure. All the values shown in the figure are in per unit. The reactive power supplied by generator G_1 and G_2 are Q_{G1} and Q_{G2} respectively. The per unit values of Q_{G1}, Q_{G2}, and line reactive power loss (Q_{loss}) respectively are

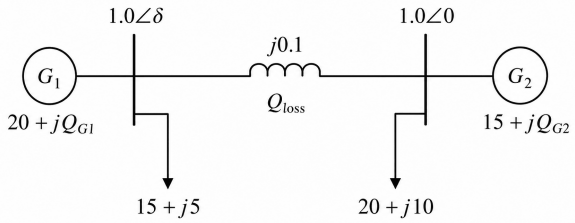


Fig. Q4.18.2

A. 5.00, 12.68, 2.68 **B.** 6.34, 10.00, 1.34

C. 6.34, 11.34, 2.68 **D.** 5.00, 11.34, 1.34

Q4.17.1 NAT 2017 1M

A 10-bus power system consists of four generator buses indexed as G_1, G_2, G_3, G_4 and six load buses indexed as L_1, L_2, L_3, L_4, L_5, L_6. The generator-bus G_1 is considered as slack bus, and the load buses L_3 and L_4 are voltage controlled buses. The generator at bus G_2 cannot supply the required reactive power demand, and hence it is operating at its maximum reactive power limit. The number of non-linear equations required for solving the load flow problem using Newton-Raphson method in polar form is _____.

Q4.17.2 MCQ 2017 1M

A 3-bus power system is shown in the figure below, where the diagonal elements of Y-bus matrix are:

$$Y_{11} = -j12 \text{ pu}, \quad Y_{22} = -j15 \text{ pu}, \quad Y_{33} = -j7 \text{ pu}$$

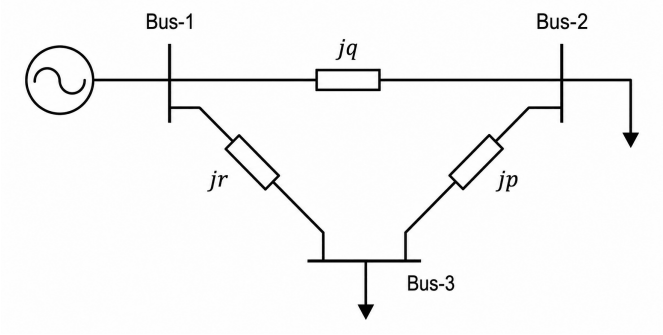


Fig. Q4.17.2

The per unit values of the line reactances p, q and r shown in the figure are

A. $p = -0.2,\ q = -0.1,\ r = -0.5$

B. $p = 0.2,\ q = 0.1,\ r = 0.5$

C. $p = -5,\ q = -10,\ r = -2$

D. $p = 5,\ q = 10,\ r = 2$

Q4.17.3 MCQ 2017 2M

The bus admittance matrix for a power system network is

$$\begin{bmatrix} -j39.9 & j20 & j20 \\ j20 & -j39.9 & j20 \\ j20 & j20 & -j39.9 \end{bmatrix} \text{ pu}$$

There is a transmission line connected between buses 1 and 3, which is represented by the circuit shown in figure.

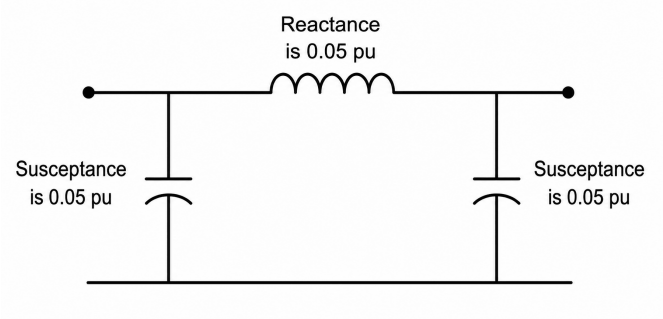


Fig. Q4.17.3

If this transmission line is removed from service, what is the modified bus admittance matrix?

A. $\begin{bmatrix} -j19.9 & j20 & 0 \\ j20 & -j39.9 & j20 \\ 0 & j20 & -j19.9 \end{bmatrix}$ pu

B. $\begin{bmatrix} -j39.95 & j20 & 0 \\ j20 & -j39.9 & j20 \\ 0 & j20 & -j39.95 \end{bmatrix}$ pu

C. $\begin{bmatrix} -j19.95 & j20 & 0 \\ j20 & -j39.9 & j20 \\ 0 & j20 & -j19.95 \end{bmatrix}$ pu

D. $\begin{bmatrix} -j19.95 & j20 & j20 \\ j20 & -j39.9 & j20 \\ j20 & j20 & -j19.95 \end{bmatrix}$ pu

Q4.17.4 MCQ 2017 1M

The figure shows the per-phase representation of a phase-shifting transformer connected between buses 1 and 2, where α is a complex number with non-zero real and imaginary parts.

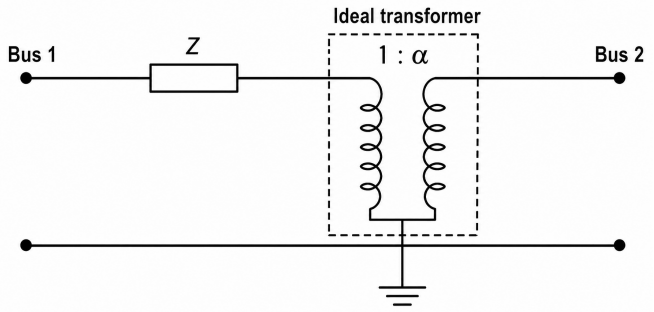


Fig. Q4.17.4

For the given circuit, Y_{bus} and Z_{bus} are bus admittance matrix and bus impedance matrix, respectively, each of size 2×2. Which one of the following statements is true?

A. Both Y_{bus} and Z_{bus} are symmetric

B. Y_{bus} is symmetric and Z_{bus} is unsymmetric

C. Y_{bus} is unsymmetric and Z_{bus} is symmetric

D. Both Y_{bus} and Z_{bus} are unsymmetric

Q4.17.5 NAT 2017 1M

In a load flow problem solved by Newton-Raphson method with polar coordinates, the size of the Jacobian is 100×100. If there are 20 PV buses in addition to PQ buses and a slack bus, the total number of buses in the system is _____.

Q4.16.1 MCQ 2016 1M

In a 100 bus power system, there are 10 generators. In a particular iteration of Newton-Raphson load flow technique (in polar coordinates), two of the PV buses are converted to PQ type. In this iteration,

A. the number of unknown voltage angles increases by two and the number of unknown voltage magnitudes increases by two.

B. the number of unknown voltage angles remains unchanged and the number of unknown voltage magnitudes increases by two.

C. the number of unknown voltage angles increases by two and the number of unknown voltage magnitudes decreases by two.

D. the number of unknown voltage angles remains unchanged and the number of unknown voltage magnitudes decreases by two.

Q4.16.2 MCQ 2016 1M

A power system has 100 buses including 10 generator buses. For the load flow analysis using Newton-Raphson method in polar coordinates, the size of the Jacobian is

A. 189×189 **B.** 100×100
C. 90×90 **D.** 180×180

Q4.15.1 MCQ 2015 2M

Determine the correctness or otherwise of the following Assertion [a] and the Reason [r].
Assertion: Fast decoupled load flow method gives approximate load flow solution because it uses several assumptions.
Reason: Accuracy depends on the power mismatch vector tolerance.

A. Both [a] and [r] are true and [r] is the correct reason for [a].

B. Both [a] and [r] are true but [r] is not the correct reason for [a].

C. Both [a] and [r] are false.

a is false and [r] is true.

Q4.15.2 MCQ 2015 1M

A 3-bus power system network consists of 3 transmission lines. The bus admittance matrix of the uncompensated system is

$$\begin{bmatrix} -j6 & j3 & j4 \\ j3 & -j7 & j5 \\ j4 & j5 & -j8 \end{bmatrix} \text{ p.u.}$$

If the shunt capacitance of all transmission lines is 50% compensated, the imaginary part of the 3rd row 3rd column element (in p.u.) of the bus admittance matrix after compensation is

A. $-j7.0$ **B.** $-j8.5$ **C.** $-j7.5$ **D.** $-j9.0$

Q4.14.1 NAT 2014 1M

A 183-bus power system has 150 PQ buses and 32 PV buses. In the general case, to obtain the load flow solution using Newton-Raphson method in polar coordinates, the minimum number of simultaneous equations to be solved is _____.

Q4.14.2 MCQ 2014 2M

A two bus power system shown in the figure supplies load of $1.0 + j0.5$ p.u. The values of V_1 in p.u. and δ_2 respectively are

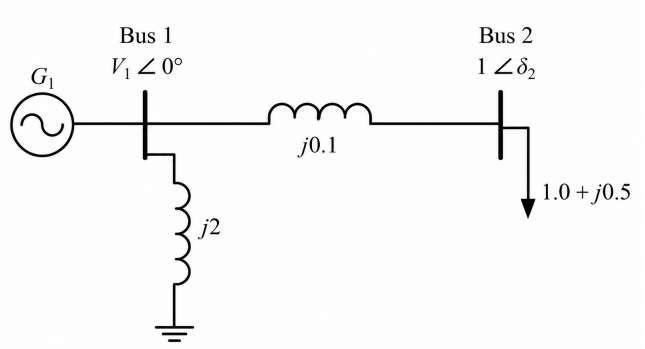


Fig. Q4.14.2

A. 0.95 and 6.00° **B.** 1.05 and −5.44°
C. 1.1 and −6.00° **D.** 1.1 and −27.12°

Q4.13.1 MCQ 2013 2M

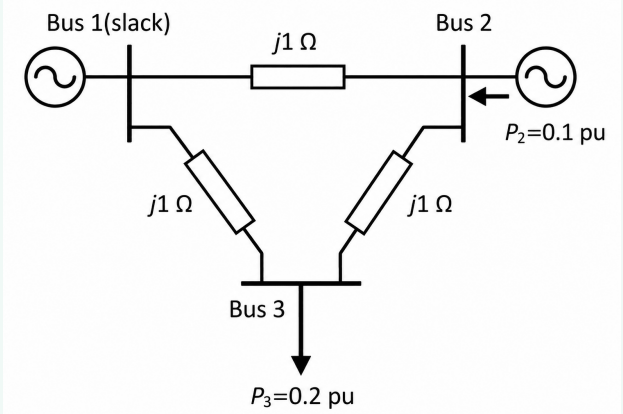
For a power system network with n nodes, Z_{33} of its bus impedance matrix is $j0.5$ per unit. The voltage at node 3 is $1.3\angle -10°$ per unit. If a capacitor having reactance of $-j3.5$ per unit is now added to the network between node 3

and the reference node, the current drawn by the capacitor per unit is

A. $0.325\angle -100°$ **B.** $0.325\angle 80°$
C. $0.371\angle -100°$ **D.** $0.433\angle 80°$

Common Data Questions

Answer the questions Q5.13.2 and Q5.13.3 using the following data.
In the following network, the voltage magnitudes at all buses are equal to 1 p.u., the voltage phase angles are very small, and the line resistances are negligible. All the line reactances are equal to $j1\ \Omega$.



Q4.13.2 MCQ 2013 2M

The voltage phase angles in rad at buses 2 and 3 are

A. $\theta_2 = -0.1,\ \theta_3 = -0.2$ **B.** $\theta_2 = 0,\ \theta_3 = -0.1$
C. $\theta_2 = 0.1,\ \theta_3 = 0.1$ **D.** $\theta_2 = 0.1,\ \theta_3 = 0.2$

Q4.13.3 MCQ 2013 2M

If the base impedance and the line-to-line base voltage are $100\ \Omega$ and 100 kV, respectively, then the real power in MW delivered by the generator connected at the slack bus is

A. -10 **B.** 0 **C.** 10 **D.** 20

Q4.12.1 MCQ 2012 1M

The bus admittance matrix of a three-bus three-line system is

$$Y = j\begin{bmatrix} -13 & 10 & 5 \\ 10 & -18 & 10 \\ 5 & 10 & -13 \end{bmatrix}$$

If each transmission line between the two buses is represented by an equivalent π-network, the magnitude of the shunt susceptance of the line connecting bus 1 and 2 is

A. 4 **B.** 2 **C.** 1 **D.** 0

Q4.11.1 MCQ 2011 2M

A three-bus network is shown in the figure below indicating the p.u. impedances of each element.

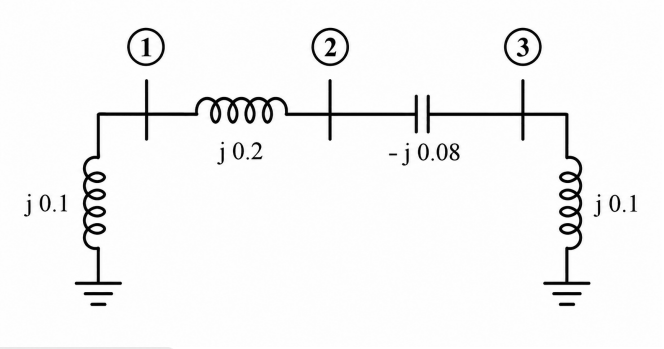


Fig. Q4.11.1

The bus admittance matrix, $\mathbf{Y}_{bus}$, of the network is

A. $j\begin{bmatrix} 0.3 & -0.2 & 0 \\ -0.2 & 0.12 & 0.08 \\ 0 & 0.08 & 0.02 \end{bmatrix}$

B. $j\begin{bmatrix} -15 & 5 & 0 \\ 5 & 7.5 & -12.5 \\ 0 & -12.5 & 2.5 \end{bmatrix}$

C. $j\begin{bmatrix} 0.1 & 0.2 & 0 \\ 0.2 & 0.12 & -0.08 \\ 0 & -0.08 & 0.10 \end{bmatrix}$

D. $j\begin{bmatrix} -10 & 5 & 0 \\ 5 & 7.5 & 12.5 \\ 0 & 12.5 & -10 \end{bmatrix}$

Q4.10.1 MCQ 2010 1M

Consider two buses connected by an impedance of $(0 + 5j)\ \Omega$. The bus '1' voltage is $100\angle 30°$ V, and bus '2' voltage is $100\angle 0°$ V. The real and reactive power supplied by bus '1' respectively are

A. 1000 W, 268 VAr **B.** −1000 W, −134 VAr
C. 276.9 W, −56.7 VAr **D.** −276.9 W, 56.7 VAr

Q4.09.1 MCQ 2009 1M

For a fixed value of complex power flow in a transmission line having a sending-end voltage V, the real power loss will be proportional to

A. V **B.** V^2 **C.** $\frac{1}{V^2}$ **D.** $\frac{1}{V}$

Q4.09.2 MCQ 2009 2M

For the Y-bus matrix of a 4-bus system given in per unit, the buses having shunt elements are

$$Y_{BUS} = j\begin{bmatrix} -5 & 2 & 2.5 & 0 \\ 2 & -10 & 2.5 & 4 \\ 2.5 & 2.5 & -9 & 4 \\ 0 & 4 & 4 & -8 \end{bmatrix}$$

A. 3 and 4 **B.** 2 and 3 **C.** 1 and 2 **D.** 1, 2 and 4

Q4.09.3 MCQ 2009 2M

Three generators are feeding a load of 100 MW. The details of the generators are

Generator	Rating (MW)	Efficiency (%)	Regulation (pu) on 100 MVA base
1	100	20	0.02
2	100	30	0.04
3	100	40	0.03

In the event of increased load power demand, which of the following will happen?

A. All the generators will share equal power

B. Generator-3 will share more power compared to Generator-1

C. Generator-1 will share more power compared to Generator-2

D. Generator-2 will share more power compared to Generator-3

Q4.08.1 MCQ 2008 1M

A three-phase transmission line is shown in the figure.

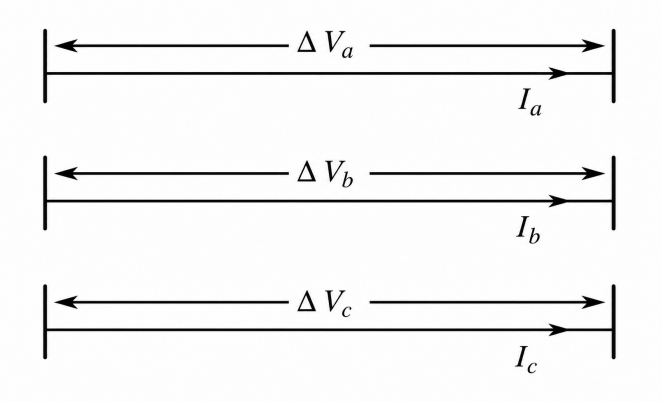


Fig. Q4.08.1

The voltage drop across the transmission line is given by

$$\begin{bmatrix} \Delta V_a \\ \Delta V_b \\ \Delta V_c \end{bmatrix} = \begin{bmatrix} Z_s & Z_m & Z_m \\ Z_m & Z_s & Z_m \\ Z_m & Z_m & Z_s \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

Shunt capacitance of the line can be neglected. If the line has positive-sequence impedance of 15 Ω and zero-sequence impedance of 48 Ω, then the values of Z_s and Z_m will be

A. $Z_s = 31.5\ \Omega,\ Z_m = 16.5\ \Omega$ **B.** $Z_s = 26\ \Omega,\ Z_m = 11\ \Omega$
C. $Z_s = 16.5\ \Omega,\ Z_m = 31.5\ \Omega$ **D.** $Z_s = 11\ \Omega,\ Z_m = 26\ \Omega$

Q4.08.2 MCQ 2008 2M

Single-line diagram of a 4-bus single-source distribution system is shown below. Branches e_1, e_2, e_3 and e_4 have equal impedances. The load current values indicated in the figure are in per unit.

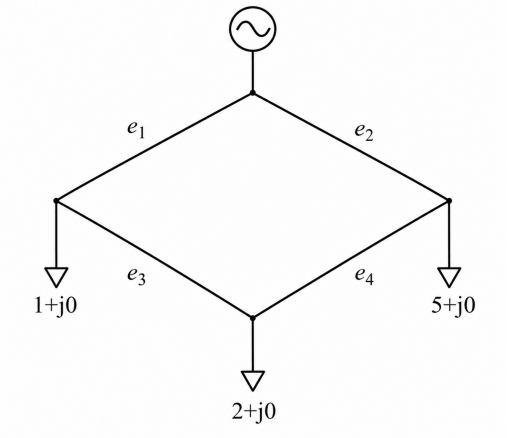


Fig. Q4.08.2

Distribution company's policy requires radial system operation with minimum loss. This can be achieved by opening

the branch

A. e_1 **B.** e_2 **C.** e_3 **D.** e_4

Q4.07.1 MCQ 2007 2M

Consider the two power systems shown in figure A below, which are initially not interconnected, and are operating in steady state at the same frequency. Separate loadflow solutions are computed individually for the two systems, corresponding to this scenario. The bus voltage phasors so obtained are indicated on figure A.

These two isolated systems are now interconnected by a short transmission line as shown in figure B, and it is found that $P_1 = P_2 = Q_1 = Q_2 = 0$.

The bus voltage phase angular difference between generator bus X and generator bus Y after the interconnection is

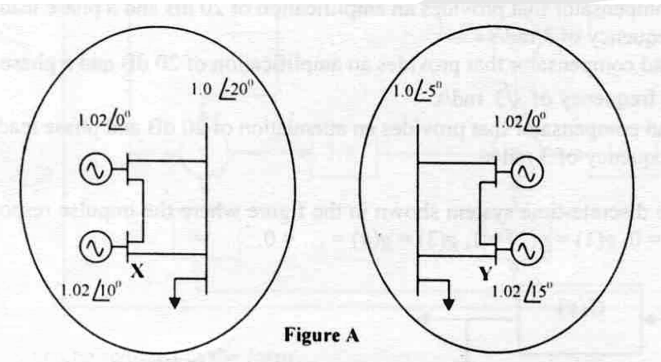


Figure A

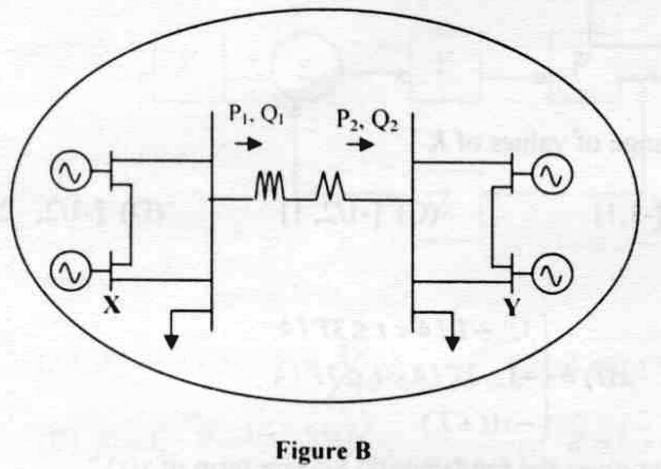


Figure B

In Figure A, the left and right load-bus angles are $-20°$ and $-5°$, respectively.

A. 10° **B.** 25° **C.** $-30°$ **D.** 30°

Q4.07.2 MCQ 2007 2M

The figure below shows a three phase self-commutated voltage source converter connected to a power system. The converter's dc bus capacitor is marked as C in the figure. The circuit is initially operating in steady state with $\delta = 0$ and the capacitor dc voltage is equal to V_{dc0}. You may neglect all losses and harmonics. What action should be taken to increase the capacitor dc voltage slowly to a new steady state value?

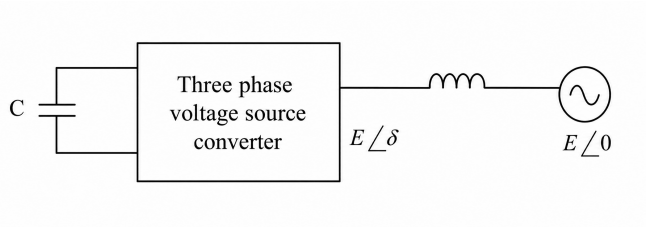


Fig. Q4.07.2

A. Make δ positive and maintain it at a positive value

B. Make δ positive and return it to its original value

C. Make δ negative and maintain it at a negative value

D. Make δ negative and return it to its original value

Q4.06.1 MCQ 2006 2M

The Gauss Seidel load flow method has following disadvantages. Tick the incorrect statement.

A. Unreliable convergence

B. A good initial guess for voltages is essential for convergence

C. Choice of slack bus affects convergence

D. Slow convergence

Common Data Questions

Answer the questions Q5.06.2 and Q5.06.3 based on the data below:

For a power system the admittance and impedance matrices for the fault studies are

$$Y_{bus} = \begin{bmatrix} -j8.75 & j1.25 & j2.50 \\ j1.25 & -j6.25 & j2.50 \\ j2.50 & -j2.50 & -j5.00 \end{bmatrix}$$

$$Z_{bus} = \begin{bmatrix} j0.16 & j0.08 & j0.12 \\ j0.08 & j0.24 & j0.16 \\ j0.12 & j0.16 & j0.34 \end{bmatrix}$$

The pre-fault voltages are 1.0 pu at all the buses. The system was unloaded prior to the fault. A solid three-phase fault takes place at bus 2.

Q4.06.2 MCQ 2006 2M

The post-fault voltages at buses 1 and 3 respectively are

A. 0.24, 0.63 **B.** 0.31, 0.76
C. 0.33, 0.67 **D.** 0.67, 0.33

Q4.06.3 MCQ 2006 2M

The per unit fault feeds from generators connected to buses 1 and 2 respectively are

A. 1.20, 2.51 **B.** 1.55, 2.61
C. 1.66, 2.50 **D.** 5.00, 2.50

Q4.05.1 MCQ 2005 2M

The network shown in figure below has impedances in p.u. as indicated. The diagonal element Y_{22} of the bus admittance matrix Y_{BUS} of the network is

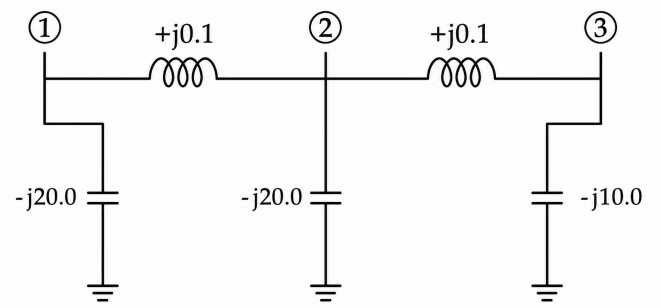


Fig. Q4.05.1

A. $-j19.8$ **B.** $+j20.0$ **C.** $+j0.2$ **D.** $-j19.95$

Q4.03.1 MCQ 2003 1M

A power system consists of 300 buses out of which 20 buses are generator buses, 25 buses are the ones with reactive power support and 15 buses are the ones with fixed shunt capacitors. All the other buses are load buses. It is proposed to perform a load flow analysis for the system using Newton-Raphson method. The size of the Newton-Raphson Jacobian matrix is

A. 553×553 **B.** 540×540
C. 555×555 **D.** 554×554

Q4.03.2 MCQ 2003 2M

The bus impedance matrix of a 4-bus power system is given by

$$Z_{bus} = \begin{bmatrix} j0.3435 & j0.2860 & j0.2723 & j0.2277 \\ j0.2860 & j0.3408 & j0.2586 & j0.2414 \\ j0.2723 & j0.2586 & j0.2791 & j0.2209 \\ j0.2277 & j0.2414 & j0.2209 & j0.2791 \end{bmatrix}$$

A branch having an impedance of $j0.2\Omega$ is connected between bus 2 and the reference. Then the values of $Z_{22,\text{new}}$ and $Z_{23,\text{new}}$ of the modified bus impedance matrix are respectively

A. $j0.5408\ \Omega$ and $j0.4586\ \Omega$

B. $j0.1260\ \Omega$ and $j0.0956\ \Omega$

C. $j0.5408\ \Omega$ and $j0.0956\ \Omega$

D. $j0.1260\ \Omega$ and $j0.1630\ \Omega$

Q4.02.1 MCQ 2002 2M

A power system consists of 2 areas (Area 1 and Area 2) connected by a single tie line. It is required to carry out a load flow study on this system. While entering the network data, the tie-line data (connectivity and parameters) is inadvertently left out. If the load flow program is run with this incomplete data

A. The load-flow will converge only if the slack bus is specified in Area 1.

B. The load-flow will converge only if the slack bus is specified in Area 2.

C. The load-flow will converge if the slack bus is specified in either Area 1 or Area 2.

D. The load-flow will not converge if only one slack bus is specified.

Q4.02.2 NAT 2002 5M

Two transposed 3-phase lines run parallel to each other. The equation describing the voltage drop in both lines is given below.

$$\begin{bmatrix}\Delta V_{a1}\\ \Delta V_{b1}\\ \Delta V_{c1}\\ \Delta V_{a2}\\ \Delta V_{b2}\\ \Delta V_{c2}\end{bmatrix} = j\begin{bmatrix}0.15 & 0.05 & 0.05 & 0.04 & 0.04 & 0.04\\ 0.05 & 0.15 & 0.05 & 0.04 & 0.04 & 0.04\\ 0.05 & 0.05 & 0.15 & 0.04 & 0.04 & 0.04\\ 0.04 & 0.04 & 0.04 & 0.15 & 0.05 & 0.05\\ 0.04 & 0.04 & 0.04 & 0.05 & 0.15 & 0.05\\ 0.04 & 0.04 & 0.04 & 0.05 & 0.05 & 0.15\end{bmatrix}\begin{bmatrix}I_{a1}\\ I_{b1}\\ I_{c1}\\ I_{a2}\\ I_{b2}\\ I_{c2}\end{bmatrix}$$

Compute the self and mutual zero-sequence impedances of this system, i.e. compute Z_{011}, Z_{012} and Z_{022} in the following equations $\Delta V_{01} = Z_{011}I_{01} + Z_{012}I_{02}$ and $\Delta V_{02} = Z_{021}I_{01} + Z_{022}I_{02}$
where ΔV_{01}, ΔV_{02}, I_{01} and I_{02} are the zero-sequence voltage drops and currents for the two lines respectively.

Q4.01.1 NAT 2001 5M

For the Y-bus matrix given in per unit values, where the first, second, third and fourth rows refer to bus 1, 2, 3 and 4 respectively, draw the reactance diagram.

$$Y_{\text{bus}} = j\begin{bmatrix}-6 & 2 & 2.5 & 0\\ 2 & -10 & 2.5 & 4\\ 2.5 & 2.5 & -9 & 4\\ 0 & 4 & 4 & -8\end{bmatrix}$$

Q4.00.1 NAT 2000 5M

In a 3-bus system, Gauss load flow method is to be used for finding the switched capacitor compensation required to maintain the voltage at bus 2 equal to 1.0 p.u. The data for the system is as follows:

Line data: $Z_{12} = Z_{13} = Z_{31} = j0.1$ p.u.

Neglect line charging.

Bus data:

Bus No.	Bus Type	Specifications
1	Slack	$V_1 = (1 + j0)$ p.u.
2	PV	Load: $P_2 + jQ_2 = (0.4 + j0.2)$ p.u.
3	PQ	Load: $P_3 + jQ_3 = (0.3 + j0.15)$ p.u.

All data are on common base values.

With the help of one iteration of load flow, explain how you will achieve the stated objective.

Q4.98.1 NAT 1998 5M

For the given network in figure below, obtain the bus admittance matrix (Y_{BUS}) using the data given:

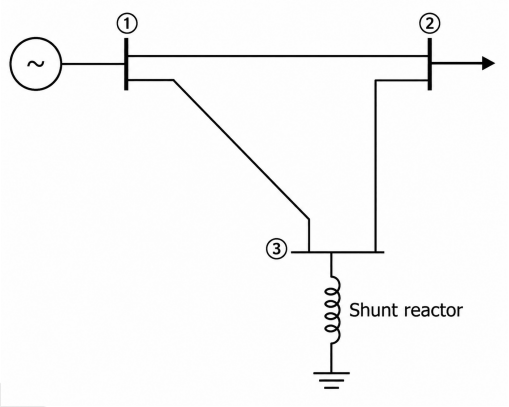


Fig. Q4.98.1

The line data, in per unit, are:

Line	Series impedance	Half-line charging admittance
1 – 2	$j0.05$	$j1.25$
1 – 3	$j0.02$	$j0.50$
2 – 3	$j0.02$	$j0.50$

The shunt reactor connected at bus 3 in the diagram has impedance $j2.0$ per unit.

Q4.97.1 MCQ 1997 1M

Gauss-Seidel iterative method can be used for solving a set of

A. linear differential equations only

B. linear algebraic equations only

C. both linear and nonlinear algebraic equations

D. both linear and nonlinear differential equations

Q4.96.1 MCQ 1996 1M

If the reference bus is changed in two load-flow runs with same system data and power obtained for reference bus taken as specified P and Q in the later run

A. the system losses will be unchanged but complex bus voltages will change

B. the system losses will change but complex bus voltages remain unchanged

C. the system losses as well as complex bus voltage will change

D. the system losses as well as complex bus voltage will be unchanged

Q4.96.2 NAT 1996 2M

The per unit voltages of two synchronous machines connected through a lossless line are $0.95\angle 10°$ and $1.0\angle 0°$. Match the following:

(a) Real power of machine 1
(b) Reactive power of machine 1
(c) Power factor of machine 1
(P) Positive real power
(Q) Positive reactive power
(R) Negative real power
(S) Negative reactive power
(T) Leading power factor
(U) Lagging power factor

Q4.95.1 MCQ 1995 1M

In load-flow analysis, a voltage-controlled bus is treated as a load bus in subsequent iterations if a reactive power limit is violated. (True/False)

Q4.93.1 MCQ 1993 1M

In load flow analysis, the load connected at a bus is represented as

A. Constant current drawn from the bus

B. Constant impedance connected at the bus

C. Voltage and frequency dependent source at the bus

D. Constant real and reactive power drawn from the bus

Q4.92.1 NAT 1992 1M

In load flow studies of a power system, the quantities specified at a voltage-controlled bus are _____ and _____.

Q4.91.1 MCQ 1991 1M

A sample power system network is shown in Figure. The reactances marked are in p.u. The p.u. value of the Bus Admittance Matrix element Y_{22} is

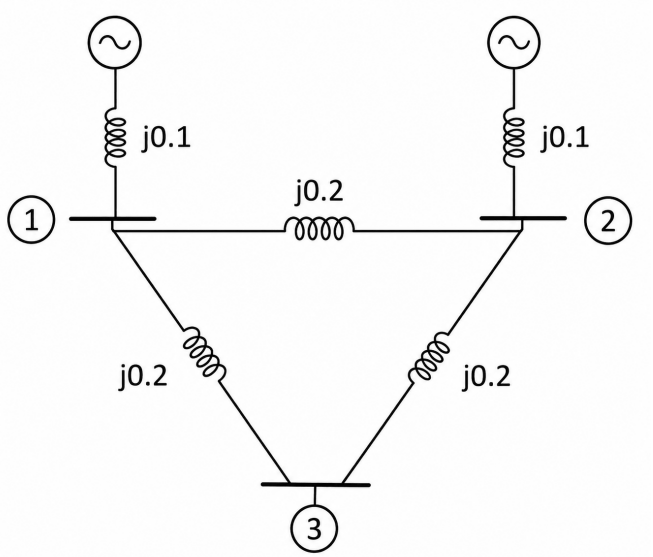


Fig. Q4.91.1

A. $j10.0$ **B.** $j0.4$ **C.** $-j0.1$ **D.** $-j20.0$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q4.26.1 B	Q4.26.2 A,D	Q4.26.3 D
Q4.25.1 0.380 to 0.400	Q4.25.2 A	Q4.24.1 A
Q4.24.2 A	Q4.23.1 A,C	Q4.23.2 D
Q4.21.1 A	Q4.21.2 C	Q4.20.1 C
Q4.20.2 A	Q4.20.3 1.11 to 1.13	Q4.19.1 0.095 to 0.105
Q4.18.1 3500	Q4.18.2 C	Q4.17.1 14

Q4.17.2 B	Q4.17.3 C	Q4.17.4 D
Q4.17.5 61	Q4.16.1 B	Q4.16.2 A
Q4.15.1 D	Q4.15.2 B	Q4.14.1 332
Q4.14.2 B	Q4.13.1 D	Q4.13.2 B
Q4.13.3 C	Q4.12.1 B	Q4.11.1 B
Q4.10.1 A	Q4.09.1 C	Q4.09.2 C
Q4.09.3 C	Q4.08.1 B	Q4.08.2 D
Q4.07.1 A	Q4.07.2 D	Q4.06.1 A
Q4.06.2 D	Q4.06.3 C	Q4.05.1 D
Q4.03.1 D	Q4.03.2 B	Q4.02.1 D
Q4.02.2 *	Q4.01.1 REACTANCE DIAGRAM	Q4.00.1 *
Q4.98.1 *	Q4.97.1 C	Q4.96.1 C
Q4.96.2 P,S,U	Q4.95.1 TRUE	Q4.93.1 D
Q4.92.1 VOLTAGE AND REAL POWER	Q4.91.1 D	

PrepFusion

UNIT

V

Performance and Models of Transmission Lines and Cables

Topics

1. Short Transmission Line
2. Medium Transmission Line
3. Long Transmission Line

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 23	**MCQ:** 12	**MSQ:** 0	**NAT:** 11
1M: 10	**2M:** 11	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2023)–2M(2019)

**See preface for how Avg Weightage and Range are calculated.*

Q5.23.1 NAT 2023 1M

A 50 Hz, 275 kV line of length 400 km has the following parameters:

Resistance, $R = 0.035\ \Omega$/km;

Inductance, $L = 1$ mH/km;

Capacitance, $C = 0.01\ \mu$F/km;

The line is represented by the nominal-π model. With the magnitudes of the sending end and the receiving end voltages of the line (denoted by V_S and V_R, respectively) maintained at 275 kV, the phase angle difference (θ) between V_S and V_R required for maximum possible active power to be delivered to the receiving end, in degree is _____ (Round off to 2 decimal places).

Q5.19.1 NAT 2019 2M

A three-phase 50 Hz, 400 kV transmission line is 300 km long. The line inductance is 1 mH/km per phase, and the capacitance is 0.01 μF/km per phase. The line is under open circuit condition at the receiving end and energized with 400 kV at the sending end, the receiving end line voltage in kV (round off to two decimal places) will be _____.

Q5.17.1 NAT 2017 1M

The nominal-π circuit of a transmission line is shown in the figure.

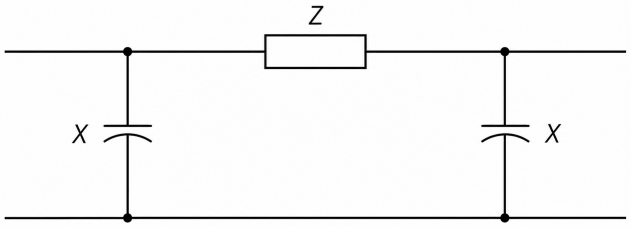


Fig. Q5.17.1

Impedance

$$Z = 100\angle 80^\circ\ \Omega$$

and reactance

$$X = 3300\ \Omega$$

The magnitude of the characteristic impedance of the transmission line, in Ω, is _____. (Give the answer up to one decimal place.)

Q5.17.2 NAT 2017 2M

A 3-phase, 50 Hz generator supplies power of 3 MW at 17.32 kV to a balanced 3-phase inductive load through an overhead line. The per phase line resistance and reactance are 0.25 Ω and 3.925 Ω respectively. If the voltage at the generator terminal is 17.87 kV, the power factor of the load is _____.

Q5.16.1 NAT 2016 2M

At no load condition, a 3-phase, 50 Hz, lossless power transmission line has sending-end and receiving-end voltages of 400 kV and 420 kV respectively. Assuming the velocity of traveling wave to be the velocity of light, the length of the line, in km, is _____.

Q5.14.1 MCQ 2014 1M

In a long transmission line with r, l, g and c are the resistance, inductance, shunt conductance and capacitance per unit length, respectively, the condition for distortionless transmission is

A. $rc = lg$ **B.** $r = \sqrt{l/c}$ **C.** $rg = lc$ **D.** $g = \sqrt{c/l}$

Q5.14.2 NAT 2014 2M

For a 400 km long transmission line, the series impedance is $(0.0 + j0.5)\ \Omega$/km and the shunt admittance is $(0.0 + j5.0)\ \mu$mho/km. The magnitude of the series impedance (in Ω) of the equivalent π circuit of the transmission line is _____.

Q5.08.1 MCQ 2008 1M

An extra high voltage transmission line of length 300 km can be approximated by a lossless line having propagation constant $\beta = 0.00127$ rad/km. Then the percentage ratio of line length to wavelength will be

A. 24.24% **B.** 12.12% **C.** 19.05% **D.** 6.06%

Q5.07.1 MCQ 2007 2M

The total reactance and total susceptance of a lossless overhead EHV line, operating at 50 Hz, are given by 0.045 pu and 1.2 pu respectively. If the velocity of wave propagation is 3×10^5 km/s, then the approximate length of the line is

A. 122 km **B.** 172 km **C.** 222 km **D.** 272 km

Q5.06.1 MCQ 2006 1M

The concept of an electrically short, medium, and long line is primarily based on the

A. physical length of the line

B. nominal voltage of the line

C. power transmitted over the line

D. wavelength of the line

Q5.06.2 MCQ 2006 2M

The A, B, C, D constants of a 220 kV line are:

$$A = D = 0.94\angle 1^\circ, \quad B = 130\angle 73^\circ, \quad C = 0.001\angle 90^\circ$$

If the sending end voltage of the line for a given load delivered at nominal voltage is 240 kV, then the percentage voltage regulation of the line is

A. 5 **B.** 9 **C.** 16 **D.** 21

Q5.04.1 MCQ 2004 2M

The generalized circuit constants of a 3-phase, 220 kV rated voltage, medium length transmission line are

$$A = D = 0.936 + j0.016 = 0.936\angle 0.98^\circ$$
$$B = 33.5 + j138 = 142.0\angle 76.4^\circ\ \Omega$$
$$C = (-5.18 + j914) \times 10^{-6}\ \mho$$

If the load at the receiving end is 50 MW at 220 kV with a power factor of 0.9 lagging, the magnitude of line sending end voltage should be

A. 133.23 kV **B.** 220.00 kV
C. 230.78 kV **D.** 246.30 kV

Q5.03.1 MCQ 2003 2M

The ABCD parameters of a 3-phase overhead transmission line are

$$A = D = 0.9\angle 0^\circ$$
$$B = 200\angle 90^\circ\ \Omega$$
$$C = 0.95 \times 10^{-3}\angle 90^\circ\ \text{S}$$

At no-load condition, a shunt inductive reactor is connected at the receiving end of the line to limit the receiving-end voltage to be equal to the sending-end voltage. The ohmic value of the reactor is

A. $\infty\ \Omega$ **B.** 2000 Ω **C.** 105.26 Ω **D.** 1052.6 Ω

Q5.02.1 NAT 2002 5M

A long lossless transmission line has a unity power factor (UPF) load at the receiving end and an AC voltage source at the sending end. The parameters of the transmission line are as follows:

Characteristic impedance $Z_c = 400\,\Omega$, propagation constant $\beta = 1.2 \times 10^{-3}$ rad/km, and length $l = 100$ km.

The equation relating sending and receiving end quantities is $V_s = V_r \cos(\beta l) + jZ_c \sin(\beta l) I_r$

Compute the maximum power that can be transferred to the UPF load at the receiving end if $|V_s| = 230$ kV

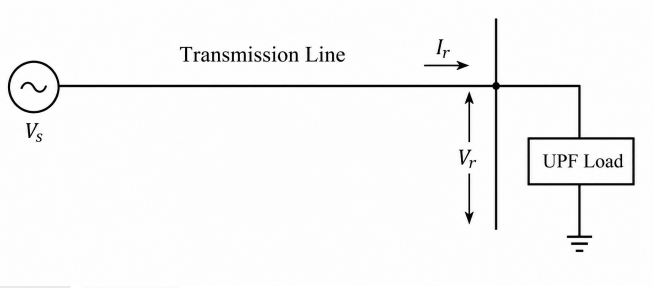


Fig. Q5.02.1

Q5.99.1 MCQ 1999 2M

A 220 kV, 20 km long, 3-phase transmission line has the following A, B, C, D constants. $A = D = 0.96\angle 3^\circ$, $B = 55\angle 65^\circ\ \Omega$/phase, $C = 0.5 \times 10^{-4}\angle 80^\circ$ S/phase. Its charging current per phase is:

A. $\frac{11}{\sqrt{3}}$ A **B.** 11 A
C. 220 A **D.** $\frac{220}{\sqrt{3}}$ A

Q5.99.2 NAT 1999 5M

A 66 kV, 3-phase, 50 Hz, 150 km long overhead transmission line is open circuited at the receiving end. Each conductor has a resistance of 0.25 Ω/km, an inductive reactance of 0.5 Ω/km and a capacitance admittance neutral of 0.04×10^{-4} S/km.

(a) Draw the nominal π-equivalent circuit and indicate the value of each parameter.

(b) Calculate the receiving end voltage if the sending end voltage is 66 kV.

Q5.98.1 MCQ 1998 1M

The reflection coefficient for the transmission line shown in figure below at P is:

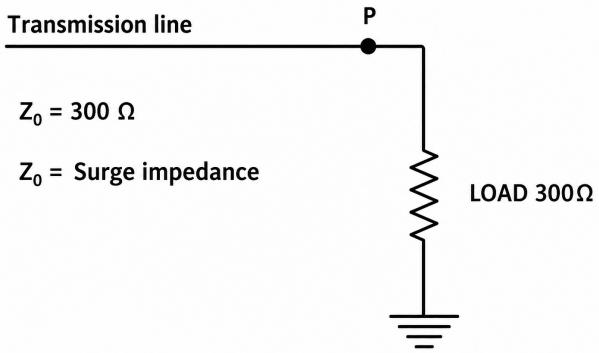


Fig. Q5.98.1

A. +1 **B.** −1 **C.** 0 **D.** 0.5

Q5.98.2 MCQ 1998 2M

A cable has the following characteristics. $L = 0.201\ \mu$H/m and $C = 196.2$ pF/m. The velocity of wave propagation through the cable is:

A. 32 m/s **B.** 159.24 m/μs
C. 0.0312 m/s **D.** 159.24 m/s

Q5.96.1 MCQ 1996 1M

For a 500 Hz frequency excitation, a 50 km short power line will be modelled as

A. Short line **B.** Medium line
C. Long line **D.** Data insufficient

Q5.95.1 MCQ 1995 1M

The surge impedance of a 400 km long overhead transmission line is 400 Ω. For a 200 km length of the same line, the surge impedance will be

A. 200 Ω **B.** 800 Ω **C.** 400 Ω **D.** 100 Ω

Q5.94.1 NAT 1994 1M

The increase in resistance due to non-uniform distribution of current in a conductor is known as _____ effect.

Q5.94.2 NAT 1994 2M

The charging current of a 400 kV transmission line is more than that of a 220 kV line of the same length. (True/False)

Q5.91.1 NAT 1991 1M

Two transmission lines are connected in cascade, whose ABCD parameters are

$$\begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} = \begin{bmatrix} 1 & 10\angle 30^\circ \\ 0 & 1 \end{bmatrix},$$

and

$$\begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0.025\angle -30^\circ & 1 \end{bmatrix}.$$

Determine the resultant ABCD parameters of the cascaded transmission line.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q5.23.1 83.00 to 84.00	Q5.19.1 414 to 423	Q5.17.1 404 to 409
Q5.17.2 0.75 to 0.85	Q5.16.1 294 to 298	Q5.14.1 A
Q5.14.2 186 to 188	Q5.08.1 D	Q5.07.1 C
Q5.06.1 A	Q5.06.2 C	Q5.04.1 C
Q5.03.1 B	Q5.02.1 556 MW	Q5.99.1 A
Q5.99.2 *	Q5.98.1 C	Q5.98.2 B
Q5.96.1 C	Q5.95.1 C	Q5.94.1 SKIN
Q5.94.2 TRUE	Q5.91.1 *	

UNIT

VI

Parameters of Transmission Lines

Topics

1. Skin Effect, Inductance due to Internal and External Flux
2. Inductance of 3 phase system- calculations
3. GMD, GMR and Capacitance of Transmission Lines
4. Travelling Waves in Transmission Lines

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 22	**MCQ:** 18	**MSQ:** 0	**NAT:** 4
1M: 8	**2M:** 14	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2022)–7M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q6.22.1 MCQ 2022 1M

The geometric mean radius of a conductor, having four equal strands with each strand of radius r, as shown in the figure below, is

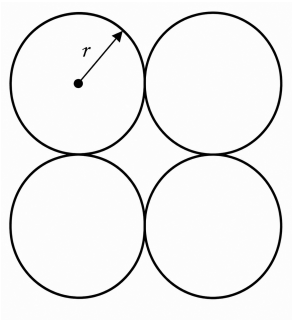


Fig. Q6.22.1

A. $4r$ **B.** $1.414\,r$ **C.** $2r$ **D.** $1.723\,r$

Q6.17.1 NAT 2017 2M

Consider an overhead transmission line with 3-phase, 50 Hz balanced system with conductors located at the vertices of an equilateral triangle of length

$$D_{ab} = D_{bc} = D_{ca} = 1 \text{ m}$$

as shown in figure below. The resistances of the conductors are neglected. The geometric mean radius (GMR) of each conductor is 0.01 m. Neglecting the effect of ground, the magnitude of positive sequence reactance in Ω/km (rounded off to three decimal places) is _____.

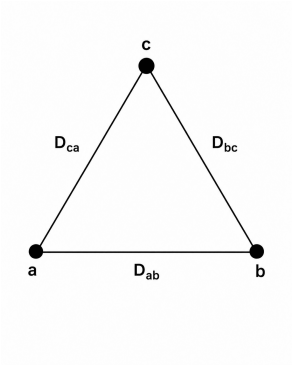


Fig. Q6.17.1

Q6.16.1 NAT 2016 2M

A single-phase transmission line has two conductors each of 10 mm radius. These are fixed at a center-to-center distance of 1 m in a horizontal plane. This is now converted to a three-phase transmission line by introducing a third conductor of the same radius. This conductor is fixed at an equal distance D from the two single-phase conductors. The positive sequence inductance per phase of the three-phase system is made 5% more than that of the inductance per conductor of the single-phase system. The distance D, in meters, is _____.

Q6.15.1 NAT 2015 2M

A composite conductor consists of three conductors of radius R each. The conductors are arranged as shown below. The geometric mean radius (GMR) (in cm) of the composite conductor is kR. The value of k is _____.

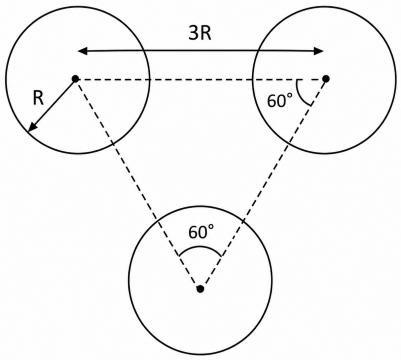


Fig. Q6.15.1

Q6.14.1 MCQ 2014 2M

The horizontally placed conductors of a single phase line operating at 50 Hz are having outside diameter of 1.6 cm, and the spacing between centers of the conductors is 6 m. The permittivity of free space is 8.854×10^{-12} F/m. The capacitance to ground per kilometer of each line is

A. 4.2×10^{-9} F **B.** 8.4×10^{-9} F
C. 4.2×10^{-12} F **D.** 8.4×10^{-12} F

Q6.10.1 MCQ 2010 2M

A 50 Hz synchronous generator is initially connected to a long lossless transmission line which is open circuited at the receiving end. With the field voltage held constant, the generator is disconnected from the transmission line. Which of the following may be said about the steady-state terminal voltage and field current of the generator?

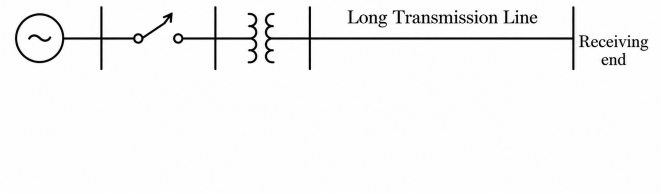


Fig. Q6.10.1

A. The magnitude of terminal voltage decreases, and the field current does not change.

B. The magnitude of terminal voltage increases, and the field current does not change.

C. The magnitude of terminal voltage increases, and the field current increases.

D. The magnitude of terminal voltage does not change and the field current decreases.

Q6.10.2 MCQ 2010 2M

Consider a three-phase, 50 Hz, 11 kV distribution system. Each of the conductors is suspended by an insulator string having two identical porcelain insulators. The self capacitance of the insulator is 5 times the shunt capacitance between the link and the ground, as shown in the figure. The voltages across the two insulators are

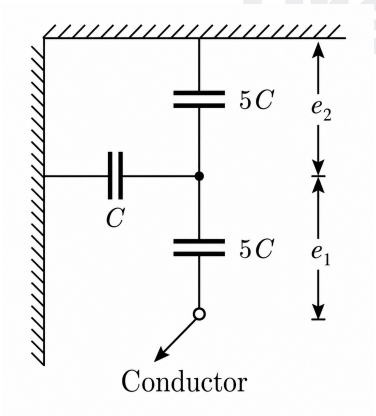


Fig. Q6.10.2

A. $e_1 = 3.74$ kV, $e_2 = 2.61$ kV

B. $e_1 = 3.46$ kV, $e_2 = 2.89$ kV

C. $e_1 = 6.0$ kV, $e_2 = 4.23$ kV

D. $e_1 = 5.5$ kV, $e_2 = 5.5$ kV

Q6.10.3 MCQ 2010 2M

Consider a three-core, three-phase, 50 Hz, 11 kV cable whose conductors are denoted as R, Y, and B as shown in the figure. The inter-phase capacitance (C_1) between each pair of conductors is 0.2 μF and the capacitance between each line conductor and the sheath is 0.4 μF. The per-phase charging current is

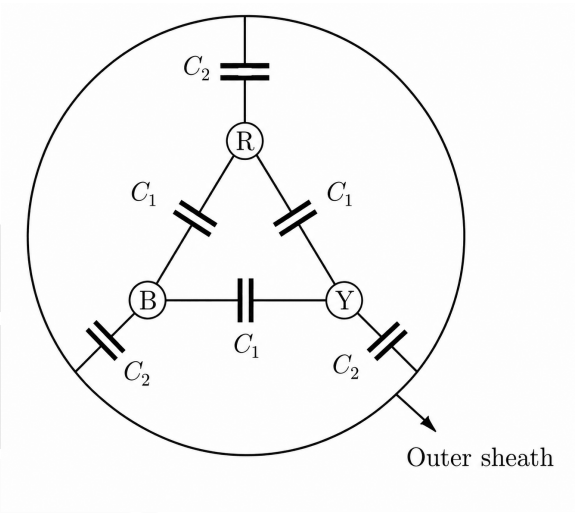


Fig. Q6.10.3

A. 2.0 A **B.** 2.4 A **C.** 2.7 A **D.** 3.5 A

Q6.10.4 MCQ 2010 1M

Consider a step voltage of magnitude 1 pu travelling along a lossless transmission line that terminates in a reactor. The voltage magnitude across the reactor at the instant travelling wave reaches the reactor is

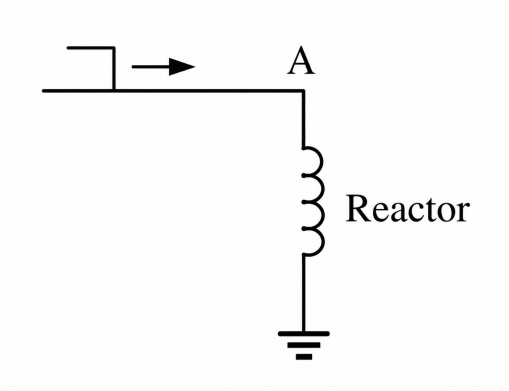


Fig. Q6.10.4

A. −1 pu **B.** 1 pu **C.** 2 pu **D.** 3 pu

Q6.04.1 MCQ 2004 2M

A 800 kV transmission line is having per phase line inductance of 1.1 mH/km and per phase line capacitance of 11.68 nF/km. Ignoring the length of the line, its ideal power transfer capability in MW is

A. 1204 **B.** 1504 **C.** 2085 **D.** 2606

Q6.03.1 MCQ 2003 1M

Bundled conductors are mainly used in high voltage overhead transmission lines to

A. reduce transmission line losses

B. reduce corona

C. increase mechanical strength of the line

D. reduce sag

Q6.03.2 MCQ 2003 2M

A surge of 20 kV magnitude travels along a lossless cable towards its junction with two identical lossless overhead transmission lines. The inductance and capacitance of the cable are 0.4 mH/km and 0.5 μF/km respectively. The inductance and capacitance of the overhead transmission lines are 1.5 mH/km and 0.015 μF/km respectively. The magnitude of the voltage at the junction due to the surge is

A. 36.72 kV **B.** 18.36 kV **C.** 6.07 kV **D.** 33.93 kV

Q6.02.1 MCQ 2002 1M

Consider a long, two-wire line composed of solid round conductors. The radius of both conductors is 0.25 cm and the distance between their centers is 1 m. If this distance is doubled, then the inductance per unit length

A. decreases but does not halve **B.** doubles
C. increases but does not double **D.** halves

Q6.01.1 MCQ 2001 2M

The conductors of a 10 km long, single-phase, two-wire line are separated by a distance of 1.5 m. The diameter of each conductor is 1 cm. If the conductors are of copper, the inductance of the circuit is

A. 50.0 mH **B.** 45.3 mH **C.** 23.8 mH **D.** 19.6 mH

Q6.99.1 MCQ 1999 2M

For a single phase overhead line having solid copper conductors of diameter 1 cm, spaced 60 cm between centers, the inductance in mH/km is:

A. $0.05 + 0.2 \ln 60$ **B.** $0.2 \ln 60$
C. $0.05 + 0.2 \ln\left(\frac{60}{0.5}\right)$ **D.** $0.2 \ln\left(\frac{60}{0.5}\right)$

Q6.99.2 MCQ 1999 1M

An overhead line having a surge impedance of 400 Ω is connected in series with an underground cable having a surge impedance of 100 Ω. If a surge of 50 kV travels from the line end towards the line-cable junctions, the value of the transmitted voltage wave at the junction is:

A. 30 kV **B.** 20 kV **C.** 80 kV **D.** −30 kV

Q6.98.1 MCQ 1998 2M

If the length of a wire of resistance R is uniformly stretched to n times its original value, its new resistance is:

A. nR **B.** $\frac{R}{n}$ **C.** n^2R **D.** $\frac{R}{n^2}$

Q6.97.1 MCQ 1997 1M

Bundled conductors are employed to improve the

A. appearance of the transmission line

B. mechanical stability of the line

C. decreases system stability

D. increases the short circuit current

Q6.97.2 NAT 1997 2M

The velocity of propagation of electromagnetic wave on an underground cable with relative permittivity of 3 will be _____ m/sec.

Q6.96.1 MCQ 1996 2M

For equilateral spacing of conductors of an untransposed 3-phase line, we have

A. balanced receiving end voltage and no communication interference

B. unbalanced receiving end voltage and no communication interference

C. balanced receiving end voltage and communication interference

D. unbalanced receiving end voltage and communication interference

Q6.93.1 MCQ 1993 1M

A three-phase overhead transmission line has its conductors horizontally spaced with spacing between adjacent conductors equal to d. If the conductors of the line are rearranged to form an equilateral triangle of side d, then

A. Average capacitance and inductance will increase

B. Average capacitance will decrease and inductance will increase

C. Average capacitance will increase and inductance will decrease

D. Surge impedance loading of the line increases

Q6.92.1 MCQ 1992 1M

The inductance of a power transmission line increases with

A. Decrease in line length

B. Increase in diameter of conductor

C. Increase in spacing between the phase conductors

D. Increase in load current carried by the conductors

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q6.22.1 D	Q6.17.1 0.271 to 0.301	Q6.16.1 1.41 to 1.47
Q6.15.1 1.85 to 1.95	Q6.14.1 B	Q6.10.1 A
Q6.10.2 B	Q6.10.3 A	Q6.10.4 C
Q6.04.1 C	Q6.03.1 B	Q6.03.2 D
Q6.02.1 C	Q6.01.1 C	Q6.99.1 C
Q6.99.2 B	Q6.98.1 C	Q6.97.1 B
Q6.97.2 *	Q6.96.1 A	Q6.93.1 C
Q6.92.1 C		

UNIT

VII

Power Factor Improvement and Voltage Control Techniques

Topics

1. Power Factor Improvement - Series / Shunt Compensation by Capacitors, SVC , TSC-TCR etc.
2. Methods to control voltage, Frequency and power factor

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 28	**MCQ:** 21	**MSQ:** 0	**NAT:** 7
1M: 9	**2M:** 16	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2014)–3M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q7.25.1 NAT 2025 1M

Consider a distribution feeder, with R/X ratio of 5. At the receiving end, a 350 kVA load is connected. The maximum voltage drop will occur from the sending end to the receiving end, when the power factor of the load is _____ *(round off to three decimal places).*

Q7.25.2 NAT 2025 2M

Using shunt capacitors, the power factor of a 3-phase, 4 kV induction motor (drawing 390 kVA at 0.77 pf lag) is to be corrected to 0.85 pf lag. The line current of the capacitor bank, in A, is _____ *(round off to one decimal place).*

Q7.16.1 MCQ 2016 1M

The inductance and capacitance of a 400 kV, three-phase, 50 Hz lossless transmission line are 1.6 mH/km/phase and 10 nF/km/phase respectively. The sending end voltage is maintained at 400 kV. To maintain a voltage of 400 kV at the receiving end, when the line is delivering 300 MW load, the shunt compensation required is

A. capacitive **B.** inductive **C.** resistive **D.** zero

Q7.16.2 NAT 2016 2M

A three-phase cable is supplying 800 kW and 600 kVAr to an inductive load. It is intended to supply an additional resistive load of 100 kW through the same cable without increasing the heat dissipation in the cable, by providing a three-phase bank of capacitors connected in star across the load. Given the line voltage is 3.3 kV, 50 Hz, the capacitance per phase of the bank, expressed in microfarads, is _____.

Q7.14.1 MCQ 2014 1M

Shunt reactors are sometimes used in high voltage transmission systems to

A. limit the short circuit current through the line.

B. compensate for the series reactance of the line under heavily loaded condition.

C. limit over-voltages at the load side under lightly loaded condition.

D. compensate for the voltage drop in the line under heavily loaded condition.

Q7.14.2 NAT 2014 2M

The complex power consumed by a constant-voltage load is given by $(P_1 + jQ_1)$, where,

$$1\,\text{kW} \le P_1 \le 1.5\,\text{kW}$$

and

$$0.5\,\text{kVAR} \le Q_1 \le 1\,\text{kVAR}$$

A compensating shunt capacitor is chosen such that

$$|Q| \le 0.25\,\text{kVAR}$$

where Q is the net reactive power consumed by the capacitor-load combination. The reactive power (in kVAR) supplied by the capacitor is _____.

Q7.12.1 MCQ 2012 2M

For the system shown below, S_{D1} and S_{D2} are complex power demands at bus 1 and bus 2 respectively. If $|V_2| = 1$ pu, the VAR rating of the capacitor (Q_{G2}) connected at bus 2 is

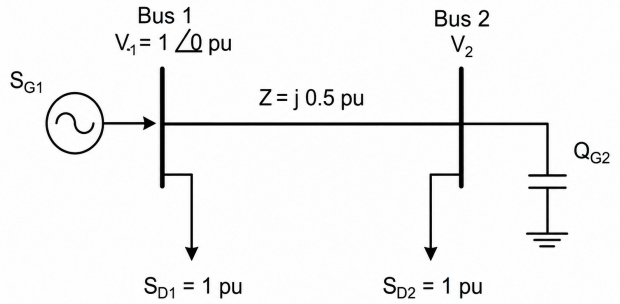


Fig. Q7.12.1

A. 0.2 pu **B.** 0.268 pu **C.** 0.312 pu **D.** 0.4 pu

Q7.11.1 MCQ 2011 1M

For enhancing the power transmission in a long EHV transmission line, the most preferred method is to connect a

A. series inductive compensator in the line

B. shunt inductive compensator at the receiving end

C. series capacitive compensator in the line

D. shunt capacitive compensator at the sending end

Q7.09.1 MCQ 2009 2M

Match the items in List-I with the items in List-II and select the correct answer using the codes given below the lists.

List-I To	List-II Use
(a) Improve power factor	(1) Shunt reactor
(b) Reduce the current ripples	(2) Shunt capacitor
(c) Increase the power flow in line	(3) Series capacitor
(d) Reduce the Ferranti effect	(4) Series reactor

A. $a \rightarrow 2,\ b \rightarrow 3,\ c \rightarrow 4,\ d \rightarrow 1$

B. $a \rightarrow 2,\ b \rightarrow 4,\ c \rightarrow 3,\ d \rightarrow 1$

C. $a \rightarrow 4,\ b \rightarrow 3,\ c \rightarrow 1,\ d \rightarrow 2$

D. $a \rightarrow 4,\ b \rightarrow 1,\ c \rightarrow 3,\ d \rightarrow 2$

Q7.08.1 MCQ 2008 2M

A lossless transmission line having Surge Impedance Loading (SIL) of 2280 MW is provided with a uniformly distributed series capacitive compensation of 30%. Then, the SIL of the compensated transmission line will be

A. 1835 MW **B.** 2280 MW
C. 2725 MW **D.** 3257 MW

Q7.07.1 MCQ 2007 2M

A 230 V (Phase), 50 Hz, three-phase, 4-wire system has a phase sequence ABC. A unity power-factor load of 4 kW is connected between phase A and neutral N. It is desired to achieve zero neutral current through the use of a pure inductor and a pure capacitor in the other two phases. The value of inductor and capacitor is

A. 72.95 mH in phase C and 139.02 μF in phase B

B. 72.95 mH in phase B and 139.02 μF in phase C

C. 42.12 mH in phase C and 240.79 μF in phase B

D. 42.12 mH in phase B and 240.79 μF in phase C

Q7.06.1 MCQ 2006 2M

A 400 V, 50 Hz, three phase balanced source supplies power to a star connected load whose rating is $12\sqrt{3}$ kVA, 0.8 pf (lag). The rating (in kVAR) of the delta connected (capacitive) reactive power bank necessary to bring the pf to unity is

A. 28.78 **B.** 21.60 **C.** 16.60 **D.** 12.47

Q7.05.1 MCQ 2005 1M

An 800 kV transmission line has a maximum power transfer capacity of P. If the line is operated at 400 kV with the series reactance unchanged, the new maximum power transfer capacity is approximately

A. P **B.** $2P$ **C.** $\frac{P}{2}$ **D.** $\frac{P}{4}$

Q7.03.1 MCQ 2003 2M

A balanced delta-connected load of $(8 + j6)\ \Omega$ per phase is connected to a 400 V, 50 Hz, 3-phase supply lines. If the input power factor is to be improved to 0.9 by connecting a bank of star-connected capacitors, the required kVAR of the bank is

A. 42.7 **B.** 10.2 **C.** 28.8 **D.** 38.4

Q7.03.2 MCQ 2003 2M

A 3-phase, 11-kV generator feeds power to a constant-power unity-power-factor load of 100 MW through a 3-phase transmission line. The line-to-line voltage at the terminals of the machine is maintained at 11 kV. The per-unit positive-sequence impedance of the line based on 100 MVA and 11 kV is $j0.2$. The line-to-line voltage at the load terminals is measured to be less than 11 kV. The total reactive power to be injected at the terminals of the load to increase the line-to-line voltage at the load terminals to 11 kV is

A. 100 MVAR **B.** 10.1 MVAR
C. -100 MVAR **D.** -10.1 MVAR

Q7.02.1 MCQ 2002 2M

A transmission line has a total series reactance of 0.2 pu. Reactive power compensation is applied at the midpoint of the line and it is controlled such that the midpoint voltage of the transmission line is always maintained at 0.98 pu. If voltage at both ends of the line are maintained at 1.0 pu, then the steady state power transfer limit of the transmission line is

A. 9.8 pu **B.** 4.9 pu **C.** 19.6 pu **D.** 5 pu

Q7.01.1 MCQ 2001 1M

A lossless radial transmission line with surge impedance loading

A. takes negative VAR at sending end and zero VAR at receiving end

B. takes positive VAR at sending end and zero VAR at receiving end

C. has flat voltage profile and unity power factor at all points along it

D. has sending end voltage higher than receiving end voltage and unity power factor at sending end

Q7.01.2 MCQ 2001 2M

Consider the model shown in figure below of a transmission line with a series capacitor at its mid-point. The maximum voltage on the line is at the location

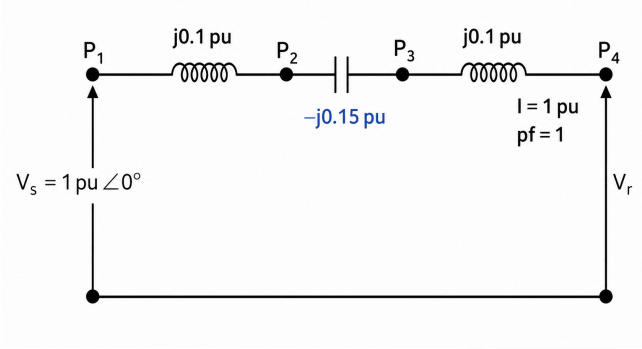


Fig. Q7.01.2

A. P_1 **B.** P_2 **C.** P_3 **D.** P_4

Q7.01.3 MCQ 2001 2M

A power system has two synchronous generators. The governor–turbine characteristics corresponding to the generators are $P_1 = 50(50 - f)$ and $P_2 = 100(51 - f)$ where f denotes the system frequency in Hz, and P_1 and P_2 are respectively the power outputs (in MW) of turbines 1 and 2. Assuming the generators and transmission network to be lossless, the system frequency for a total load of 400 MW is

A. 47.5 Hz **B.** 48.0 Hz **C.** 48.5 Hz **D.** 49.0 Hz

Q7.00.1 MCQ 2000 2M

A transmission line has equal voltages at the two ends, maintained constant by two sources. A third source is to be provided to maintain constant voltage (equal to end voltages) at either the midpoint of the line or at 75% of the distance from the sending end. Then the maximum power transfer capabilities of the line in the original case and the other two cases respectively will be in the following ratios.

A. 1:1:1 **B.** $1 : 2 : \frac{1}{0.75}$ **C.** 1:2:4 **D.** 1:4:16

Q7.00.2 NAT 2000 5M

A 275 kV, 3-phase, 50 Hz, 400 km lossless transmission line has the following parameters:

$$x = 0.05\ \Omega/\text{km}$$

$$y = 3.0 \times 10^{-6}\ \text{S/km}$$

(a) Calculate the receiving-end voltage under open-circuit conditions using appropriate and justified assumptions.

(b) Determine the receiving-end load that results in a flat voltage profile along the transmission line.

(c) If a flat voltage profile is to be maintained at 1.2 times the loading obtained in part (b), determine the type and magnitude of the uniformly distributed compensation required.

Q7.99.1 MCQ 1999 2M

A 3-phase, 11 kV, 50 Hz, 200 kW load has a power factor of 0.8 lag. A delta connected 3-phase capacitor is used to improve the power factor to unity. The capacitance per phase of the capacitor in microfarads is

A. 3.948 **B.** 1.316 **C.** 0.439 **D.** 11.844

Q7.98.1 MCQ 1998 1M

Series capacitive compensation in EHV transmission lines is used to

A. reduce the line loading

B. improve the stability of the system

C. reduce the voltage profile

D. improve the protection of the line

Q7.98.2 MCQ 1998 2M

A shunt reactor of 100 MVAr is operated at 98% of its rated voltage and at 96% of its rated frequency. The reactive power absorbed by the reactor is:

A. 98 MVAr **B.** 104.02 MVAr
C. 96.04 MVAr **D.** 100.04 MVAr

Q7.97.1 NAT 1997 5M

The system shown in figure below feeds two loads:

(a) a factory load of 15 MVA consisting of induction motors and

(b) a domestic load of 10 MW at 0.8 p.f. lagging at 6 kV.

If the induction motors are rated at 6 kV and take 5 times their rated current at zero p.f. at starting, calculate the dip in voltage at the domestic load bus bar when all the induction motors are started at the same time.

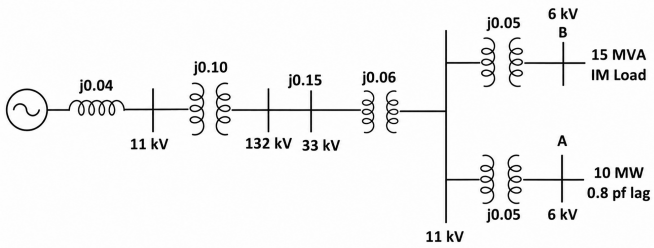


Fig. Q7.97.1

Q7.97.2 NAT 1997 5M

A factory draws 100 kW at 0.7 p.f lagging from a 3-phase, 11 kV supply. It is desired to raise the p.f. to 0.95 lagging using series capacitors. Calculate the rating of the capacitor required.

Q7.94.1 MCQ 1994 1M

In a 400 kV network, 360 kV is recorded at a 400 kV bus. The reactive power absorbed by a shunt reactor rated for 50 MVAR at 400 kV connected at the bus is

A. 61.73 MVAR **B.** 55.56 MVAR
C. 45 MVAR **D.** 40.5 MVAR

Q7.91.1 MCQ 1991 1M

When a fixed amount of power is to be transmitted, the efficiency of transmission increases when

A. Voltage decreases, power factor remains constant

B. Voltage increases, power factor increases

C. Voltage decreases, power factor decreases

D. Voltage remains constant, power factor decreases

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q7.25.1 0.975 to 0.985	Q7.25.2 8.5 to 10	Q7.16.1 B
Q7.16.2 47 to 49	Q7.14.1 C	Q7.14.2 0.75
Q7.12.1 B	Q7.11.1 C	Q7.09.1 B
Q7.08.1 C	Q7.07.1 B	Q7.06.1 D
Q7.05.1 D	Q7.03.1 B	Q7.03.2 B
Q7.02.1 A	Q7.01.1 C	Q7.01.2 C
Q7.01.3 B	Q7.00.1 B	Q7.00.2 *
Q7.99.1 B	Q7.98.1 B	Q7.98.2 D
Q7.97.1 *	Q7.97.2 7.76 μ F	Q7.94.1 D
Q7.91.1 B		

UNIT

VIII

Economic Load Dispatch

Topics

1. Economic Load Dispatch with transmission losses
2. Economic Load Dispatch without transmission losses

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 18	**MCQ:** 11	**MSQ:** 0	**NAT:** 7
1M: 5	**2M:** 11	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2024)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q8.26.1 NAT 2026 2M

A system with two generators G1 and G2 (without generator limits) is shown.

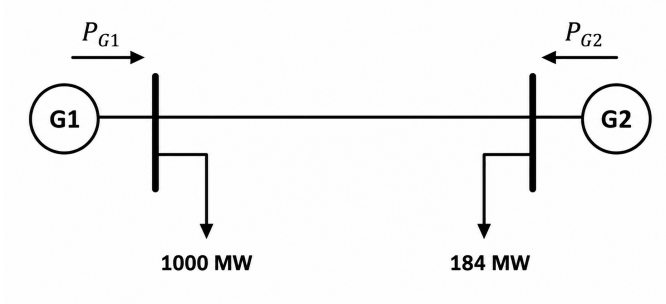


Fig. Q8.26.1

The total load on the system is 1184 MW. The expressions for the cost of generation (C_1 and C_2) and real power loss (P_{Loss}) are as follows:

$$C_1(P_{G1}) = 1000 + 50P_{G1} + 0.01(P_{G1})^2 \quad \text{Rs}$$
$$C_2(P_{G2}) = 2000 + 50P_{G2} + 0.001(P_{G2})^2 \quad \text{Rs}$$
$$P_{\text{Loss}} = 0.001(P_{G2} - 50)^2 \quad \text{MW}$$

When the generators are operating at their optimal generation, meeting the total load requirement, the real power loss in the system is _____ MW *(Round off to one decimal place)*

Consider the Lagrange multiplier $\lambda = 70.25$ for optimal generation.

Q8.25.1 NAT 2025 2M

Two units, rated at 100 MW and 150 MW, are enabled for economic load dispatch. When the overall incremental cost is 10,000 Rs./MWh, the units are dispatched to 50 MW and 80 MW respectively. At an overall incremental cost of 10,600 Rs./MWh, the power output of the units are 80 MW and 92 MW, respectively. The total plant MW-output (without overloading any unit) at an overall incremental cost of 11,800 Rs./MWh is _____ *(round off to the nearest integer).*

Q8.24.1 NAT 2024 1M

The incremental cost curves of two generators (Gen A and Gen B) in a plant supplying a common load are shown in the figure. If the incremental cost of supplying the common load is Rs. 7400 per MWh, then the common load in MW is _____ (rounded off to the nearest integer).

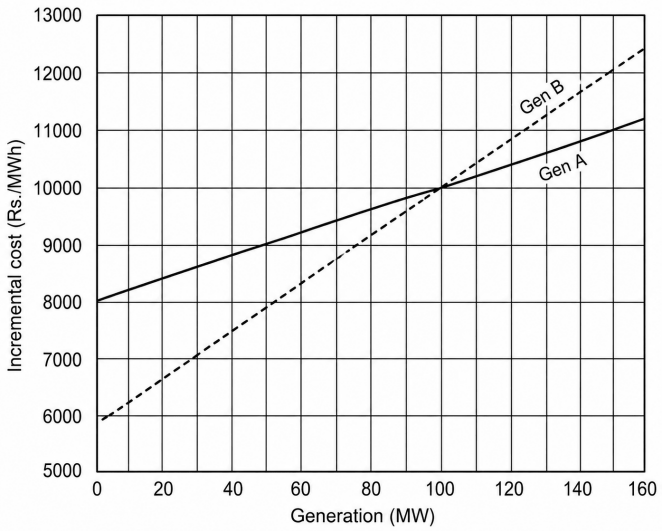


Fig. Q8.24.1

Q8.23.1 MCQ 2023 1M

The expressions of fuel cost of two thermal generating units as a function of the respective power generation P_{G1} and P_{G2} are given as

$$F_1(P_{G1}) = 0.1aP_{G1}^2 + 40P_{G1} + 120 \text{ Rs./hour}$$
$$0 \text{ MW} \le P_{G1} \le 350 \text{ MW}$$
$$F_2(P_{G2}) = 0.2P_{G2}^2 + 30P_{G2} + 100 \text{ Rs./hour}$$
$$0 \text{ MW} \le P_{G2} \le 300 \text{ MW}$$

where a is a constant. For a given value of a, optimal dispatch requires the total load of 290 MW to be shared as $P_{G1} = 175$ MW and $P_{G2} = 115$ MW. With the remaining load unchanged, the value of a is increased by 10% and optimal dispatch is carried out. The changes in P_{G1} and the total cost of generation, F ($= F_1 + F_2$) in Rs./hour will be as follows.

A. P_{G1} will decrease and F will increase

B. Both P_{G1} and F will increase

C. P_{G1} will increase and F will decrease

D. Both P_{G1} and F will decrease

Q8.22.1 MCQ 2022 2M

The fuel cost functions in rupees/hour for two 600 MW thermal power plants are given by

$$C_1 = 350 + 6P_1 + 0.004P_1^2$$

$$C_2 = 450 + aP_2 + 0.003P_2^2$$

where P_1 and P_2 are power generated by plant 1 and plant 2, respectively, in MW and a is constant. The incremental cost of power (λ) is 8 rupees per MWh. The two thermal power plants together meet a total power demand of 550 MW. The optimal generation of plant 1 and plant 2 in MW, respectively, are

A. 200, 350 **B.** 250, 300
C. 325, 225 **D.** 350, 200

Q8.21.1 MCQ 2021 1M

Two generators have cost functions F_1 and F_2. Their incremental-cost characteristics are

$$\frac{dF_1}{dP_1} = 40 + 0.2P_1$$

$$\frac{dF_2}{dP_2} = 32 + 0.4P_2$$

They need to deliver a combined load of 260 MW. Ignoring the network losses, for economic operation, the generations P_1 and P_2 (in MW) are

A. $P_1 = P_2 = 130$ **B.** $P_1 = 160,\ P_2 = 100$
C. $P_1 = 140,\ P_2 = 120$ **D.** $P_1 = 120,\ P_2 = 140$

Q8.15.1 NAT 2015 2M

Consider the economic dispatch problem for a power plant having two generating units. The fuel costs in Rs/MWh along with the generation limits for the two units are given below:

$$C_1(P_1) = 0.01P_1^2 + 30P_1 + 10$$

$$100\,\text{MW} \le P_1 \le 150\,\text{MW}$$

$$C_2(P_2) = 0.05P_2^2 + 10P_2 + 10$$

$$100\,\text{MW} \le P_2 \le 180\,\text{MW}$$

The incremental cost (in Rs/MWh) of the power plant when it supplies 200 MW is _____.

Q8.15.2 NAT 2015 2M

The incremental costs (in Rupees/MWh) of operating two generating units are functions of their respective powers P_1 and P_2 in MW, and are given by

$$\frac{dC_1}{dP_1} = 0.2P_1 + 50$$

$$\frac{dC_2}{dP_2} = 0.24P_2 + 40$$

where

$$20\,\text{MW} \le P_1 \le 150\,\text{MW}$$

$$20\,\text{MW} \le P_2 \le 150\,\text{MW}$$

For a certain load demand, P_1 and P_2 have been chosen such that

$$\frac{dC_1}{dP_1} = 76\,\text{Rs/MWh}$$

and

$$\frac{dC_2}{dP_2} = 68.8\,\text{Rs/MWh}$$

If the generations are rescheduled to minimize the total cost, then P_2 is _____.

Q8.14.1 MCQ 2014 2M

The fuel cost functions of two power plants are

$$\text{Plant } P_1: \quad C_1 = 0.05P_{g1}^2 + AP_{g1} + B$$

$$\text{Plant } P_2: \quad C_2 = 0.10P_{g2}^2 + 3AP_{g2} + 2B$$

where, P_{g1} and P_{g2} are the generated powers of two plants, and A and B are the constants. If the two plants optimally share 1000 MW load at incremental fuel cost of 100 Rs/MWh, the ratio of load shared by plants P_1 and P_2 is

A. 1 : 4 **B.** 2 : 3 **C.** 3 : 2 **D.** 4 : 1

Q8.12.1 MCQ 2012 1M

The figure shows a two-generator system supplying a load of $P_D = 40$ MW, connected at bus 2. The fuel cost of generators G_1 and G_2 are:

$$C_1(P_{G1}) = 10{,}000 \text{ Rs/MWh}$$

and

$$C_2(P_{G2}) = 12{,}500 \text{ Rs/MWh}$$

and the loss in the line is

$$P_{\text{loss(pu)}} = 0.5P_{G1(\text{pu})}^2$$

where the loss coefficient is specified in pu on a 100 MVA base. The most economic power generation schedule in MW is

A. $P_{G1} = 20,\ P_{G2} = 22$ **B.** $P_{G1} = 22,\ P_{G2} = 20$
C. $P_{G1} = 20,\ P_{G2} = 20$ **D.** $P_{G1} = 0,\ P_{G2} = 40$

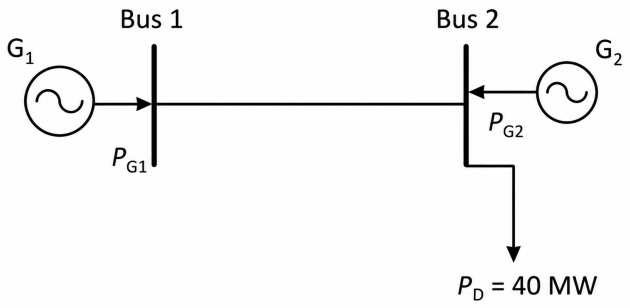


Fig. Q8.12.1

Q8.11.1 MCQ 2011 2M

A load center of 120 MW derives power from two power stations connected by 220 kV transmission lines of 25 km and 75 km as shown in the figure below. The three generators G_1, G_2 and G_3 are of 100 MW capacity each and have identical fuel cost characteristics. The minimum loss generation schedule for supplying the 120 MW load is

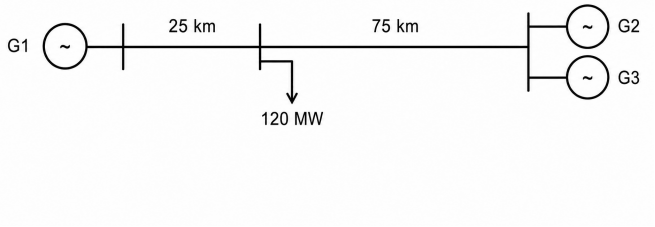


Fig. Q8.11.1

A. $P_1 = 80$ MW + losses, $P_2 = 20$ MW, $P_3 = 20$ MW

B. $P_1 = 60$ MW, $P_2 = 30$ MW + losses, $P_3 = 30$ MW

C. $P_1 = 40$ MW, $P_2 = 40$ MW, $P_3 = 40$ MW + losses

D. $P_1 = 30$ MW + losses, $P_2 = 45$ MW, $P_3 = 45$ MW

Q8.08.1 MCQ 2008 2M

A lossless power system has to serve a load of 250 MW. There are two generators (G_1 and G_2) in the system with cost curves C_1 and C_2 respectively defined as follows:

$$C_1(P_{G1}) = P_{G1} + 0.055P_{G1}^2$$

$$C_2(P_{G2}) = 3P_{G2} + 0.03P_{G2}^2$$

where P_{G1} and P_{G2} are the MW injections from generator G_1 and G_2 respectively. Then, the minimum cost dispatch will be

A. $P_{G1} = 250$ MW, $P_{G2} = 0$ MW

B. $P_{G1} = 150$ MW, $P_{G2} = 100$ MW

C. $P_{G1} = 100$ MW, $P_{G2} = 150$ MW

D. $P_{G1} = 0$ MW, $P_{G2} = 250$ MW

Q8.07.1 MCQ 2007 1M

The incremental cost curves in Rs/MWhr for two generators supplying a common load of 700 MW are shown in the figure. The maximum and minimum generation limits are also indicated.

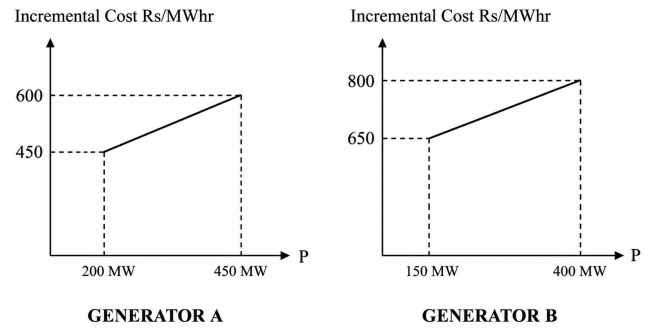


Fig. Q8.07.1

The optimum generation schedule is

A. Generator A: 400 MW, Generator B: 300 MW

B. Generator A: 350 MW, Generator B: 350 MW

C. Generator A: 450 MW, Generator B: 250 MW

D. Generator A: 425 MW, Generator B: 275 MW

Q8.05.1 MCQ 2005 2M

A load centre is at an equidistant from the two thermal generating stations G_1 and G_2 as shown in figure below. The fuel cost characteristics of the generating stations are given by

$$F_1 = a + bP_1 + cP_1^2 \quad \text{Rs/hour}$$

$$F_2 = a + bP_2 + 2cP_2^2 \quad \text{Rs/hour}$$

where P_1 and P_2 are the generation in MW of G_1 and G_2, respectively. For most economic generation to meet 300 MW of load, P_1 and P_2, respectively, are

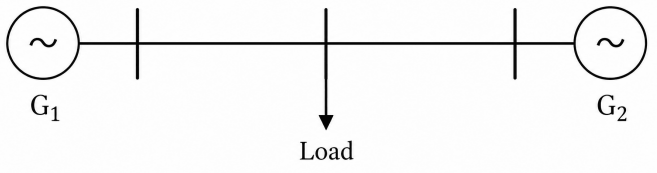


Fig. Q8.05.1

A. 150, 150 **B.** 100, 200 **C.** 200, 100 **D.** 175, 125

Q8.03.1 MCQ 2003 2M

Incremental fuel costs (in some appropriate unit) for a power plant consisting of three generating units are

$$IC_1 = 20 + 0.3P_1$$

$$IC_2 = 30 + 0.4P_2$$

$$IC_3 = 30$$

where P_i is the power in MW generated by unit i, for $i = 1, 2$ and 3. Assume that all the three units are operating all the time. Minimum and maximum loads on each unit are 50 MW and 300 MW respectively. If the plant is operating on economic load dispatch to supply the total power demand of 700 MW, the power generated by each unit is

A. $P_1 = 242.86$ MW, $P_2 = 157.14$ MW, $P_3 = 300$ MW

B. $P_1 = 157.14$ MW, $P_2 = 242.86$ MW, $P_3 = 300$ MW

C. $P_1 = 300$ MW, $P_2 = 300$ MW, $P_3 = 100$ MW

D. $P_1 = 233.3$ MW, $P_2 = 233.3$ MW, $P_3 = 233.4$ MW

Q8.01.1 NAT 2001 5M

A power system has two generators with the following cost curves:

$$C_1(P_{G1}) = 0.006P_{G1}^2 + 8P_{G1} + 350$$

$$C_2(P_{G2}) = 0.009P_{G2}^2 + 7P_{G2} + 400$$

where the costs are in thousand rupees per hour. The generator limits are $100 \text{ MW} \le P_{G1} \le 650$ MW

$50 \text{ MW} \le P_{G2} \le 500$ MW.

A load demand of 600 MW is supplied by the generators in an optimal manner. Neglecting losses in the transmission network, determine the optimal generation of each generator.

Q8.00.1 MCQ 2000 2M

The incremental cost characteristics of two generators delivering 200 MW are as follows

$$\frac{dF_1}{dP_1} = 2.0 + 0.01P_1$$

$$\frac{dF_2}{dP_2} = 1.6 + 0.2P_2$$

For economic operation, the generations P_1 and P_2 should be

A. $P_1 = P_2 = 100$ MW

B. $P_1 = 80$ MW, $P_2 = 120$ MW

C. $P_1 = 200$ MW, $P_2 = 0$ MW

D. $P_1 = 120$ MW, $P_2 = 80$ MW

Q8.98.1 NAT 1998 5M ◯ ◯ ◯

In a power system, the fuel inputs per hour of plants 1 and 2 are given as

$$F_1 = 0.20P_1^2 + 30P_1 + 100$$

$$F_2 = 0.25P_2^2 + 40P_2 + 150$$

Rs per hour

The limits of generators are $20 \leq P_1 \leq 80$ MW $40 \leq P_2 \leq 200$ MW. Find the economic operating schedule of generation, if the load demand is 130 MW, neglecting transmission losses.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q8.26.1 **20**	Q8.25.1 **216**	Q8.24.1 **35**
Q8.23.1 **A**	Q8.22.1 **B**	Q8.21.1 **B**
Q8.15.1 **20**	Q8.15.2 **135 to 137**	Q8.14.1 **D**
Q8.12.1 **A**	Q8.11.1 **A**	Q8.08.1 **C**
Q8.07.1 **C**	Q8.05.1 **C**	Q8.03.1 **A**
Q8.01.1 *	Q8.00.1 **MTA**	Q8.98.1 *

UNIT

IX

Overhead Transmission Lines (O.H.T.L.) and Underground Cables (UG Cable)

Topics

1. Overhead Transmission Lines
2. Underground Cables
3. Electric Field Distribution and Insulators

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 15	**MCQ:** 8	**MSQ:** 0	**NAT:** 7
1M: 9	**2M:** 2	**Descriptive:** 4	**Avg Weightage** ~1M	**Range:** 0M(2014)–1M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q9.22.1 NAT 2022 1M

If only 5% of the supplied power to a cable reaches the output terminal, the power loss in the cable, in decibels, is _____ (round off to nearest integer)

Q9.21.1 NAT 2021 1M

Two single-core power cables have total conductor resistances of 0.7 Ω and 0.5 Ω, respectively, and their insulation resistances (between core and sheath) are 600 MΩ and 900 MΩ, respectively. When the two cables are joined in series, the ratio of insulation resistance to conductor resistance is _____ $\times 10^6$

Q9.14.1 MCQ 2014 1M

The undesirable property of an electrical insulating material is

A. high dielectric strength

B. high relative permittivity

C. high thermal conductivity

D. high insulation resistivity

Q9.07.1 MCQ 2007 1M

Consider a bundled conductor of an overhead transmission line consisting of three identical sub-conductors placed at the corners of an equilateral triangle as shown in the figure. Neglect the charges on the other phase conductors and ground. Assume that the spacing between sub-conductors is much larger than their radius.

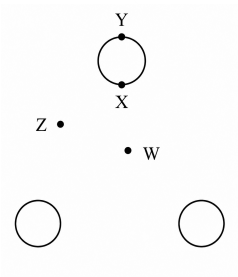


Fig. Q9.07.1

The maximum electric field intensity is experienced at

A. Point X **B.** Point Y **C.** Point Z **D.** Point W

Q9.05.1 MCQ 2005 1M

The insulation strength of an EHV transmission line is mainly governed by

A. load power factor **B.** switching over-voltages

C. harmonics **D.** corona

Q9.04.1 MCQ 2004 2M

A 110 kV, single core coaxial, XLPE insulated power cable delivering power at 50 Hz, has a capacitance of 125 nF/km. If the dielectric loss tangent of XLPE is 2×10^{-4}, the dielectric power loss in this cable in W/km is

A. 5.0 **B.** 31.7 **C.** 37.8 **D.** 189.0

Q9.04.2 MCQ 2004 2M

A lightning stroke discharges impulse current of 10 kA (peak) on a 400 kV transmission line having surge impedance of 250 Ω. The magnitude of transient over-voltage travelling in either direction assuming equal distribution from the point of lightning strike will be

A. 1250 kV **B.** 1650 kV **C.** 2500 kV **D.** 2900 kV

Q9.02.1 MCQ 2002 1M

A long wire composed of a smooth round conductor runs above and parallel to the ground (assumed to be a large conducting plane). A high voltage exists between the conductor and the ground. The maximum electric stress occurs at

A. the upper surface of the conductor

B. the lower surface of the conductor

C. the ground surface

D. midway between the conductor and ground

Q9.01.1 NAT 2001 5M

A step voltage surge of magnitude 100 kV travels on an overhead transmission line having a surge impedance of 300 Ω and reaches a junction with an underground cable having a surge impedance of 50 Ω. Determine the magnitude of the transmitted voltage wave entering the cable.

Q9.99.1 NAT 1999 5M

A 6.6 kV, 50 Hz, single core lead-sheathed cable has the following data:

- Conductor diameter: 1.5 cm
- Length: 4 km
- Internal diameter of the sheath: 3 cm
- Resistivity of insulation: 1.3×10^{12} Ω-m
- Relative permittivity of insulation: 3.5

Calculate:
(a) the insulation resistance
(b) the capacitance
(c) the maximum electric stress in the insulation

Q9.99.2 NAT 1999 5M

In a transmission line each conductor is at 20 kV and is supported by a string of 3 suspension insulators. The air capacitance between each cap-pin junction and tower is one-fifth of the capacitance C of each insulator unit. A guard ring, effective only over the line-end insulator unit is fitted so that the voltages on the two units nearest the line-end are equal.
(a) Calculate the voltage on the line-end unit.
(b) Calculate the value of capacitance C_x required.

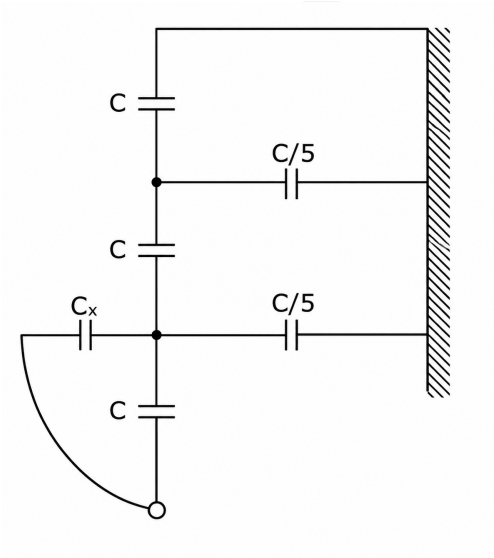


Fig. Q9.99.2

Q9.95.1 MCQ 1995 1M

The main criterion for selection of the size of a distributor for a radial distribution system is

A. Voltage drop **B.** Corona loss
C. Temperature rise **D.** Capital cost

Q9.95.2 MCQ 1995 1M

The insulation resistance of a cable of length 10 km is 1 MΩ.
For a length of 100 km of the same cable, the insulation resistance will be

A. 1 MΩ **B.** 10 MΩ **C.** 0.1 MΩ **D.** 0.01 MΩ

Q9.95.3 NAT 1995 1M

The rated load of an underground cable is always _____ than its natural load.

Q9.95.4 NAT 1995 5M

Two identical three-phase transmission lines are connected in parallel to supply a total load of 100 MW at 132 kV and 0.8 p.f lagging at the receiving end. The ABCD constants of each transmission line are $A = D = 0.98\angle 1°, B = 100\angle 75°$ Ω/phase, $C = 0.0005\angle 90°$ S/phase Determine:

(a) The ABCD constants of the combined network.

(b) The sending-end power factor.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q9.22.1 13	Q9.21.1 300	Q9.14.1 B
Q9.07.1 B	Q9.05.1 B	Q9.04.1 B
Q9.04.2 A	Q9.02.1 B	Q9.01.1 28.57 kV
Q9.99.1 *	Q9.99.2 *	Q9.95.1 A
Q9.95.2 C	Q9.95.3 LESS	Q9.95.4 *

UNIT

X

Distribution Systems, HVDC, Sag and Corona

Topics

1. Types of Distribution Systems
2. DC Distribution
3. High Voltage DC (HVDC) Transmission
4. Sag
5. Corona

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 21	**MCQ:** 20	**MSQ:** 0	**NAT:** 1
1M: 14	**2M:** 6	**Descriptive:** 1	**Avg Weightage** ~1M	**Range:** 1M(2014)–1M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Q10.14.1 MCQ 2014 2M

A distribution feeder of 1 km length having resistance, but negligible reactance, is fed from both the ends by 400 V, 50 Hz balanced sources. Both voltage sources S_1 and S_2 are in phase. The feeder supplies concentrated loads of unity power factor as shown in the figure. The contributions of S_1 and S_2 in 100 A current supplied at location P respectively, are

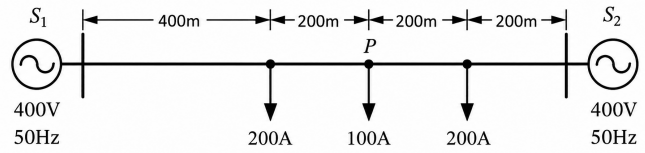


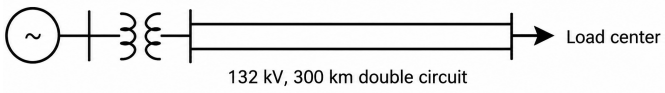
Fig. Q10.14.1

A. 75 A and 25 A
B. 50 A and 50 A
C. 25 A and 75 A
D. 0 A and 100 A

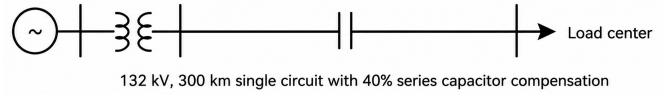
Q10.11.1 MCQ 2011 1M

A nuclear power station of 500 MW capacity is located at 300 km away from a load center. Select the most suitable power evacuation transmission configuration among the following options.

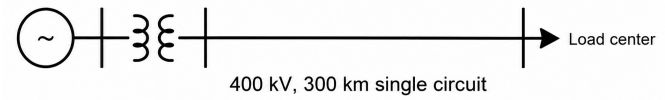
A.



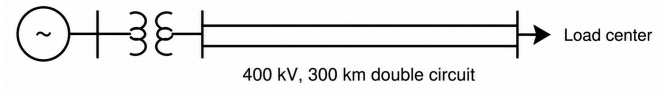
B.



C.



D.



Q10.10.1 MCQ 2010 1M

Power is transferred from system A to system B by an HVDC link as shown in the figure. If the voltage V_{AB} and V_{CD} are as indicated in figure, and $I > 0$, then

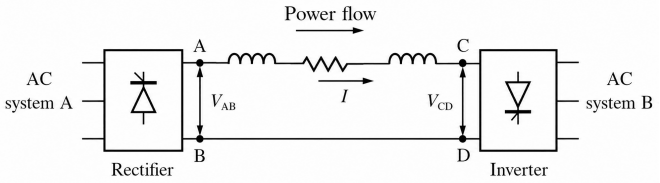


Fig. Q10.10.1

A. $V_{AB} < 0,\ V_{CD} < 0,\ V_{AB} > V_{CD}$

B. $V_{AB} > 0,\ V_{CD} > 0,\ V_{AB} < V_{CD}$

C. $V_{AB} > 0,\ V_{CD} > 0,\ V_{AB} > V_{CD}$

D. $V_{AB} > 0,\ V_{CD} < 0$

Q10.07.1 MCQ 2007 1M

Two regional systems, each having several synchronous generators and loads, are interconnected by an AC line and an HVDC link as shown in the figure.

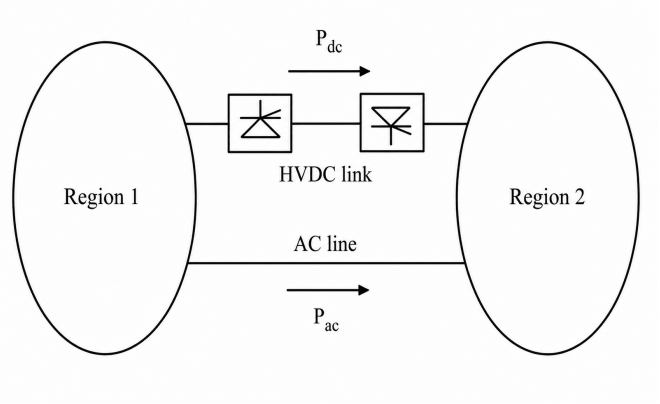


Fig. Q10.07.1

Which of the following statements is true in the steady state?

A. Both regions need not have the same frequency

B. The total power flow between the regions $(P_{ac} + P_{dc})$ can be changed by controlling the HVDC converters alone

C. The power sharing between the AC line and the HVDC link can be changed by controlling the HVDC converters alone

D. The direction of power flow in the HVDC link (P_{dc}) cannot be reversed

Q10.06.1 MCQ 2006 1M

An HVDC link consists of rectifier, inverter, transmission line and other equipments. Which one of the following is true for this link?

A. The transmission line produces/supplies reactive power

B. The rectifier consumes reactive power and the inverter supplies reactive power from/to the respective connected AC systems

C. Rectifier supplies reactive power and the inverter consumes reactive power to/from the respective connected AC systems

D. Both the converters (rectifier and inverter) consume reactive power from the respective connected AC systems

Q10.06.2 MCQ 2006 2M

A single phase transmission line and a telephone line are both symmetrically strung one below the other, in horizontal configurations, on a common tower. The shortest and longest distances between the phase and telephone conductors are 2.5 m and 3 m respectively. The voltage (volt/km) induced in the telephone circuit, due to 50 Hz current of 100 amps in the power circuit is

A. 4.81 **B.** 3.56 **C.** 2.29 **D.** 1.27

Q10.05.1 MCQ 2005 1M

High Voltage DC (HVDC) transmission is mainly used for

A. bulk power transmission over very long distances

B. inter-connecting two systems with the same nominal frequency

C. eliminating reactive power requirement in the operation

D. minimizing harmonics at the converter stations

Q10.03.1 MCQ 2003 1M

Choose two appropriate auxiliary components of a HVDC transmission system from the following:

P. D.C. line inductor

Q. A.C. line inductor

R. Reactive power sources

S. Distance relays on D.C. line

T. Series capacitance of A.C. line

A. P and Q **B.** P and R **C.** Q and S **D.** S and T

Q10.03.2 MCQ 2003 2M

A dc distribution system is shown in figure below with load currents as marked. The two ends of the feeder are fed by voltage sources such that

$$V_P - V_Q = 3 \text{ V}$$

The value of the voltage V_P for a minimum voltage of 220 V at any point along the feeder is

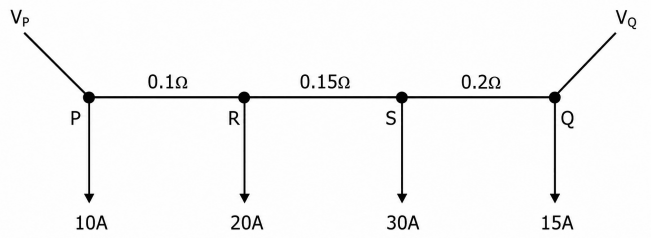


Fig. Q10.03.2

A. 225.89 V **B.** 222.89 V **C.** 220.0 V **D.** 228.58 V

Q10.02.1 MCQ 2002 2M

A three phase thyristor bridge rectifier is used in a HVDC link. The firing angle α (as measured from the point of natural commutation) is constrained to lie between 5° and 30°. If the dc side current and ac side voltage magnitudes are constant, which of the following statements is true (neglect harmonics in the ac side currents and commutation overlap in your analysis)

A. Reactive power absorbed by the rectifier is maximum when $\alpha = 5°$.

B. Reactive power absorbed by the rectifier is maximum when $\alpha = 30°$.

C. Reactive power absorbed by the rectifier is maximum when $\alpha = 15°$.

D. Reactive power absorbed by the rectifier is maximum when $\alpha = 10°$.

Q10.00.1 MCQ 2000 1M

Out of the considerations (i) to (iv) listed below,

(i) no distance limitation related to steady state stability

(ii) no reactive power requirement from the system at the two terminals

(iii) no substantial effect on fault level of the two systems at the terminals in spite of the interconnection

(iv) no corona problems

The considerations, which constitute advantages of HVDC transmission, are

A. all of the above **B.** (i) and (iii)
C. (iii) and (iv) **D.** (ii) and (iv)

Q10.00.2 MCQ 2000 2M

The corona loss on a particular system at 50 Hz is 1 kW/km per phase. The corona loss at 60 Hz would be

A. 1 kW/km per phase **B.** 0.83 kW/km per phase
C. 1.2 kW/km per phase **D.** 1.13 kW/km per phase

Q10.99.1 MCQ 1999 1M

Corona losses are minimized when

A. conductor size is reduced

B. smooth conductor is reduced

C. sharp points are provided in the line hardware

D. current density in conductors is reduced

Q10.99.2 MCQ 1999 1M

In a DC transmission line

A. it is necessary for the sending end and receiving end to be operated in synchronism.

B. the effects of inductive and capacitive reactances are greater than in an AC transmission line of the same rating.

C. there are no effects due to inductive and capacitive reactances

D. power transfer capability is limited by stability considerations.

Q10.99.3 MCQ 1999 2M

A single phase AC distributor supplies two single phase loads as shown in figure below. The voltage drop from A to C is:

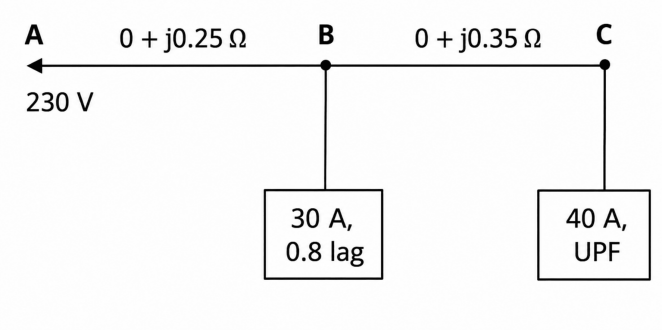


Fig. Q10.99.3

A. 4.5 V **B.** 31.5 V **C.** 30 V **D.** 20 V

Q10.98.1 MCQ 1998 1M

Bulk power transmission over long HVDC lines are preferred, on account of

A. low cost of HVDC terminals **B.** no harmonic problems
C. minimum line power losses **D.** simple protection

Q10.95.1 MCQ 1995 1M

The insulation level of a 400 kV EHV overhead transmission line is decided on the basis of

A. Lightning over voltage

B. Switching over voltage

C. Corona inception voltage

D. Radio and TV interference

Q10.95.2 NAT 1995 5M

In a string of three identical suspension insulator units supporting a transmission line conductor, if the self capacitance of each unit is denoted as C Farads, the capacitance of each connector pin to ground can be taken as $0.1C$ Farads. Determine the voltage distribution across the string if the maximum permissible voltage per unit is 20 kV. Also determine the string efficiency.

Q10.94.1 MCQ 1994 1M

HVDC transmission is preferred to EHV-AC because

A. HVDC terminal equipment are inexpensive

B. VAR compensation is not required in HVDC systems

C. System stability can be improved

D. Harmonics problem is avoided

Q10.92.1 MCQ 1992 1M

For the same cross-sectional area and conductor material, a distributor requires a larger cross-section than a feeder because the distributor is designed based on

A. Current carrying capacity only

B. Mechanical strength only

C. Corona loss limitation

D. Permissible voltage drop

Q10.91.1 MCQ 1991 1M

Ring main distribution system is preferred to a radial system, because

A. It is less expensive

B. Voltage drop in the feeder is less

C. Power factor is higher

D. Supply is more reliable

Q10.94.1	Q10.92.1	Q10.91.1
C	D	D

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q10.14.1 D	Q10.11.1 D	Q10.10.1 C
Q10.07.1 C	Q10.06.1 B	Q10.06.2 C
Q10.05.1 A	Q10.03.1 B	Q10.03.2 A
Q10.02.1 B	Q10.00.1 B	Q10.00.2 D
Q10.99.1 D	Q10.99.2 C	Q10.99.3 C
Q10.98.1 C	Q10.95.1 B	Q10.95.2 *

UNIT

XI

Relays Circuit Breakers

Topics

1. Working Principle of Relay
2. Types of Relay, Plug Setting Multiplier and Time Setting Multiplier- PSM, TSM
3. Directional Relay, Distance Relay
4. Differential Relay
5. Protection Zones, Distance Protection
6. Introduction and ARC interruption
7. Current Chopping

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 39	**MCQ:** 29	**MSQ:** 1	**NAT:** 9
1M: 26	**2M:** 10	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2023)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q11.26.1 MCQ 2026 2M

The operating characteristic of a reactance relay is given by $X \leq 1\,\Omega$, where X is the reactance calculated by the relay. Its operating characteristic in the admittance plane (G-B plane, where G and B denote conductance and susceptance, respectively, expressed in $\mho$) is given by:

A. $G^2 + (B + 0.5)^2 \geq \frac{1}{4}$ **B.** $B \geq 1$

C. $(G - 1)^2 + B^2 \leq \frac{1}{2}$ **D.** $G^2 + (B - 1)^2 \geq \frac{1}{2}$

Q11.23.1 MCQ 2023 1M

For the three-bus power system shown in the figure, the trip signals to the circuit breakers B_1 to B_9 are provided by overcurrent relays R_1 to R_9, respectively, some of which have directional properties also. The necessary condition for the system to be protected for short circuit fault at any part of the system between bus 1 and the R-L loads with isolation of minimum portion of the network using minimum number of directional relays is

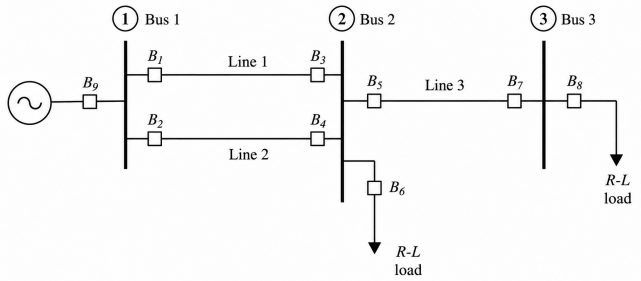


Fig. Q11.23.1

A. R_3 and R_4 are directional overcurrent relays blocking faults towards bus 2

B. R_3 and R_4 are directional overcurrent relays blocking faults towards bus 2 and R_7 is directional overcurrent relay blocking faults towards bus 3

C. R_3 and R_4 are directional overcurrent relays blocking faults towards Line 1 and Line 2, respectively, R_7 is directional overcurrent relay blocking faults towards Line 3 and R_5 is directional overcurrent relay blocking faults towards bus 2

D. R_3 and R_4 are directional overcurrent relays blocking faults towards Line 1 and Line 2, respectively.

Q11.22.1 MCQ 2022 1M

The most commonly used relay, for the protection of an alternator against loss of excitation, is

A. offset Mho relay **B.** over current relay

C. differential relay **D.** Buchholz relay

Q11.20.1 MCQ 2020 1M

A lossless transmission line with 0.2 pu reactance per phase uniformly distributed along the length of the line, connecting a generator bus to a load bus, is protected up to 80% of its length by a distance relay placed at the generator bus. The generator terminal voltage is 1 pu. There is no generation at the load bus. The threshold pu current for operation of the distance relay for a solid three phase-to-ground fault on the transmission line is closest to:

A. 1.00 **B.** 3.61 **C.** 5.00 **D.** 6.25

Q11.19.1 NAT 2019 2M

In a 132 kV system, the series inductance up to the point of circuit breaker location is 50 mH. The shunt capacitance at the circuit breaker terminal is 0.05 μF. The critical value of resistance in ohms required to be connected across the circuit breaker contacts which will give no transient oscillation is _____.

Q11.16.1 MCQ 2016 1M

A power system with two generators is shown in the figure below. The system (generators, buses and transmission lines) is protected by six overcurrent relays R_1 to R_6. Assuming a mix of directional and nondirectional relays at appropriate locations, the remote backup relays for R_4 are

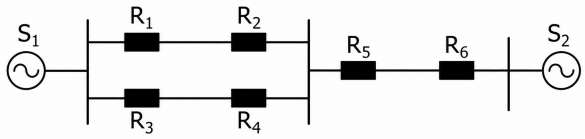


Fig. Q11.16.1

A. R_1, R_2 **B.** R_2, R_6 **C.** R_2, R_5 **D.** R_1, R_6

Q11.14.1 NAT 2014 2M

The overcurrent relays for the line protection and loads connected at the buses are shown in the figure.

The relays are IDMT in nature having the characteristic

$$t_{op} = \frac{0.14 \times \text{Time Multiplier Setting}}{(\text{Plug Setting Multiplier})^{0.02} - 1}$$

The maximum and minimum fault currents at bus B are 2000 A and 500 A respectively. Assuming the time multiplier setting and plug setting for relay R_B to be 0.1 and 5 A respectively, the operating time of R_B (in seconds) is _____.

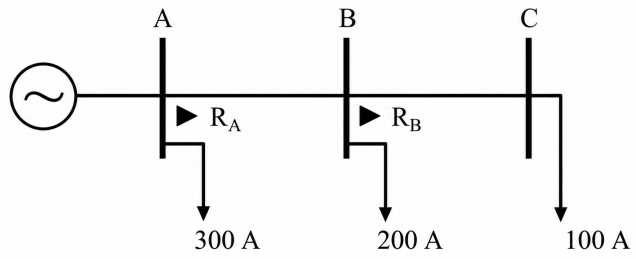


Fig. Q11.14.1

Q11.11.1 MCQ 2011 1M

A negative sequence relay is commonly used to protect

A. an alternator **B.** a transformer
C. a transmission line **D.** a bus bar

Q11.10.1 MCQ 2010 1M

A three-phase, 33 kV oil circuit breaker is rated 1200 A, 2000 MVA, 3 s. The symmetrical breaking current is

A. 1200 A **B.** 3600 A **C.** 35 kA **D.** 104.8 kA

Q11.10.2 MCQ 2010 1M

Consider a stator winding of an alternator with an internal high-resistance ground fault. The currents under the fault condition are as shown in the figure. The winding is protected using a differential current scheme with current transformers of ratio 400/5 A as shown. The current through the operating coils is

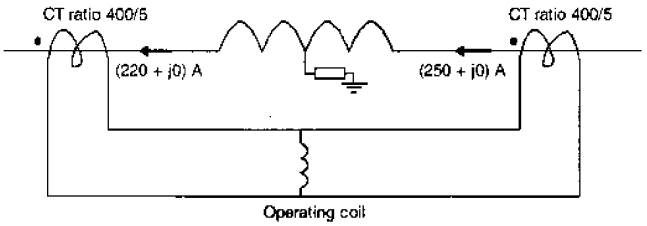


Fig. Q11.10.2

A. 0.1875 A **B.** 0.2 A **C.** 0.375 A **D.** 60 kA

Q11.09.1 MCQ 2009 2M

Match the items in List-I with the items in List-II and select the correct answer using the codes given below the lists.

List-I	**List-II**
Type of transmission line	**Type of distance relay preferred**
(a) Short Line	(1) Ohm Relay
(b) Medium Line	(2) Reactance Relay
(c) Long Line	(3) Mho Relay

A. $a \rightarrow 2,\ b \rightarrow 1,\ c \rightarrow 3$ **B.** $a \rightarrow 3,\ b \rightarrow 2,\ c \rightarrow 1$
C. $a \rightarrow 1,\ b \rightarrow 2,\ c \rightarrow 3$ **D.** $a \rightarrow 1,\ b \rightarrow 3,\ c \rightarrow 2$

Q11.08.1 MCQ 2008 1M

A two-machine power system is shown below. Transmission line XY has positive-sequence impedance of $Z_1\ \Omega$ and zero-sequence impedance of $Z_0\ \Omega$.

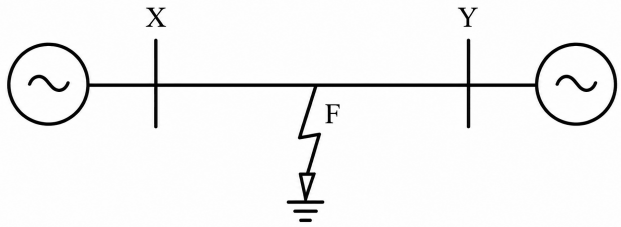


Fig. Q11.08.1

An 'a' phase-to-ground fault with zero fault impedance occurs at the centre of the transmission line. Bus voltage at X and line current from X to F for phase 'a' are given by V_a volts and I_a amperes, respectively. Then, the impedance measured by the ground distance relay located at terminal X of line XY will be given by

A. $Z_1/2\ \Omega$ **B.** $Z_0/2\ \Omega$
C. $(Z_0 + Z_1)/2\ \Omega$ **D.** $V_a/I_a\ \Omega$

Q11.08.2 MCQ 2008 2M

Voltage phasors at the two terminals of a transmission line of length 70 km have a magnitude of 1.0 per unit but are 180° out of phase. Assuming that the maximum load current in the line is $1/5^{th}$ of the minimum 3-phase fault current, which one of the following transmission-line protection schemes will NOT pick up for this condition?

A. Distance protection using mho relays with zone-1 set to 80% of the line impedance

B. Directional overcurrent protection set to pick up at 1.25 times the maximum load current

C. Pilot relaying system with directional comparison scheme

D. Pilot relaying system with segregated phase comparison scheme

Q11.07.1 MCQ 2007 2M

Consider the protection system shown in the figure below. The circuit breakers numbered from 1 to 7 are of identical type. A single line-to-ground fault with zero fault impedance occurs at the midpoint of the line (at point F), but circuit breaker 4 fails to operate (stuck breaker). If the relays are coordinated correctly, a valid sequence of circuit-breaker operation is

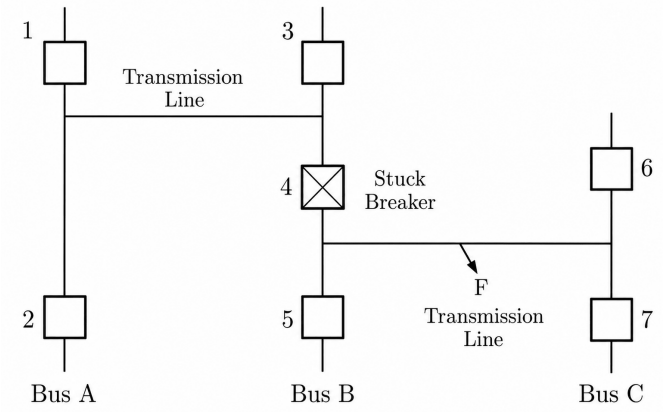


Fig. Q11.07.1

A. 1, 2, 6, 7, 3, 5
B. 1, 2, 5, 6, 7, 3
C. 5, 6, 7, 3, 1, 2
D. 5, 1, 2, 3, 6, 7

Q11.06.1 MCQ 2006 1M

Keeping in view the cost and overall effectiveness, the following circuit breaker is best suited for capacitor bank switching

A. vacuum **B.** air blast **C.** SF_6 **D.** oil

Q11.06.2 MCQ 2006 1M

In a biased differential relay, the bias is defined as a ratio of

A. number of turns of restraining and operating coil

B. operating coil current and restraining coil current

C. fault current and operating coil current

D. fault current and restraining coil current

Q11.04.1 MCQ 2004 1M

The transmission line distance protection relay having the property of being inherently directional is

A. impedance relay **B.** MHO relay
C. OHM relay **D.** reactance relay

Q11.03.1 MCQ 2003 1M

The interrupting time of a circuit breaker is the period between the instant of

A. initiation of short circuit and the arc extinction on an opening operation

B. energizing of the trip circuit and the arc extinction on an opening operation

C. initiation of short circuit and the parting of primary arc contacts

D. energizing of the trip circuit and the parting of primary arc contacts

Q11.03.2 MCQ 2003 2M

A list of relays and the power-system components protected by the relays are given in Group I and Group II respectively.

Group I	Group II
P. Distance relay	1. Transformers
Q. Under-frequency relay	2. Turbines
R. Differential relay	3. Busbars
S. Buchholz relay	4. Shunt capacitors
	5. Alternators
	6. Transmission lines

Choose the correct match.

A. P–6, Q–5, R–3, S–1 **B.** P–4, Q–3, R–2, S–1
C. P–5, Q–2, R–1, S–6 **D.** P–6, Q–4, R–5, S–3

Q11.02.1 MCQ 2002 1M

Consider the problem of relay co-ordination for the distance relays R1 and R2 on adjacent lines of a transmission system as shown below. The Zone 1 and Zone 2 settings for both the relays are indicated on the diagram. Which of the following indicates the correct time setting for the Zone 2 of relays R1 and R2?

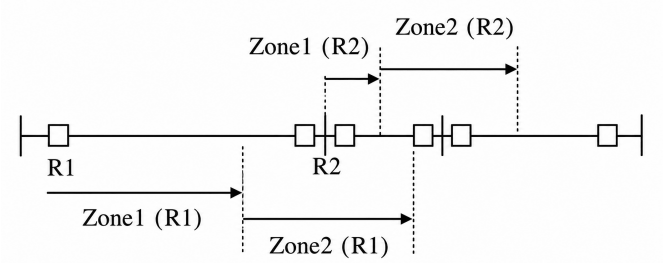


Fig. Q11.02.1

A. $TZ2_{R1} = 0.6\,s,\ TZ2_{R2} = 0.3\,s$

B. $TZ2_{R1} = 0.3\,s,\ TZ2_{R2} = 0.6\,s$

C. $TZ2_{R1} = 0.3\,s,\ TZ2_{R2} = 0.3\,s$

D. $TZ2_{R1} = 0.1\,s,\ TZ2_{R2} = 0.3\,s$

Q11.01.1 MCQ 2001 1M

In the protection of transformers, harmonic restraint is used to guard against

A. magnetizing inrush current **B.** unbalanced operation
C. switching over-voltages **D.** lightning

Q11.00.1 MCQ 2000 1M

In an inverse definite minimum time, electromagnetic type over-current relay the minimum time feature is achieved because of

A. saturation of the magnetic circuit

B. proper mechanical design

C. appropriate time delay element

D. electromagnetic damping

Q11.00.2 MCQ 2000 1M

In a 3-step distance protection, the reach of the three zones of the relay at the beginning of the first line typically extends up to

A. 100% of the first line, 50% of the second line and 20% of the third line

B. 80% of the first line, 50% of the second line and 20% of the third line

C. 80% of the first line, 20% of the second line and 10% of the third line

D. 50% of first line, 50% of second line and 20% of the third line

Q11.00.3 MCQ 2000 2M

The plug setting of a negative sequence relay is 0.2 A. The current transformer ratio is 5 : 1. The minimum value of line-to-line fault current for the operation of the relay is

A. 1 A **B.** $\frac{1}{1.732}$ A **C.** 1.732 A **D.** $\frac{0.2}{1.732}$ A

Q11.99.1 MCQ 1999 1M

Three sections of a feeder are provided with circuit breakers CB1, CB2, CB3, CB4, CB5 and CB6. For a fault F as indicated in figure below.

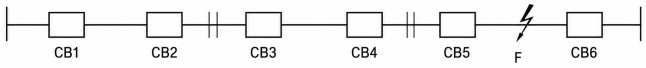


Fig. Q11.99.1

A. CB5 must be set to trip after CB1 trips

B. CB5 must be set to trip after CB3 and CB4 trips

C. CB5 must be set to trip after CB2 trip

D. CB5 must be set to trip before CB1, CB2, CB3 and CB4 trips

Q11.99.2 NAT 1999 5M

Determine the required MVA rating of the circuit breaker CB for the system shown in figure below. Consider 6 MVA as base.
Transformer: 3-phase, 33/11 kV, 6 MVA, $0.01 + j0.08$ p.u. impedance.
Load: 3-phase, 11 kV, 5800 kVA, 0.8 lag, $j0.2$ p.u.impedance.
Impedance of each feeder = $9 + j18\ \Omega$.

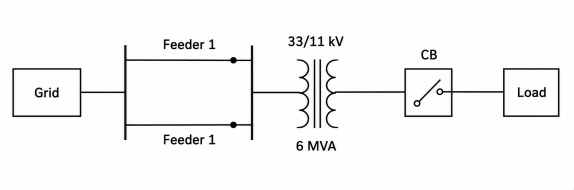


Fig. Q11.99.2

Q11.97.1 MCQ 1997 1M

Reactance relay is normally preferred for protection against

A. earth faults **B.** phase faults
C. open-circuit faults **D.** none of these

Q11.97.2 MCQ 1997 1M

The use of high-speed circuit-breakers

A. reduces the short circuit current

B. improves system stability

C. decreases system stability

D. increases the short circuit current

Q11.96.1 MCQ 1996 1M

If the fault current is 2000 A, the relay setting is 50% and CT ratio is 400:5, the plug setting multiplier will be

A. 25 **B.** 15 **C.** 50 **D.** 10

Q11.96.2 MCQ 1996 1M

Resistance switching is normally employed in

A. All breakers **B.** Bulk oil breakers
C. Minimum oil breakers **D.** Air blast circuit breakers

Q11.95.1 NAT 1995 1M

If the inductance and capacitance of a power system network up to a circuit breaker location are 1 H and 0.01 μ F respectively, the value of the shunt resistor across the circuit breaker required for critical damping of the re-striking voltage is _____

Q11.95.2 NAT 1995 1M

The distance relay with inherent directional property is known as _____ relay.

Q11.95.3 MSQ 1995 5M

Match the following relay applications.

(a) Buchholz relay	(P) Feeder
(b) Translay relay	(Q) Transformer
(c) Carrier current phase comparison relay	(R) Radial distributor
(d) Directional overcurrent relay	(S) Generator
(e) Negative sequence relay	(T) Ring main distributor
	(U) Long overhead transmission line

Q11.95.4 NAT 1995 5M

A three-phase delta-wye connected 30 MVA, 33 kV/11 kV transformer is protected by a simple differential relaying scheme. The CT ratio on the primary side is 500:5 and that on the secondary side is 2000:5. Sketch the CT connection diagram for the relaying scheme. Also calculate the relay current setting for faults drawing up to 200% of the rated current.

Q11.93.1 MCQ 1993 2M

The distribution system shown in the figure is to be protected by an overcurrent system of protection. For proper fault discrimination, directional overcurrent relays will be required at locations

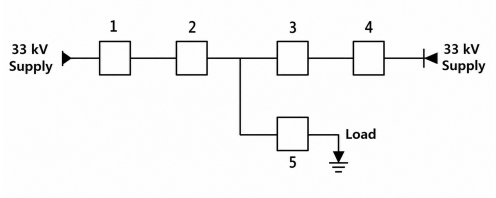


Fig. Q11.93.1

A. 1 and 4 **B.** 2 and 3
C. 1, 4 and 5 **D.** 2, 3 and 5

Q11.92.1 MCQ 1992 1M

A Buchholz relay is used for

A. Protection of a transformer against all internal faults

B. Protection of a transformer against external faults

C. Protection of a transformer against both internal and external faults

D. Protection of induction motors

Q11.92.2 NAT 1992 1M

Match the following:

Equipment	Function
(a) Circuit breaker	(P) Voltage control
(b) Lightning arrester	(Q) Power control
(c) Governor	(R) Over-voltage protection
(d) Exciter	(S) Over-current protection

Q11.91.1 NAT 1991 1M

In a 4-bus system the circuit breakers and the various zones of protection are shown in Figure. If the circuit breakers E, F and G trip, the location of the fault is on _____.

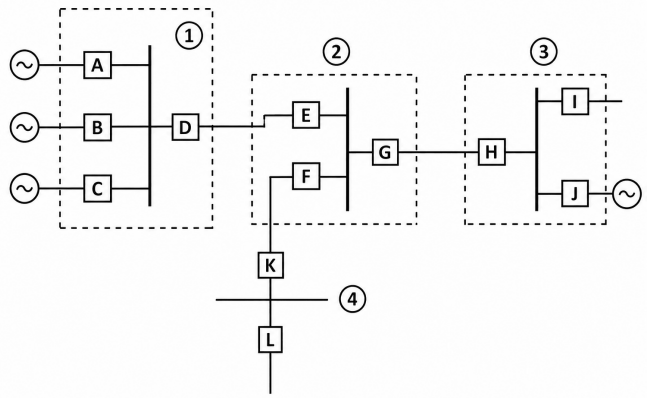


Fig. Q11.91.1

Q11.91.2 NAT 1991 2M

A 50 MVA, 132/66 kV, Δ/Y, three-phase power transformer is protected by percentage differential relays. The current transformers (CTs) located on the delta and wye sides of the power transformer are 300/5 A and 1200/5 A respectively. Determine

(a) the output current at full load,

(b) the relay current at full load,

(c) the minimum relay current setting to permit 25% overload.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q11.26.1 A	Q11.23.1 A	Q11.22.1 A
Q11.20.1 D	Q11.19.1 500	Q11.16.1 D
Q11.14.1 0.21 to 0.23	Q11.11.1 A	Q11.10.1 C
Q11.10.2 C	Q11.09.1 A	Q11.08.1 A
Q11.08.2 A	Q11.07.1 C	Q11.06.1 A
Q11.06.2 B	Q11.04.1 B	Q11.03.1 B
Q11.03.2 A	Q11.02.1 C	Q11.01.1 A
Q11.00.1 A	Q11.00.2 B	Q11.00.3 C
Q11.99.1 D	Q11.99.2 *	Q11.97.1 A
Q11.97.2 B	Q11.96.1 D	Q11.96.2 D
Q11.95.1 5000	Q11.95.2 MHO	Q11.95.3 Q,P,U,T,S
Q11.95.4 *	Q11.93.1 B	Q11.92.1 A
Q11.92.2 *	Q11.91.1 BUS 2	Q11.91.2 *

Also in this Series

Every PrepFusion GATE title is built the same way: the real previous-year papers, nothing paraphrased or reconstructed, arranged so the pattern the exam repeats is visible on the page. Each questions volume has a companion solutions volume that reuses the same question identifiers, so you can move between the two without a lookup table.

NexusX	Network Theory, Analog Electronics, Digital Electronics, Control Systems and Signals & Systems — EC, EE, IN
SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.

PrepFusion